Banks as Firms:
The Macroeconomics of Financial Firm Dynamics*

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Abstract

Why do certain banks grow large while others remain small, and how do these dynamics shape economic activity and policy? I first develop a model in which banks grow by expanding across regional loan markets and capturing market share based on differences in their ability to screen risk, attract borrowers, and source capital. Guided by this theory, I estimate U.S. bank-level fundamentals using a dynamic state-space model and embed the estimates in a quantitative multi-region New Keynesian setting. I find that banks' screening ability is the primary driver of bank dynamics, accounting for three-quarters of a bank's idiosyncratic loan growth. This is because, unlike other fundamentals, screening ability directly affects loan losses, which can trigger an accelerating balance-sheet contraction. These dynamics shape the macroeconomy through two channels. First, the distribution of screening ability determines how effectively credit is allocated to productive firms. Changes in screening ability therefore generate short-run fluctuations in aggregate labor productivity, while the level and dispersion of screening ability lower U.S. long-run labor productivity by roughly 1 and 5 percent, respectively. Second, heterogeneity in bank fundamentals generates asymmetric regional transmission of monetary policy, creating a tradeoff between aggregate stabilization and regional inequality.

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Bank lending is central to both short-run fluctuations and long-run growth because it determines which borrowers receive credit and on what terms. At the same time, banks are heterogeneous firms that vary along several margins. For example, banks differ in how well they can identify productive borrowers to lend to (Hauswald and Marquez, 2003) or in how forcefully they transmit monetary policy (Kashyap and Stein, 2000). Consequently, aggregate outcomes depend not only on how much credit is supplied, but also on which banks supply it.

However, the composition of the banking sector, that is, which banks end up doing the lending, is itself an equilibrium outcome that depends on how individual banks grow, expand across credit markets, and capture market share. Existing macroeconomic models of banking generally abstract from this endogenous evolution of the composition of lenders. They instead treat banks as pass-through entities that transmit aggregate shocks through reduced-form spreads or as a representative intermediary subject to balance-sheet constraints. This abstraction reflects two limitations. Conceptually, banking does not map cleanly into standard firm-production frameworks. Practically, the resulting absence of a well-defined notion of bank productivity has hindered the construction of firm-level measures that could discipline models of lending heterogeneity, competition, and growth.

This paper studies the firm dynamics of banks and how these dynamics shape the allocation of credit, economic activity, and policy.

I proceed in three steps. I first develop a framework for bank dynamics that combines insights from the non-financial firm dynamics literature with a model of banking competition. In this setting, banks grow by expanding across regional credit markets and capturing market share based on differences in their ability to screen risk, attract borrowers, and source capital funding. Guided by this theory, I estimate these bank-level fundamentals using a structurally identified dynamic state-space model. This measurement exercise yields bank-level estimates of lending productivity and bank-specific demand, objects that are central to the study of non-financial firms but have not previously been measured for banks. Finally, I embed these estimates in a quantitative multi-region New Keynesian model to quantify how bank dynamics shape the macroeconomy.

I document two key results. First, screening ability emerges as the primary driver of bank dynamics, explaining three-quarters of idiosyncratic loan growth within a bank and half of the variation across banks. This is because, unlike other fundamentals, screening ability directly affects loan losses and balance-sheet size, which can trigger an accelerating balance-sheet contraction. Second, this sorting along screening ability shapes the allocation of credit and, in turn, the broader economy through two channels: non-financial firm dynamics and the transmission of monetary policy. Through the first channel, the distribution of screening ability governs how effectively credit is routed to productive firms. Through the second channel, monetary tightenings generate larger lending contractions in regions exposed to policy-sensitive banks, those with weaker fundamentals. This generates a policy tradeoff between aggregate stabilization and regional inequality.

I first formalize the forces that shape bank dynamics in a novel theoretical framework. The economic setting begins with a standard regional New Keynesian DSGE model but with a financial sector made up of heterogeneous banks. Banks are ex ante identical, established in a headquarters region, and endogenously decide to enter (or exit) other oligopolistic regional lending markets, subject to regional market frictions and leverage-based regulatory costs. Banks differ across three fundamentals: screening ability, capital costs, and loan appeal. A bank's *screening ability* is its ability to assess whether a loan applicant will default if given a loan, and is analogous to the bank's lending productivity. A bank's *capital cost* is the premium the bank pays for capital funding relative to the federal funds rate, analogous to the bank's funding productivity. A bank's *loan appeal* captures firm-specific demand generated by the non-pecuniary reasons a borrower would want to work with a specific bank (for example, the ease of loan covenants, distribution of bank branches, or advertising), and is analogous to a non-financial firm's product appeal.

Bank-level fundamentals govern *bank dynamics*, a bank's regional market entry decision, market share expansion, and balance sheet growth. I characterize these relationships with three analytical results. First, a bank's market share increases as its fundamentals strengthen relative to its regional competitors. Second, a bank enters a region when its expected profits exceed the fixed costs of operating in the region. As a result, banks are more likely to enter markets that are large, protected by high frictions, or close to their headquarters, and more productive banks enter a greater number of markets overall. Third, a bank's profit margin, lending volume, and number of operating markets (a measure of product or geographic scope) in turn determine balance sheet growth.

The model then guides the empirical estimation of these bank-level fundamentals. I use a theoretically disciplined dynamic state-space model to jointly estimate the bank-level fundamentals (screening ability, capital cost, bank-region-specific loan appeal), region-level fundamentals (borrower quality and non-financial productivity), loan market frictions, and regulatory costs. Structural equations map the unobservable fundamentals to observable bank- or region-level data. For example, bank-level screening ability is identified by residual variation in a bank's loan default rate, while bank-region-specific loan appeal is identified by the residual variation in a bank's regional market share. I estimate the model at the bank- and state-level using balance sheet and income statements from regulatory U.S. Call Report filings, geographic footprints from the FDIC Summary of Deposits, and state-level non-financial firm dynamics from the U.S. Census Bureau.

The estimated fundamentals have evolved over time. The average screening ability was stable from 1995 to 2023 outside of a sharp, asymmetric decline during the global financial crisis (GFC), when banks in the bottom half of the screening distribution saw ability fall by more than three percent, compared to almost no change among the best screeners. Capital cost premia have compressed sharply over the same period while dispersion in loan appeal remained narrow.

Having measured the fundamentals and frictions that theoretically drive bank-level outcomes,

I next turn to studying the forces that shape bank dynamics. To do so, I use the newly estimated bank- and region-level fundamentals to discipline a quantitative calibration of the model. In this quantitative setting, I use simulations of the economy to trace out the effect of bank-level fundamental shocks on bank dynamics, and then put the sources of these dynamics in context with a full within- and across-bank variance decomposition.

All three fundamentals shift loan rates and lending, but only screening ability directly impacts loan losses. Higher capital costs raise loan rates, weaker loan appeal depresses lending, and a decline in screening ability, once recognized, raises loan rates to reflect higher expected losses. However, screening ability has an additional and more direct effect: because banks learn of a decline in their screening ability only after issuing loans and realizing losses, screening ability shocks decrease net worth immediately, before any repricing takes place. This activates a *balance-sheet accelerator* effect. As net worth falls, regulatory costs rise, pushing loan rates higher and further slowing lending and balance-sheet growth.

As a result, screening ability shapes bank dynamics. A negative one standard deviation bank-specific screening ability shock first costs 0.75 percent of a bank's net worth on impact before increasing its spreads by 6 percentage points while slowing lending and market share growth by approximately 25 percent. In comparison, neither capital cost nor loan appeal shocks have an economically or statistically significant impact on bank-level dynamics. The variance decomposition reveals that screening ability shocks are the key driver of fluctuations in a bank's dynamics over time, and the primary source of the heterogeneity in size and lending growth across banks. For example, screening ability shocks account for 78 percent of fluctuations in a simulated bank's lending growth over time and 44.5 percent of the simulated across-bank variation in lending growth.

Banks that grow large and capture market share are therefore those that are good at screening and pricing risk in their loan portfolios. This sorting along screening ability affects the allocation of credit, and, as a result, the broader economy through two simultaneous channels.

First, bank dynamics shape non-financial firm dynamics along both intensive and extensive margins. Banks act on the intensive margin of production through the effective price of factor inputs. When screening ability deteriorates, loan rates increase, the financing cost of the wage bill increases, and production slows. Meanwhile, banks act on the extensive margin of production via a selection effect. When a bank's screening ability falls, it provides more resources to firms destined to exit before producing any goods. As more resources are diverted to firms that will not produce, the output-to-input ratio (i.e., productivity) falls. In the quantitative setting, a negative one standard deviation screening ability shock decreases measured labor productivity by 2.74 percent on impact, and this effect is almost entirely due to the selection channel.

The composition of lenders therefore determines labor productivity in the long run. The intuition works through two offsetting effects. First, there is a loss from realized dispersion in bank

screening ability. A bank that screens badly funds projects that eventually fail and destroys real output, while a bank that screens well cannot increase a borrower's productivity. As a result, the marginal damage from the left tail exceeds the marginal gain from the right tail, so heterogeneity destroys aggregate credit quality holding loan market shares fixed. Second, there is a gain from heterogeneous allocation of lending across banks (i.e., heterogeneity in market share). This is because the banks that grow large are those that are good at screening risk. As the banking sector becomes more concentrated, a larger share of lending is accounted for by banks with strong screening abilities. In a series of counterfactual simulations, I find the existing level of heterogeneity in screening ability across U.S. banks to dampen labor productivity by 4.83 percent relative to a benchmark economy with no heterogeneity in banks' screening ability. The loss from heterogeneity in banks' realized screening ability is 4.87 percent, offset slightly by a 0.04 percent gain from the allocative effect of heterogeneity in market share.

Second, bank dynamics shape the transmission of monetary policy. The *credit elasticity channel* captures how weak fundamentals or high frictions amplify a region's response to shocks: a decline in screening ability or a rise in capital costs, for example, steepens a bank's marginal cost curve and magnifies its loan-rate response. Screening ability's level and dispersion govern the size of this effect. For example, lending falls 2.7 percent more in economies in the bottom quintile of the simulated aggregate screening ability distribution than in economies in the top quintile when facing a 25 basis point tightening shock.

Finally, I use my estimated bank fundamentals to create both a national and state-level measure of monetary policy transmission. On one hand, aggregate amplification is minimal, under one percent through the GFC and slightly above thereafter. On the other hand, regional dispersion is substantial and rising, spanning 4.5 percentage points across states by 2022. Monetary policy can therefore stabilize aggregate conditions, but only at the cost of raising regional income inequality.

Literature review. This paper's primary contribution is to bridge the two important but largely separate literatures on firm dynamics and the macroeconomics of banking and financial frictions.

Macroeconomic models have evolved in their treatment of financial intermediation. Early frameworks abstracted from intermediation, consistent with the frictionless benchmark of Modigliani and Miller (1958). Subsequent work introduced financial intermediaries as a source of amplification of non-financial shocks through the so-called financial accelerator mechanism, formalized in Bernanke et al. (1999) and Kiyotaki and Moore (1997). Following the GFC, banks became explicit economic agents and potential sources of aggregate fluctuations. Seminal contributions such as Gertler and Kiyotaki (2010) and Brunnermeier and Sannikov (2014) incorporated balance-sheet constraints and intermediary net worth into dynamic general equilibrium frameworks, while others, including Jermann and Quadrini (2012) and Ajello (2016), identified intermediation shocks as distinct drivers

of aggregate business cycles. More recent work emphasizes the role of cross-bank heterogeneity in determining macroeconomic outcomes, as in Jamilov and Monacelli (2025); Bellifemine et al. (2026), Coimbra and Rey (2024), and Villa (2026). Despite this progress, models with banks still typically abstract from endogenous entry and exit, regional expansion, competitive interactions, and firm-level growth. Key exceptions include Corbae and D'Erasmo (2021, 2024, 2025) and Corbae et al. (2026), which study banking *sector* dynamics and incorporate oligopolistic competition and regional dispersion among banks. However, these works focus on sector-wide dynamics and take the initial regional and size distribution (abstracted to a dominant-fringe paradigm) as given.

In contrast, I model banks as heterogeneous firms, applying Hopenhayn (1992), Hopenhayn and Rogerson (1993), Melitz (2003) and their quantitative extensions Hottman et al. (2016) and Eslava et al. (2024) to oligopolistic regional loan markets. Bank market shares, footprints, and competitive interactions thus evolve endogenously, linking bank dynamics to the allocation of credit, and in turn to regional and aggregate economic outcomes.

This paper's second contribution is to construct bank-level measures of screening ability and loan appeal. I am the first to structurally model and estimate loan appeal, an object analogous to product appeal in the firm dynamics literature (see for example Foster et al. 2008, Hottman et al. 2016, or Eslava et al. 2024). Meanwhile, the estimation of screening ability builds on a large micro-oriented literature on credit market competition that began with symmetric screening technologies (Broecker, 1990) and later allowed asymmetric screening ability across banks (Hauswald and Marquez, 2003). Other contributions endogenize screening ability through costly information acquisition (Hauswald and Marquez, 2006; Karapetyan and Stacescu, 2014), introduce fixed costs of screening (Yannelis and Zhang, 2023), or model screening as the outcome of multiple information inputs (Blickle et al., 2024; He et al., 2024). Despite their insights into the role of informational frictions in credit markets, these studies are primarily theoretical and neither attempts to measure bank-level screening ability in the data nor study its macroeconomic impact.

A related macro-finance literature interprets monitoring costs (isomorphic to screening ability) as financial sector productivity, but focuses primarily on aggregate measures. For example, Philippon (2015, 2022) document the long-run stability of U.S. financial intermediation costs, building on earlier efforts to measure financial-sector value added (Wang et al., 2008; Haldane et al., 2010; Basu et al., 2011), with complementary European evidence in Bazot (2018, 2024). These approaches are informative about long-run trends but abstract from cross-bank heterogeneity and leave bank-level productivity unmeasured.

This paper's third contribution is to characterize the short-run sullying effect of bank dynamics and its long-run cost to productivity through misallocation. A literature on whether downturns cleanse or sully the productivity distribution through selective entry and exit (Caballero and Hammour, 1994; Barlevy, 2002) has more recently connected this selection to credit market conditions

(such as in Midrigan and Xu 2014 and Khan and Thomas 2013). Ates and Saffie (2021) find that financial disruptions lead banks to be more selective funders, a fewer but better effect. I instead document a fewer and weaker effect, in which a negative screening ability shock degrades the quality of funded firms. Meanwhile, dispersion in firm-level distortionary wedges is a well-known determinant of cross-country productivity (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009), with firm-side financial frictions a leading source of these wedges in Buera et al. (2011), Midrigan and Xu (2014), and Moll (2014).[1] Closer to this paper, Bau and Matray (2023) and Neuhann and Sockin (2024) link misallocation to bank regulation and banking sector concentration, respectively.

This paper's fourth contribution is to the literature on the transmission of monetary policy. Canonical quantitative macro-banking and financial accelerator models, such as Gertler and Kiyotaki (2010) and Christiano et al. (2014), summarize the financial sector through aggregate net worth alone and assign no independent role to heterogeneity across banks. I show a distinct role for dispersion in fundamentals, and in turn bank dynamics, for shaping the amplification of policy —an effect that models with representative banks cannot capture. Similarly, a quantitative literature documents regional heterogeneity in monetary policy transmission through household balance sheets or industrial composition, including Beraja et al. (2019), Jones et al. (2022), and Hazell et al. (2022)I therefore complement these works by quantifying the extent to which heterogenous bank dynamics generate the asymmetric regional transmission of monetary policy .

Roadmap. Section 2 presents the theoretical framework, Section 3 analytically characterizes bank dynamics, Section 4 measures fundamentals, Section 5 quantifies bank dynamics, Sections 6 and 7 trace the macroeconomic consequences of these dynamics, and Section 8 concludes.

2 A Theoretical Framework for Bank Dynamics

I first develop a theoretical framework for bank dynamics. The following section will describe the economic environment before Section 3 analytically characterizes the resulting bank dynamics.

2.1 Summary of the economic setting

The model is a regional New Keynesian DSGE. Time is discrete and infinite. The economy is a set of independent regions that are home to immobile labor (households) and idiosyncratic production technology (firms and retailers), but are spanned by a financial sector that owns capital and lends to regional industries to enable production. Figure 1 summarizes the flow of resources in the economy.

[1] See Ghatak and Mookherjee (2025) for a recent survey.

Banks decide each period which oligopolistic regional lending markets to enter subject to a fixed cost of operating that increases with market distance from the bank's headquarters. After deciding which markets to serve in a given period, banks pay capital and regulatory costs to borrow funds from the Central Bank and issue risky loans with an imperfect screening technology.

A regional market is home to a continuum of households that save in local bond markets and supply labor to local firms. The market also has a continuum of firms that use bank loans as working capital to hire labor and produce a homogeneous good that is sold to retailers in a perfectly competitive goods market. However, firms are subject to an exogenous exit shock that results in them defaulting on their loans and ceasing production. Retailers differentiate the intermediate goods and sell the new variety of consumption goods to households, setting prices subject to a standard menu-cost nominal rigidity. A central bank conducts monetary policy to stabilize aggregate inflation and output across regions in the monetary union.

The financial block applies a firm dynamics modeling approach to the financial sector of heterogeneous banks. The imperfectly competitive regional loan markets and loan "exporting" dynamics position my treatment of banks closest to the Melitz (2003) tradition of firm dynamics models, although extensions to oligopolistic competition and the inclusion of firm-specific appeal bring this model in line with modern firm dynamic frameworks, such as Hottman et al. (2016) and Eslava et al. (2024). The non-financial block of the model combines the heterogeneous firm structure of Ottonello and Winberry (2020) and a regional monetary union setting similar to Nakamura and Steinsson (2014), Beraja et al. (2019), and Jones et al. (2022).

A presentation of the firms and the monetary policy authority follows before moving to describe the key agents in the economy, banks.

2.2 Regional economies and the monetary authority

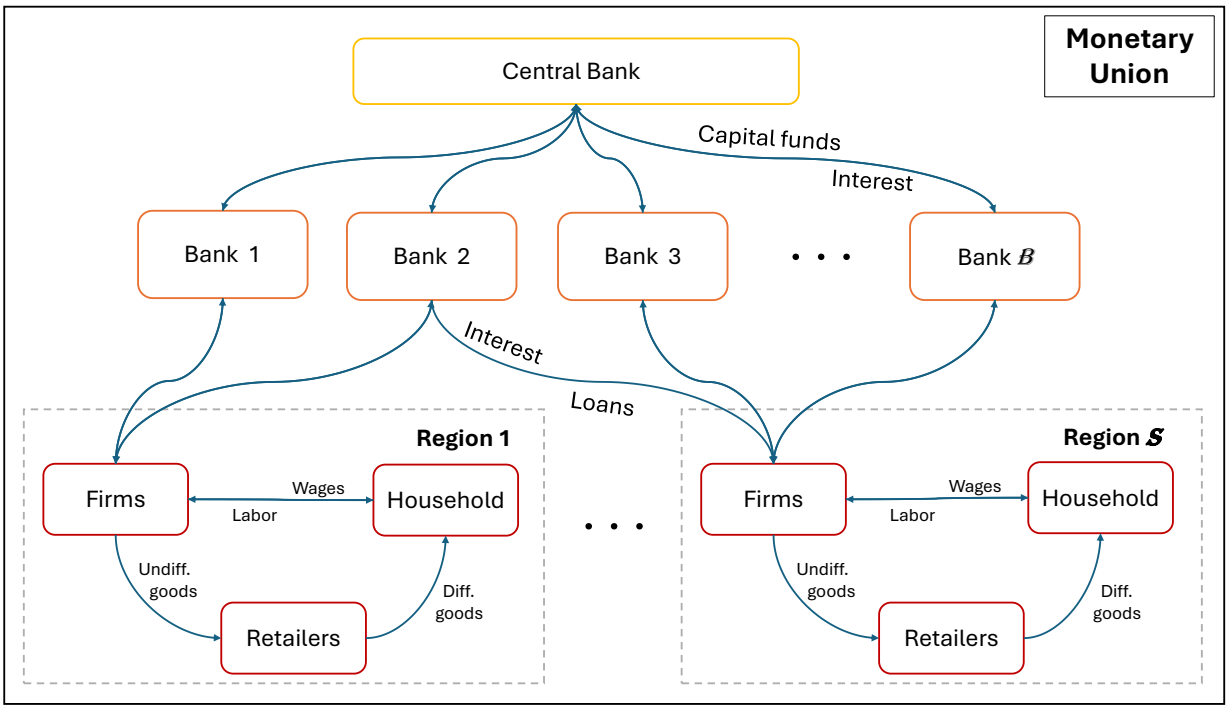
I first describe the regional economies, in particular the production sector that demands loans from banks and the monetary authority that supplies banks with funds to lend.

The region's production sector, borrower quality, and loan demand. There is a continuum of individual firms in each region. However, each firm potentially receives an exogenous exit shock. The probability of surviving is time-varying and region specific, with law of motion: $q_{s,t} = \rho^q q_{s,t-1} + \epsilon^q_{s,t}$, where ρ^q is the regional persistence and $\epsilon^q_{s,t}$ is the stochastic exit shock.

Each firm is matched with a bank and applies for a loan to fund its labor bill before starting production.[2] Firms denied a loan do not produce. If a firm is granted a loan but then forced to exit,

[2]Because firms submit loan applications to a single bank each period there is no multiple equilibria or "winner's curse" present in this economy. These outcomes can arise in the presence of sticky borrower-lender relationships and asymmetric screening ability, as analyzed by works recently reviewed by Blickle et al. (2024).

Figure 1: Map of the economy



it defaults on the entire principal of its loan. For this reason, $q_{s,t}$ can be interpreted as a measure of a regional credit market's borrower quality.

The individual firms can be studied as a single regionally representative firm or production sector. The representative firm aggregates bank-specific loans into a composite credit product $L_{s,t}$ using CES aggregator technology:

$$L_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} \right)^{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}}, \qquad \sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t} = 1, \tag{1}$$

with standard cost dual $R_{s,t} = \left(\sum_i \eta_{i,s,t} R_{i,s,t}^{1-\sigma_{s,t}} \right)^{\frac{1}{1-\sigma_{s,t}}}$, where $\sigma_{s,t} > 1$ is the elasticity of substitution across bank loans, and $\eta_{i,s,t} \geq 0$ is the firm's idiosyncratic preference weight for bank i's loans.

The bank's loan appeal $\eta_{i,s,t}$ subsume bank-specific demand shocks, as well as accounts for non-pecuniary differences that create persistent differences in bank-specific demand (such as the strictness of loan terms, branch locations within a region, or advertising).[3] Simultaneously, the elasticity of loan substitution may be considered a stand-in for local frictions that make substituting

[3]Loan appeal may also capture banks with multiple business streams, like large investment banks, tying loans to other products, such as financial advising, to attract borrowers by acting as a "one-stop-shop".

across bank loans more difficult for borrowers. These frictions include search and matching costs that generate sticky borrower-lender relationships.[4]

The firm uses the composite loan $L_{s,t}$ to hire labor as a flexible input to produce output with standard decreasing returns to scale and time-varying technology:

$$Y_{s,t} = A_{s,t} \cdot M_{s,t}^{\alpha}, \quad \text{with } \alpha \in (0,1), \quad a_{s,t} = \rho^{a} a_{s,t-1} + \epsilon_{s,t}^{a} \qquad a_{s,t} \equiv \log A_{s,t}$$

where $A_{s,t}$ is region-specific productivity, $M_{s,t}$ is employment, and $\epsilon_{s,t}^{a}$ is a productivity shock.

Combining the loan aggregation and output production technology, the firm's problem is to then maximize profits by choosing labor inputs $M_{s,t}$, taking wages $W_{s,t}$ as given. The result is a standard isoelastic loan demand curve for each bank,

$$L_{i,s,t} = \eta_{i,s,t} \left(\frac{R_{i,s,t}}{R_{s,t}} \right)^{-\sigma_{s,t}} L_{s,t},$$

which banks will take as given.

Regional retailers and households. The remaining regional agents, retailers and households, are standard and I leave their formal problems and discussion to Appendix A.

Central Bank. The Central Bank sets the nominal interest rate following a Taylor Rule:

$$I_t = I_{t-1}^{\rho_i} \left(\hat{Y}_t^{\psi_y} \hat{\Pi}_t^{\psi_\pi} \right)^{(1-\rho_i)} U_t \tag{2}$$

where $\hat{Y}_t$ is the aggregate output gap (with respect to steady state $\bar{Y}$) and $\hat{\Pi}_t$ is the inflation gap (with respect to an exogenously determined inflation target). The Central Bank interacts with the economy primarily through lending capital funds directly to banks at the nominal policy rate. I assume the Central Bank is a deep-pocketed investor and can lend as much capital as is demanded by banks at the policy rate it sets.

2.3 Banks

Banks operate across a set of oligopolistic (Cournot) competitive regional loan markets, $s \in S$, with the goal of maximizing their expected present discounted franchise value (i.e. net worth). Banks compete in these loan markets by issuing uncollateralized, one-period, risky loans.

[4] See Hottman et al. (2016) or Eslava et al. (2024) for a more thorough description of firm-specific appeal in a similar oligopolistic competition and CES demand setting, but in a non-financial firm application. See Freixas and Rochet (2008) for a textbook treatment of credit market frictions.

2.3.1 The bank's problem

Bank i's formal dynamic programming problem is:

$$\max_{\{L_{i,s,t}\}_{s\in S}} \mathbf{V}(N_{i,t}, \mathcal{M}_t, \Omega_t) = \mathbb{E}_t \Pi_{i,t} + (1-m)\beta \mathbb{E}_t \mathbf{V}(N_{i,t+1}, \mathcal{M}_{t+1}, \Omega_{t+1}) \quad \text{s.t.} \tag{3}$$

$$\Pi_{i,t} = \sum_{s\in S} \left\{ R^*_{i,s,t} L_{i,s,t} - f(d_{i,s}) \right\} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma(\sigma^R \frac{L_{i,t}}{N_{i,t}}) \qquad \text{(Profit function)}$$

$$R^*_{i,s,t} = (1 - \lambda_{i,s,t}) R_{i,s,t} \qquad \text{(Realized gross return)}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \qquad \text{(Net worth LoM)}$$

where the bank maximizes profits, and in turn net worth, by issuing a portfolio of region-specific loans $\{L_{i,s,t}\}_{s\in S}$, that yield the region-specific gross realized portfolio return on loans, $R^*_{i,s,t}$. Total profits, $\Pi_{i,t}$, are realized at the end of period t after $\lambda_{i,s,t}$ percent of borrowers default. The bank's net worth in period t is denoted $N_{i,t}$ and is simply the previous period's net worth plus realized gains or losses from the current period's loan portfolio. Future net worth is discounted at rate $\beta < 1$. Banks exogenously exit with probability m at the end of the period after profits are realized.[5] Banks take as given the cross-sectional distribution of banks' balance sheets and fundamentals, $\mathcal{M}_t$, (which includes their own states) and the distribution of region-specific fundamentals, Ω_t.

2.3.2 The bank's costs

Banks issue loans after paying capital, regulatory, and trade costs.

Capital costs. Banks can use their net worth, $N_{i,t}$, to fund loans and can borrow additional capital from the Central Bank. The cost of capital funding for banks will be the federal funds rate and a multiplicative markup, so that $C_{i,t} = \kappa_{i,t} I_t$, where $C_{i,t}$ is the bank's marginal cost of capital, $\kappa_{i,t}$ is the bank's capital cost premium, and I_t is the nominal federal funds rate. Capital funds are fungible and can be used in any market to make loans.

The assumption that the capital cost of forming loans is bank-specific is motivated by the classical Monti-Klein oligopolistic banking model and empirical work such as McAllister and McManus (1993), or recent work on banking market power in deposit markets, such as Drechsler et al. (2017), and deliberate spatial sorting into markets with cheap deposits, such as Oberfield et al. (2024).

[5] I will make the simplifying assumption that banks are sufficiently well funded (that is, have a sufficiently large $N_{i,t}$) to cover loan losses so that I may abstract away from the possibility of endogenous bank failures and inability to pay depositors or resulting bank runs.

Regulatory costs. Banks also pay a regulatory cost based on their risk-adjusted leverage ratio.[6] The regulatory cost function, $\Gamma(\cdot)$ will take the increasing and weakly convex linear functional form, $\Gamma(\sigma^R \frac{L_{i,t}}{N_{i,t}}) = \zeta_t(\sigma^R \frac{L_{i,t}}{N_{i,t}})$, for analytical tractability. I model regulation as a convex cost rather than a hard constraint, as in Gabaix and Maggiori 2015, to preserve differentiability and to allow regulatory tightness to map continuously into loan pricing.

The regulatory costs acknowledge the real presence of such costs borne by banks (due to Basel III and Dodd-Frank regulations, for example), as well as establish a connection between a bank's net worth, $N_{i,t}$, and ability to make loans. Moreover, by explicitly incorporating the expected portfolio return risk, the regulatory cost term captures reduced-form, risk-averse asset-pricing dynamics, mirroring those highlighted by the intermediary asset pricing literature (namely the seminal works by He and Krishnamurthy (2013) and Brunnermeier and Sannikov (2014), or more recent work such as Akinci et al. (2022)).

Trade costs. There is a fixed cost of operating in markets outside a bank's headquarters market, $f(d_{i,s})$, that is increasing in the distance $d_{i,s}$ between the headquarters and non-headquarters market. The cost of operating in a bank's own headquarter region is zero. These spatial fixed costs are central to generating endogenous regional concentration and time-varying loan market structure. The distance-based cost is similar in spirit to an iceberg trade cost common in the trade models of endogenous regional market entry and exit, such as Melitz (2003), and recent spatial banking models, such as Oberfield et al. (2024), where it is a result of managerial span-of-control problems.

2.3.3 The bank's screening technology

Banks form their loan portfolio each period by screening loan applicants and offering credit to those they predict will not default. However, with imperfect screening technology, a bank is not able to guarantee that its borrowers will not default. As a result, banks must charge a risk premium to compensate for expected losses. That is, the bank's expectations for, and ultimately realizations of, regional loan losses, $\lambda_{i,s,t}$, are determined by its ability to solve a loan-level screening problem. I next turn to describing this problem in detail, which is motivated by the seminal Hauswald and Marquez (2003) model of banking competition with asymmetric screening ability.

Loan applicants. Loan applicants are of two types, $\tau \in \{l, h\}$, such that l type borrowers default and h type do not. Each individual applicant in market s will receive an exit shock and default with probability $1 - q_{s,t}$. By the law of large numbers, $1 - q_{s,t}$ percent of loans in the bank's regional loan portfolio will default if all applicants are given loans.

[6] Various authors take up a similar object of interest when formulating regulatory costs and constraints. For example, Gabaix and Maggiori (2015) focus on a liquidity ratio, while Coimbra and Rey (2024) employ a leverage ratio.

Screening technology and productivity. Banks have access to an imperfect screening device that creates a private noisy signal, ι, that labels a loan applicant as an h or l type. Banks only extend loans to applicants with an h type signal, $\iota = h$.

The probability that a borrower is actually type h, given that its signal indicates type h, is determined by the bank's *screening ability*, which is the accuracy of the screening device and denoted ψ, such that:

$$Pr(\tau = h|\iota = h) = \frac{\psi q}{\psi q + (1-\psi)(1-q)}, \quad \psi \in [0.5, 1], \qquad q \in [0, 1] \tag{4}$$

where q is the *borrower quality*, the true proportion of the population that will not default. With $\psi = 1/2$, the signal is uninformative, and the bank's posterior belief collapses to its prior, q. When $\psi = 1$, the signal is perfectly accurate and $Pr(\tau = h|\iota = h) = 1$.

I interpret a bank's screening ability as its lending productivity. Higher screening ability reduces expected loan losses by reallocating funds away from borrowers that are likely to default and toward solvent firms. In doing so, effective screening preserves financial capital that would otherwise be dissipated through defaults, thereby lowering aggregate financial risk and expanding the volume of funds that can be intermediated. This role is directly analogous to technological productivity in non-financial firms, which measures the extent to which a given set of inputs can generate a larger effective resource set for the economy. Improvements in screening technology similarly enlarge the economy's usable financial resources, making bank screening ability the financial-sector counterpart to standard notions of non-financial productivity.[7]

Loan application and offer timing. Banks do not know their own screening ability before making lending decisions, which creates a source of risk that the banks must bear. The exact timing of the "morning" lending sub-period follows:

- Potential borrowers come to the bank and submit loan applications, becoming applicants.
- Banks do not know the current efficacy of their screening ability, so they form $E\psi_{i,t}$.
- Banks evaluate all applicants and formulate their loan offer rate based on the expected regional portfolio default rate $\mathbb{E}\lambda_{i,s,t}$.[8]

[7] Philippon (2015) makes a closely related argument for banks' monitoring costs, which are nearly isomorphic to my notion of screening ability. The key distinction is that he assumes perfect screening with a resource cost of monitoring that banks incur to assess risk and pass on to borrowers, whereas in my framework screening is imperfect and borrower costs arise endogenously as compensation for expected loan losses.

[8] Recall that all applicants that will receive a loan offer are the same h signal type, thus pose the same perceived credit risk to the bank and will receive identical offer rates, so I will abstract pricing from the loan to portfolio level.

- Applicants with h type signals are offered loans, while those with l type signals are rejected.
- Approved applicants accept or reject the bank's take-it-or-leave-it loan rate offer.

Banks and borrowers then learn the realized state of the world in the "afternoon" sub-period, which also reveals the bank's technological efficiency for the given period. A full timeline of events within a period is outlined at the end of this section.

2.3.4 Banking sector evolution

The banking sector is the time-varying set of banks $\mathcal{B}_t$. Along the intensive margin of the banking sector, individual banks are characterized by their endogenous net worth $N_{i,t}$, exogenous region-specific appeal $\bar{\eta} = \{\eta_{i,s,t}\}_{s \in S}$, screening ability $\psi_{i,t}$, and capital cost premium $\kappa_{i,t}$. The banking sector can be characterized by the joint distribution of these four bank-level and bank-region-level characteristics, $\mathcal{M}(N, \kappa, \psi, \bar{\eta})$, where the density function $m(N, \kappa, \psi, \bar{\eta})$ is the population share of banks characterized by the specific tuple of values $\{N, \kappa, \psi, \bar{\eta}\}$.

Bank birth and death. Along the extensive margin, the cardinality of the banking sector $\mathcal{B}_t$ is held fixed for tractability. Banks are subject to an exogenous exit shock and are replaced one-for-one by new entrants upon exit.[9] This replacement mechanism prevents unbounded accumulation of bank net worth and ensures a stationary cross-sectional distribution of banks. Banks are ex ante identical at birth, draw a headquarters location uniformly across regions, and begin operating as incumbents upon entry. Entrants are endowed with initial net worth N_0.

Evolution. With banks entering, exiting, and growing, the banking sector distribution evolves along both the extensive and intensive margin each period. The banking sector evolves along with aggregate state $(\mathcal{M}_t, \Omega_t, \Upsilon_t)$ —the distribution of banks, the distribution of regional fundamentals, and the monetary policy states, respectively— via time-invariant transition kernel Φ induced by individual policies, realized loan and default outcomes, and bank exit and replacement.

2.4 Timeline and equilibrium

I conclude the description of the economic environment by summarizing the within-period timing and summarizing the economy's equilibrium.

Timeline within a Period. Each period t is divided into a financing morning and a real-activity afternoon. The sequence of events is as follows:

[9] This is similar to the exogenous banker exit and replacement strategy in Gertler and Kiyotaki (2010, 2015).

- *Partial states are realized.* Regional productivity $A_{s,t}$, regional loan market frictions $\sigma_{s,t}$, and banks' capital cost premiums $\kappa_{i,t}$ are drawn and observed. Banks' screening abilities $\psi_{i,t}$ and firms' survival rates $q_{s,t}$ are drawn but not observed.

- *Morning: participation, funding, and lending.* Given the policy rate and the observed states, banks choose which regions to serve, pay the associated fixed costs, and regional appeal $\{\eta_{i,s,t}\}$ is realized. Firms submit loan applications. Banks form expectations $\mathbb{E}_t\psi_{i,t}$ and $\mathbb{E}_t q_{s,t}$, screen applicants, and choose regional loan quantities $\{L_{i,s,t}\}_{s\in S}$, funding the resulting portfolio from net worth and Central Bank borrowing. Loan rates $\{R_{i,s,t}\}$ are the rates at which the CES demand system clears at those quantities, and are offered to applicants on a take-it-or-leave-it basis.

- *Afternoon: production and repayment.* Financed firms hire labor and produce output. Exit shocks are realized, defaults occur, and surviving firms repay their loans. Banks' screening abilities $\psi_{i,t}$ and firms' survival rates $q_{s,t}$ are realized. Retailers set prices subject to nominal rigidities and sell to households. Banks realize profits, net worth is updated, and exogenous bank exit and replacement occur.

This timing ensures that banks' loan pricing and market entry decisions are forward-looking, embedding expectations over borrower risk and screening ability.

Equilibrium. The equilibrium concept has two components. Within a period, banks compete for regional market share in an aggregative game: because loans enter the representative firm's problem only through the CES aggregator, each bank's profit depends on rivals only through the regional loan index, which it internalizes when choosing $\{L_{i,s,t}\}$. Under the linear regulatory cost specification, this separates into $|S|$ independent regional Cournot games linked only through net worth $N_{i,t}$. Across time, entry requires forecasting every other bank's state in every region it might serve —a dynamic game (Ericson and Pakes, 1995) that is intractable at this scale. I instead invoke the large-market simplification of Weintraub et al. (2008) and in the spirit of Krusell and Smith (1998). Because participation costs are per-period and non-sunk, and regions (U.S. states) are large enough that no single bank materially moves the regional aggregate, each bank treats its evolution as an exogenous, correctly anticipated process, reducing entry to single-agent optimization (Krusell and Smith, 1998). The aggregate state $(\mathcal{M}_t, \Omega_t, \Upsilon_t)$ then evolves via a transition kernel Φ induced by individual policies, realized loan and default outcomes, and bank exit and replacement. A full description of the equilibrium is presented in Appendix A.3.

3 Analytically Characterizing Bank Dynamics

Having outlined the economic environment, I next turn to analytically characterizing bank dynamics and the fundamentals and frictions that shape them. Additional bank-level outcomes, along with regional outcomes, and all proofs are discussed in Appendix A.

3.1 Loan losses

A bank knows neither $\psi_{i,t}$ nor $q_{s,t}$ while screening applicants and must use $\mathbb{E}\psi_{i,t}$ and $\mathbb{E}q_{s,t}$ to form expectations over the level of loan losses that will be realized in its portfolio. This leads to the probability that the bank's signal (of type h) is correct:

$$\mathbb{E}[Pr(\tau = h|\iota = h)] = \frac{\mathbb{E}\psi_{i,t}\mathbb{E}q_{s,t}}{\mathbb{E}\psi_{i,t}\mathbb{E}q_{s,t} + (1 - \mathbb{E}\psi_{i,t})(1 - \mathbb{E}q_{s,t})} \tag{5}$$

implying the expected default rate $\mathbb{E}\lambda_{i,s,t} = 1 - \mathbb{E}[Pr(\tau = h|\iota = h)]$.

The realized default rate will then reflect the realized screening accuracy across the bank's loans, and assuming a "large enough" portfolio that the law of large numbers applies, will be:

$$\lambda_{i,s,t} = 1 - \mathbb{E}[Pr(\tau = h|\iota = h)|\psi] = 1 - \frac{\psi_{i,t}q_{s,t}}{\psi_{i,t}q_{s,t} + (1 - \psi_{i,t})(1 - q_{s,t})} \tag{6}$$

so that the resulting difference between the expected and realized loan default rate is a function of the difference in expected and realized screening ability and borrower quality. Moreover, Equation 6 shows that as $\psi_{i,t}$ approaches one, the screening device is perfect and the loan default rate falls to zero as banks are able to filter out all would-be defaulting borrowers. Similarly, as the borrower quality $q_{s,t}$ approaches one, loan defaults fall to zero because no firms are exiting.

3.2 Loan rates

A bank solves Equation 3 by issuing the loan portfolio $\{L_{ist}\}_{s\in\mathcal{S}}$, which leads to charging, in market clearing equilibria, the region-specific loan rate:

Proposition 3.1 (Loan rate)**.** *Bank i's equilibrium loan rate in region s at time t is*

$$R_{i,s,t} = \underbrace{\frac{1}{1 - \mathbb{E}\lambda_{i,s,t}}}_{\text{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t} - 1}\frac{1}{1 - s_{i,s,t}}}_{\text{markup}} \cdot \underbrace{\left(C_{i,t} + \frac{\sigma^R}{N_{i,t}}\Gamma'\left(\sigma^R\frac{L_{i,t}}{N_{i,t}}\right)\right)}_{\text{marginal cost}}. \tag{7}$$

Expanding the expected default rate, regulatory cost, and capital cost terms gives the equivalent expression

$$R_{i,s,t} = \underbrace{\left[1 + \mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right) \cdot \mathbb{E}_t\left(\frac{1-\psi_{i,t}}{\psi_{i,t}}\right)\right]}_{\textit{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}\frac{1}{1-s_{i,s,t}}}_{\textit{markup}} \cdot \underbrace{\left(I_t\kappa_{i,t} + \zeta_t\frac{\sigma^R}{N_{i,t}}\right)}_{\textit{marginal cost}} \tag{8}$$

where the regulatory cost function is $\Gamma(\sigma^R\frac{L_{i,t}}{N_{i,t}}) = \zeta_t(\sigma^R\frac{L_{i,t}}{N_{i,t}})$.

As Equation 7 makes clear, the loan rate increases as the expected default rate $\mathbb{E}_t\lambda_{i,s,t}$, bank-region market share $s_{i,s,t}$, cost of capital $C_{i,t}$, or regional market friction term $(\sigma_{s,t}/(\sigma_{s,t}-1))$ increases. Equation 8 further shows that loan rates decrease as expected borrower quality $q_{s,t}$ or the bank's expected ability $\psi_{i,t}$ increases, while increasing with the bank's funding cost premium $\kappa_{i,t}$ or the policy rate I_t. Conversely, as the size of the bank increases the loan rate decreases and the quantity supplied increases, in line with standard macro-banking models (for example the seminal Gertler and Kiyotaki 2010 model) and empirical evidence (Ottonello and Song 2025).

Finance view. I rewrite the equation one last time into a "finance view" of the loan rate to emphasize that even though the rate is born of a firm dynamics model, it still captures the common features of loan rates explored in the finance and IO literature. The loan rate is equivalently:

$$R_{i,s,t} = \left[\underbrace{\frac{\sigma_{s,t}}{(\sigma_{s,t}-1)}}_{\substack{\text{exog.}\\\text{markup}}} \cdot \underbrace{\frac{1}{(1-s_{i,s,t})}}_{\substack{\text{endog.}\\\text{markup}}}\right] \cdot \left[\underbrace{C_{i,t}}_{\substack{\text{capital}\\\text{cost}}} \cdot \underbrace{\mathbb{E}_t\left(\frac{1}{1-\lambda_{i,s,t}}\right)}_{\substack{\text{risk neutral comp.}\\\text{for expected defaults}}} + \underbrace{\frac{\sigma^R}{1-\mathbb{E}\lambda_{i,s,t}}}_{\text{quantity of risk}} \cdot \underbrace{\frac{\zeta_t}{N_{i,t}}}_{\substack{\text{price}\\\text{of risk}}}\right] \tag{9}$$

which emphasizes two key features of banks' pricing behavior.

Markups. Profitable markups exist and are both exogenous and endogenous to the bank's behavior. The exogenous component arises from the premium extracted from sticky lending relationships generated by regional market frictions. The endogenous component arises from the oligopolistic rent that banks extract from borrowers in an imperfectly competitive loan market, and is a key feature of banking sector dynamics models (Corbae and D'Erasmo, 2021, 2025).

Compensation for risk. The compensation for risk is a function of the level of expected defaults, the quantity of risk (the loan portfolio return variance divided by the expected return, here more primitively the solvency rate) and the price of risk (a time-varying coefficient divided by the wealth of the investor, in this case, the net worth of the bank). In this view, the time-varying regulatory cost coefficient would in fact be interpreted as the bank's time-varying risk aversion. The presence of the price and quantity of risk means that a bank prices its loan rates as if it were a risk-averse investor

pricing risky assets. This captures the key dynamics of the risk-averse intermediary asset pricing literature. The compensation for risk term also highlights that the bank requires compensation for risk over and above the risk-neutral compensation for expected defaults. This means that the price and quantity of risk term may also be interpreted as the Excess Loan Premium, analogous Excess Bond Premium of Gilchrist and Zakrajšek (2012).

3.3 Market share

I next turn from prices to quantities and analytically characterize banks' competitive outcomes.

Proposition 3.2 (Market share)**.** *A bank's log regional loan market share (i.e. relative regional lending volume) is*

$$\log(s_{i,s,t}) \approx \frac{\sigma_{s,t}\log\bar{s}_{s,t}}{\sigma_{s,t}\log\bar{s}_{s,t}+1}\Big[(1/\sigma_{s,t})\big(\log(\eta_{i,t})-\tilde{\eta}_{s,t}\big)+\big(\tilde{C}_{s,t}-\log(C_{i,t})\big) + \big(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t})\big)+\big(\mathbb{E}\tilde{\chi}^{\psi}_{s,t}-\mathbb{E}\log(\chi^{\psi}_{i,t})\big)\Big], \quad (10)$$

where $s_{i,s,t} = L_{i,s,t}/L_{s,t}$, $\bar{s}_{s,t} = (1/B_{s,t})\sum_{i\in\mathcal{B}_{s,t}} s_{i,s,t}$, $\tilde{x}_{s,t} = (1/B_{s,t})\sum_{i\in\mathcal{B}_{s,t}}\log x_{i,s,t}$, $B_{s,t} = |\mathcal{B}_{s,t}|$, *and* $\chi^{\psi}_{i,t} = (1-\psi_{i,t})/\psi_{i,t}$.

A bank's market share is a function of its fundamentals relative to the (geometric) average fundamentals of other banks competing within the regional loan market. The bank's market share is an increasing function of its appeal relative to the average appeal and of its ability relative to the average ability, but a decreasing function of its capital costs relative to average capital costs. Borrower quality does not enter this comparison because it is common to every bank operating in region s, and therefore cancels out of any within-region relative comparison. The fact that banks with the strongest fundamentals (i.e. the most productive banks) will garner the largest market share is in line with the firm dynamics in models of imperfect export competition (Melitz, 2003).

However, a bank's fundamentals are not the only factors that matter for determining its market share. The bank's relative regulatory costs appear in Equation 10, meaning that a bank's pre-existing size (i.e. balance sheet conditions) also determines its ability to compete for market share. The greater the bank's size, the smaller its regulatory costs, and the more market share it will claim.

3.4 Regional market entry

A bank will then enter a regional market if the expected profits of operating in the region are positive. Given the oligopolistic structure of the regional loan markets, the condition in turn depends on

whether the bank can make a large enough profit per loan to pay its fixed cost of operating in the given region.

Proposition 3.3 (Regional market entry)**.** *Under the linear regulatory cost* $\Gamma(\cdot) = \zeta_t(\sigma^R \frac{L_{i,t}}{N_{i,t}})$*, bank* i *is indifferent between entering and not entering region* s *at time* t *when*

$$\mathbb{E}_t \log \Pi_{i,s,t} \;=\; \mathbb{E}_t \left\{ \log \frac{1 + (\sigma_{s,t} - 1)\, s_{i,s,t}}{\sigma_{s,t}} + \log s_{i,s,t} + \log(1 - \lambda_{i,s,t}) + \log \alpha + y_{s,t} \right\} \;=\; \log f_{i,s,t}, \tag{11}$$

where $\Pi_{i,s,t}$ *is bank* i*'s operating profit in region* s*, gross of the fixed cost* $f_{i,s,t}$*, and* $s_{i,s,t}$ *is the market share the bank obtains upon entry. The bank enters if and only if the left-hand side weakly exceeds* $\log f_{i,s,t}$*.*

This condition is based on four primary forces: the bank's expected competitiveness, the depth of the regional market, the distance from headquarters, and local market frictions.

First, market entry is a function of the bank's expected market share (and oligopolistic profits), thus the more competitive a bank is expected to be, the more likely it will enter the market. This is the productive exporter motivation that is seen in trade models such as Melitz (2003). Second, the larger the regional economy, as measured by its log output, $y_{s,t}$, the more likely a bank will be to enter the market. Third, the farther the region is from the bank's headquarters, the more expensive it will be to start operations there, and the less likely the bank will be to move into the new region. The second force represents an agglomeration effect while the third force may represent a reduced-form managerial span-of-control cost, both of which are commonly present in spatial sorting models such as Oberfield et al. (2024). Lastly, the greater the local market frictions, the greater the exogenous markups and profit the potential entrant can expect to make in the region. Therefore, as local market frictions rise, the more likely banks are to enter the region, all else equal.

3.5 Bank growth

Lastly, banks grow based on the accumulation of profits accrued across the credit markets:

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1}, \qquad \text{where}$$

$$\Pi_{i,t} = \sum_{s \in S} \left\{ (1 - \lambda_{i,s,t})(s_{i,s,t} L_{s,t}) R_{i,s,t} - f(d_{i,s}) \right\} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma\left(\sigma^R \frac{L_{i,t}}{N_{i,t}} \right). \tag{12}$$

A bank's growth can therefore be described as the product of an intensive and extensive margin. The intensive margin is the profits a bank captures within each market and depends upon: its

lending volume within the region —thus its ability to capture market share in that region— its ability to minimize losses in its loan portfolio, and the risk-adjusted markups —reflecting local market frictions and the bank's market power. The extensive margin is then the number of regions or credit markets that the bank has entered and is lending in during a given period (i.e., its geographic or product scope).

In summary, bank-level growth is determined by how competitive the bank's fundamentals are relative to its peers, its ability to limit losses, local credit market frictions, and the size of the markets it chooses to compete in.

4 Measuring Bank Fundamentals and Market Frictions

I next turn to estimating the unobservable fundamentals that drive bank-level dynamics and lending outcomes. To do so, I develop a theoretically disciplined state-space model that uses equilibrium conditions from the structural model to map observable bank- and region-level outcomes to unobservable frictions, distortions, and fundamentals. The measurement procedure will jointly estimate these objects of interest with an Expectation-Maximization (EM) algorithm, which effectively estimates the unobservable fundamentals in the E-step, and the regional frictions and distortions in the M-step. The following will outline these two steps in turn. I leave a formal description of the algorithm and estimating the non-financial regional productivity wedge, $A_{s,t}$, to Appendix C.

4.1 E-step: estimating fundamentals

I use a state-space model to jointly estimate bank fundamentals, $\psi_{i,t}$, $\kappa_{i,t}$, region fundamentals $q_{s,t}$, and bank-region fundamental $\eta_{i,s,t}$. The state transition equation is a standard AR(1) process with *i.i.d.* Gaussian errors:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \qquad \epsilon_t \sim N(0, Q), \tag{13}$$

where the state vector is the stacked partitioned vector $X_t = \{\log \chi_t^\phi \ \log \kappa_t \ \log \chi_t^q \ \log \eta_t\}$ in which $\chi_t^\phi = \{\chi_{i,t}^\phi\}_{i\in\mathcal{B}}$, $\kappa_t = \{\kappa_{i,t}\}_{i\in\mathcal{B}}$, $\chi_t^q = \{\chi_{s,t}^q\}_{s\in S}$, $\eta_t = \{\eta_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; Γ is the conformable matrix of auto-regressive coefficients, and ϵ_t is the analogously stacked vector of fundamental shocks. The shocks are *i.i.d.* across and within banks and regions, thus Q is a diagonal matrix. Note that I use the odds transformation of both screening ability $\chi^\phi = \frac{1-\phi}{\phi}$ and borrower quality $\chi^q = \frac{1-q}{q}$ rather than the direct objects of interest $\phi_{i,t}$ and $q_{s,t}$. This transformation of the latent states is necessary to maintain the linear structure of the measurement equations.

The unobservable fundamentals are then related to a set of observable bank and region characteristics: the regional firm survival odds, $E_{s,t} = \frac{1-q_{s,t}}{q_{s,t}}$, bank loan losses, $\lambda_{i,t}$, bank capital costs, $\log C_{i,t}$, and bank-region market share, $s_{i,s,t}$. The measurement equation relating the latent states

(i.e. unobservable fundamentals) to the observable data then follows:

$$Y_t = \gamma + \underbrace{H_t X_t}_{\substack{\text{bank/region} \\ \text{fundamentals}}} + \underbrace{\tilde{H}_t X_t}_{\substack{\text{regional} \\ \text{banking sector} \\ \text{characteristics}}} + \underbrace{B_t Z_t}_{\substack{\text{exogenous} \\ \text{terms}}} + \underbrace{\nu_t}_{\substack{\text{meas.} \\ \text{error}}} \qquad \nu_t \sim N(0, R) \tag{14}$$

where the stacked vector of observables is $Y = \{\log E_t \ \log \lambda_t \ \log C_t \ \log s_t\}$ in which $\lambda_t = \{\lambda_{i,t}\}_{i\in\mathcal{B}}$, $C_t = \{C_{i,t}\}_{i\in\mathcal{B}}$, $E_t = \{E_{s,t}\}_{s\in S}$, $s_t = \{s_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; the matrix H_t relates the latent fundamentals to the observable outcomes; $\tilde{H}_t$ is the conformable matrix such that $\tilde{H}_t X_t$ creates a stacked vector of region-specific banking industry geometric average fundamentals for calculating the bank-region market share following Equation 10; the vector Z_t is the stacked vector of exogenous variables (the bank-level marginal regulatory costs and nominal federal funds rate), while ν_t captures the possible measurement error and γ the vector of constants.[10] The measurement matrices are time-varying functions of regional frictions, $\sigma_{s,t}$ (Equation 10).

Consistent with the model structure, estimation proceeds in two steps: an aggregate component that identifies the laws of motion for economy-wide states, and an idiosyncratic component that disciplines bank- and region-level deviations from these aggregates. The bifurcated estimation is discussed in detail in Appendix C.

Identification. The identification of bank and region fundamentals comes from the theoretically disciplined measurement equation. For exposition, consider the model with only one bank and one region so that the system would follow as:

$$\mathbf{X}_t = \begin{bmatrix} \log C_{i,t} \\ \log \eta_{i,s,t} \\ \log \chi^{\psi}_{i,t} \\ \log \chi^{q}_{s,t} \end{bmatrix}, \quad \mathbf{Y}_t = \begin{bmatrix} \log C_{i,t} \\ \log \lambda_{i,t} \\ \log E_{s,t} \\ \log s_{i,s,t} \end{bmatrix} = \gamma_t + \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ \bar{\sigma}_{s,t} & \frac{\bar{\sigma}_{s,t}}{\sigma_{s,t}} & \bar{\rho}^{\psi}_{s,t} & 0 \end{bmatrix}}_{\equiv \mathbf{H}_t} \mathbf{X}_t + \tilde{H}_t X_t + B_t Z_t + \nu_t \tag{15}$$

$$\bar{\sigma}_{s,t} \equiv \frac{\sigma_{s,t} \log \bar{s}_{s,t}}{\sigma_{s,t} \log \bar{s}_{s,t} + 1} \qquad \rho^{\psi}_{i,t} \equiv \rho^{\psi} \bar{\sigma}_{s,t}$$

where both s and i are equal to one. The observable variation used to identify each fundamental then becomes clear. First, the observable proxies for the bank's capital cost and the regions log-odds exit rate directly identify the bank's capital cost premium and region's borrower quality, respectively

[10] The regional banking sector matrix $\tilde{H}_t$ additionally introduces the possibility of regional spillovers, and resembles the spatial autocorrelation term present in spatial vector-autoregression models, Beenstock and Felsenstein (2019).

(and after explicitly accounting for measurement error in the state-space model). The structural equation for a bank's loan default rate, Equation 6, then shows that the log loan default rate is a function of both the regional exit rate, $1 - q_{s,t}$, and the bank's screening ability $\psi_{i,t}$. However, with the regional exit rate pinned down by the observable firm survival odds, the residual loan default rate is accounted for by the bank's screening ability. Lastly, with all three other fundamentals already uniquely identified, the residual regional market share is determined by, and thus measures, the bank-region loan appeal.

Threat to identification. Because the variation identifying screening ability, $\psi_{i,t}$, comes from loan losses residualized against state-level borrower quality, $q_{s,t}$, the resulting estimate is biased if a bank's loan portfolio is exposed to sub-state or industry-specific borrower risk that diverges systematically from the state aggregate. I address this threat in two steps. first, I replace $q_{s,t}$ with a bank-specific borrower quality proxy constructed from the deposit-weighted average county-level firm survival rate across a bank's geographic footprint. Second, I instrument for this local borrower quality measure with Känzig (2021) oil supply news shocks —which satisfy both the relevance condition (pooled first-stage F-stat of 18.48) and the exclusion restriction, since bank screening technology is not plausibly affected by oil price news— and use the resulting loan losses, residualized of their borrower-quality component, as the observable in the state-space measurement equation. Appendix C.4 presents the two-step procedure, the formal relevance and exclusion conditions, first-stage diagnostics, and the resulting correction to the measurement equation.

4.2 M-step: estimating regional frictions and industry distortions

Regional loan market frictions, $\sigma_{s,t}$, and the regulatory cost coefficient, ζ_t, are recovered in a two stage procedure utilizing the cross-section of bank-level net interest income margins, residualized against the fundamentals estimated in the E-step. First, the size-based regulatory cost is identified and estimated based on the conditional correlation between banks' residualized returns and size. Second, market frictions are identified and measured with by the common regional bias in local banks residualized returns, following the demand-function approach to markup estimation widely used in the industrial organization literature (De Loecker et al., 2020). Appendix C.4 presents the underlying structural theory, full two-step estimation procedure, and identification arguments.

4.3 Data

This estimation algorithm requires data spanning banks' balance sheets and income statements, loan market shares, and non-financial firm dynamics. The following briefly describes these data while Appendix B presents a more detailed discussion.

Table 1: Summary statistics of bank size, capital costs, and loan characteristics

Variable	Units	Mean	P(5)	P(25)	P(50)	P(75)	P(95)	P(99)
Avg. capital cost	%	6.39	0.34	1.01	2.56	4.75	7.25	11.38
Loan rate	%	6.93	3.90	5.43	6.74	8.58	10.15	11.44
Loan default rate	%	0.36	0.00	0.01	0.10	0.33	1.46	3.99
Net worth	$ Bn.	0.21	0.00	0.01	0.01	0.03	0.19	1.98
Loans	$ Bn.	1.13	0.01	0.03	0.08	0.20	1.24	11.78

Notes: This table presents the summary statistics for the panel of banks used to estimate bank-level fundamentals, regulatory costs, and regional loan market frictions. There are 13,924 unique banks in the unbalanced panel. Data are annual from 1995 to 2024.

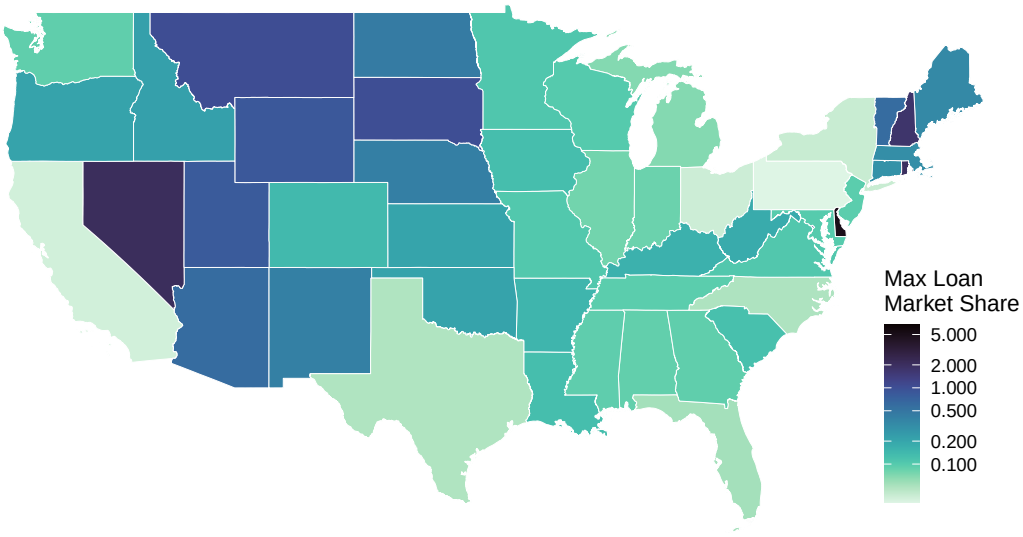
Bank balance sheets and income statements. Bank balance sheet and income statement data are obtained from U.S. regulatory Call Reports. Banks' net worth and the size of their loan portfolios are directly observable from the balance sheet. Other key objects of interest, including capital costs, loan default rates, realized returns, and loan rates, are not directly observed and must be constructed from available accounting variables. Capital costs are proxied by interest expenses divided by total loans, capturing the average interest expense required to finance loans on the bank's balance sheet. A bank's loan default rate is defined as net loan charge-offs divided by total loan assets.[11] Realized loan returns are interest income from loans divided by total loans, capturing the loan-weighted portfolio return. Lastly, loan rates can be written as a function of charge-offs and realized returns, $R = \frac{R^*}{1-\lambda}$, and are measured accordingly.

The panel displays substantial heterogeneity in size, income, and cost behavior. Table 1 summarizes the pooled distribution of key observable characteristics in the bank sample. The median bank holds $80 million in loans and $10 million in net worth. At the 99th percentile, loans reach $11.78 billion and net worth $1.98 billion—nearly an order of magnitude above the 95th percentile in both cases. The median loan rate is 6.74 percent, and the median default rate is 0.10 percent, rising to 3.99 percent at the 99th percentile.

Loan market share. A bank's regional market share is its share of all loans in a U.S. state. Banks with branches in a single state are assumed to lend entirely within that state. Banks with branches across multiple states have their loan portfolio allocated by deposit share, following Nguyen (2019) (for example, a bank raising 3/4 of its deposits in Maryland has 3/4 of its loan

[11] Charge-offs are measured net of recoverable assets, thus reflect the net losses to the bank due to the default of a given loan. Therefore, where the analytical model may be unrealistic in ignoring the possibility of recoverable collateral or liens, the empirical exercises allow for this realistic possibility.

Figure 2: Maximum bank loan market share by state



Notes: This plot shows the maximum loan market share of any bank in a given U.S. state. Market share is expressed in percent of state-level loan market. Data are annual from 1995 through 2024.

portfolio assigned to Maryland). Regional portfolios are then summed across all banks operating in a state to construct total regional loan market size, and a bank's market share is its portfolio divided by this total. Branch location and deposit data are from the FDIC's Summary of Deposits (SOD).

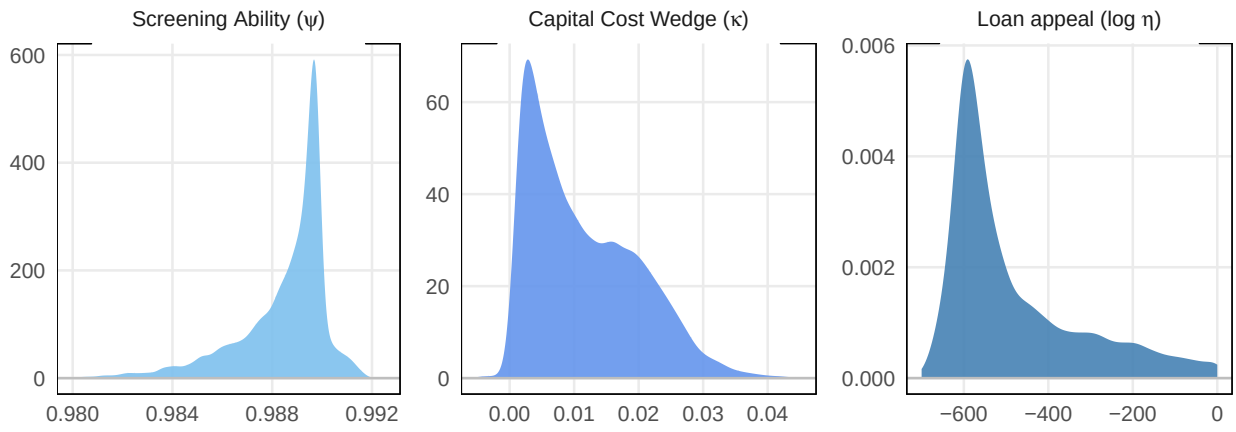
The key identifying assumption is that banks lend where they raise deposits, as required by the Community Reinvestment Act. Oberfield et al. (2024) show banks also sort into localities with excess deposits relative to lending to fund loans in higher-margin markets. I test this assumption in Appendix B and find a high spatial correlation between banks' small business and mortgage lending and their deposits, supporting its use here.

The estimated banking sector concentration is muted at the state level. The greatest state-level market share reaches approximately five percent of local loans (Figure 2), justifying the assumption that regional loan markets are "large."

Sample. The sample covers nearly the universe of U.S. commercial banks with FDIC-insured deposits and at least one physical branch, across all 50 states, and covers 99 percent of loans reported by national accounts (Appendix Figure 15). Regional firm survival rates and state-level wages are proxied by the U.S. Census Bureau's Business Dynamics Statistics (BDS) establishment exit rates and real median household income, respectively.

Data are annual from 1995 through 2023, covering 13,924 unique banks in an unbalanced panel, merger-adjusted to the bank holding company level (or the standalone bank level, absent a holding

Figure 3: Marginal distribution of bank fundamentals



Notes: This figure shows the marginal distribution of bank-level fundamentals. Each panel shows the distribution of bank-specific average fundamentals. Screening ability is presented in screening efficiency, ψ, points. The capital cost wedge is presented as a percentage point wedge between realized capital costs and the policy rate. Loan appeal is presented in log normalized CES weights. Data are at the bank level.

company), following standard practice (see for example Corbae and D'Erasmo 2021). The sample begins in 1995, when SOD data become reliably available and the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 began ending restrictions on interstate banking, and ends in 2023, limited by the Census BDS.

4.4 Describing fundamentals

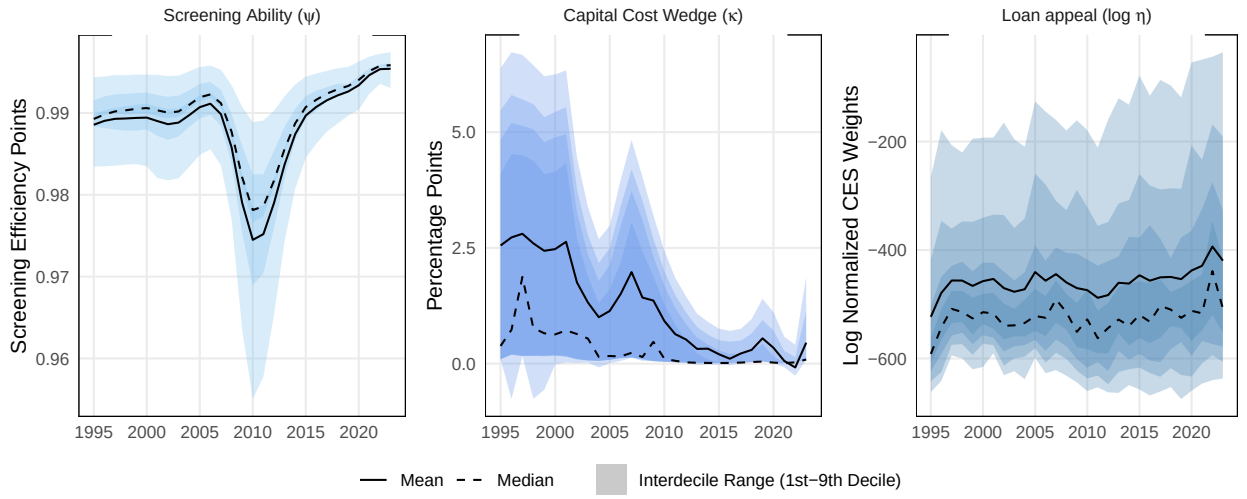
I will lastly describe the estimated bank-level fundamentals. The estimated regional fundamentals, regulatory costs, and loan market frictions are described in Appendix D.1.

Banks' average fundamentals. The marginal distributions of bank-level fundamentals jointly describe a banking sector in which most firms are relatively homogeneous in securing cheap capital inputs, highly proficient in screening risk in their loan portfolios, and poorly differentiated by non-pecuniary characteristics. Figure 3 shows the marginal distribution of the bank-level average fundamentals (i.e. the distribution of the average fundamental of a bank over its lifetime).

The distribution of bank screening ability is narrow and near its upper bound. The mode is approximately 0.99, which indicates a 1.01 percentage point increase in the modal bank's log loan default relative to perfect screening ability. Meanwhile the distribution is skewed with a fat left tail, with more than 97.5 percent of all banks exhibiting screening ability greater than 0.98 on average over their lifetime (equivalent to adding a 2.04 percentage point increase in excess loan losses).

In comparison, the distribution of the capital cost premium is more dispersed than screening

Figure 4: The evolution of bank fundamentals



Notes: This figure shows the time series evolution of the marginal distribution of bank fundamentals. Each panel shows the distribution of bank fundamentals over time. Screening ability is presented in screening efficiency points (ψ). The capital cost wedge is presented as a percentage point wedge between realized capital costs and the policy rate. Loan appeal is presented in log normalized CES weights. Data are at the bank-year level.

ability. The minimum is below zero, indicating a persistent subsidy in banks' input cost relative to the three-month Treasury. However, the large majority (greater than 95 percent of banks) have a positive capital cost wedge, paying a premium over the 3-month Treasury. The range in premiums is wide, from zero to greater than 50 basis points on average over the lifetime of the bank.

Lastly, the distribution in (log) loan appeal is the most dispersed of the three fundamentals.[12] The mode of $\log(\eta)$ is approximately -600 —a level, in the untransformed η, economically indistinguishable from zero— while a fat right tail of banks operate across the entire range up to zero, i.e., η approaching 1, indicating banks that capture all of the non-pecuniary advantage within a regional credit market. However, the vast majority of bank shave log appeal that is economically insignificant in levels and their dispersion is likewise economically meaningless.

To the extent that banks do differ in their fundamentals, Table 3 shows that a bank's fundamentals co-move: banks with stronger screening ability tend to have lower capital costs and higher demand. At the same time, bank fundamentals are also positively correlated with bank-level lending: banks with stronger screening ability, higher demand, or lower capital costs tend to lend more, as predicted by the model.

Fundamentals over time. Bank fundamentals are not stable over time. Table 2 shows the estimated parameters that govern the laws of motion for both idiosyncratic and aggregate shocks.

[12] Demand is normalized within a region-year, so η is bounded on $(0, 1]$, while $\log(\eta)$ is unbounded below.

Table 2: Estimated laws of motion for bank- and region-level fundamentals

	ρ_κ	ρ_ψ	ρ_η	$\text{Var}(\epsilon^\kappa)$	$\text{Var}(\epsilon^\psi)$	$\text{Var}(\epsilon^\eta)$
Idiosyncratic shocks						
Micro	−0.347	0.821	0.937	0.156	0.403	9138
Aggregate shocks						
Macro	0.927	1.020	—	0.00542	0.159	—

Notes: This table reports the estimated autoregressive coefficients and innovation variances for micro- and macro-level fundamentals. Dashes indicate parameters not estimated at the aggregate level. The variance of ϵ^η is the variance of shocks before bank-region-level loan appeal is normalized.

Table 3: Correlations across bank fundamentals and lending

	ψ	κ	η	Total Loans
ψ	1.000			0.268
κ	−0.094	1.000		−0.176
η	0.130	0.249	1.000	0.124

Notes: This table reports correlations computed across bank-year observations of screening ability (ψ), the capital cost wedge (κ), demand (η), and total lending, weighted by each bank's average net worth over the sample period.

Aggregate shocks exhibit greater persistence, while micro-level shocks display substantially higher volatility —a pattern also found among estimates of non-financial firm shock processes (see for example Bloom et al. 2018). Moreover, as individual banks are buffeted by both idiosyncratic and common shocks, the full distribution of banks' realized fundamentals evolves as well.

The distribution of bank-level screening ability was relatively stable through 2006 (in line with Philippon (2015, 2022) finding that the average monitoring cost of the US financial sector has been relatively stable since the early 1900s) before exhibiting a large and coordinated decline during the GFC. That is, the financial crisis coincided with a large and widespread decline in banks' ability to assess risk in their loan portfolios, an observation echoed by regulatory narrative accounts of the crisis.[13] After the financial crisis screening ability began a secular recovery that persists to the end of the sample, leaving the average screening ability greater in 2022 than in the pre-crisis period.

Meanwhile, the dispersion of banks' screening ability is not stable through time, and the evolution is not symmetrical. There is a right tail of high-performing banks that throughout the sample have close to a perfect screening ability. However, the left tail, and in particular the bottom decile of banks, has a much lower and more variable screening ability through time. For example, while the top decile of banks saw screening ability fall by less than 0.1 percent during the GFC, the bottom decile fell from 0.975 to 0.945 —a decline of roughly 3 percent in ψ that corresponds to an increase in implied excess loan losses from 2.56 to 5.82 percentage points, more than doubling. Appendix D.3 presents exercises exploring a possible source of fluctuations and differentiation in bank-level screening ability, expenditures on IT and loan officers.

The evolution of the capital cost distribution is also asymmetric. The right tail of capital costs

[13] "Faced with uncertainty about the value of specific instruments and mindful of the higher volatility of assets more generally, lenders demanded substantial cushions... on the assets they were willing to finance." (Senior Supervisors Group, 2009; Risk Management Lessons from the Global Banking Crisis of 2008)

Table 4: Persistence in banks' fundamental rankings

	Panel A: Full sample (deciles 1–10)				Panel B: Interior deciles (2–9)			
	P(Up)	P(Same)	P(Down)	Net	P(Up)	P(Same)	P(Down)	Net
ψ	0.369	0.227	0.403	−0.034	0.385	0.206	0.410	−0.025
κ	0.285	0.401	0.314	−0.029	0.294	0.381	0.325	−0.030
η	0.335	0.300	0.365	−0.030	0.329	0.304	0.367	−0.038

Notes: This table reports the estimated probability of a bank transitioning across deciles of the marginal fundamental distribution. P(Up), P(Same), and P(Down) denote the pooled probability that a bank's within-year decile rank for the given fundamental rises, stays the same, or falls between consecutive sample years. Net denotes net upward mobility, $P(\text{Up}) - P(\text{Down})$.

is positive and declining over time, while the left tail of the distribution is approximately zero for the entire sample. Dispersion in capital costs is positively correlated with the monetary policy tightening cycle, reflecting heterogeneity in banks' reliance on interbank and repo lending markets relative to cheap deposits (Drechsler et al. 2017). Appendix D.2 examines the sources of capital cost heterogeneity, including its relationship to banks' funding mix.

The large negative numbers in $\log(\eta)$ are indicative of banks with loan appeal near zero, indicating little non-pecuniary differentiation among competitors. The loan appeal distribution is relatively stable over time, although with the upper 75th and 90th quantiles beginning a positive secular trend after the GFC. This increasing dispersion in appeal coincides with the secular decline in the number of banks (a possible source of banks' appeal).

Ranking analysis. Even though the distribution of bank fundamentals moves over time, banks' position within the distribution are largely static. Table 4 shows the probability of a bank transitioning from its current decile for a given fundamental to a higher or lower position within the marginal distribution. On average the net upward mobility of a bank is roughly −3% for the full distribution (and −3% to −4% when restricting to interior deciles only). So while banks are more likely to see a relative deterioration in their ranking compared to competitors than a rise in their standing, this drift is quantitatively small. It is therefore accurate to interpret a bank's rank in a given fundamental as relatively persistent, with only small net changes in rank over time.

5 Quantifying Bank Dynamics

With empirical measures for bank fundamentals and market frictions now in hand, I can turn to asking which forces and processes shape bank dynamics and the resulting heterogeneity across banks. To do so, I rely on two exercises. I first trace the response of a bank's loan spreads,

lending, and net worth to an unexpected change in its fundamentals. These impulse response functions will characterize how each fundamental impacts a bank's lending behavior and growth opportunities. I then decompose the within- and across-bank variation in these dynamics into the relative contributions made by each shock in the economy. These variance decompositions in turn identify which shocks are responsible for individual banks' ability to grow over time and the heterogeneity observed across banks' lending behavior.

5.1 Quantitative setting and tools

To isolate the causal relationship between economic fundamentals and bank dynamics, I turn to a simulation-based quantitative exercise where I can directly observe the changes in bank outcomes in response to structural shocks. To do so, I embed the empirically estimated processes from Section 4 in a quantitative calibration of the regional New Keynesian DSGE outlined in Section 2.

I quantify the model in three steps. I first calibrate the model. The laws of motion for bank- and region-level fundamentals are set to the empirical estimates in Section 4, while deep parameters take standard values from the New Keynesian DSGE and macro-banking literatures. I then solve the banks' regional entry problem under a computationally tractable bounded-rationality rule, following Krusell and Smith (1998) and Weintraub et al. (2008). Conditional on entry and exit decisions, bank- and region-level outcomes are available analytically. Finally, I simulate the economy, beginning with 50 ex ante identical regions and 500 banks headquartered uniformly across regions. Because the fundamentals' laws of motion are estimated from U.S. bank- and state-level data, the model is disciplined by the micro-data, and Appendix E.3 shows that this bottom-up approach reproduces aggregate and regional business cycle dynamics reasonably well. Appendix E presents the calibration, solution method, and simulation in detail.

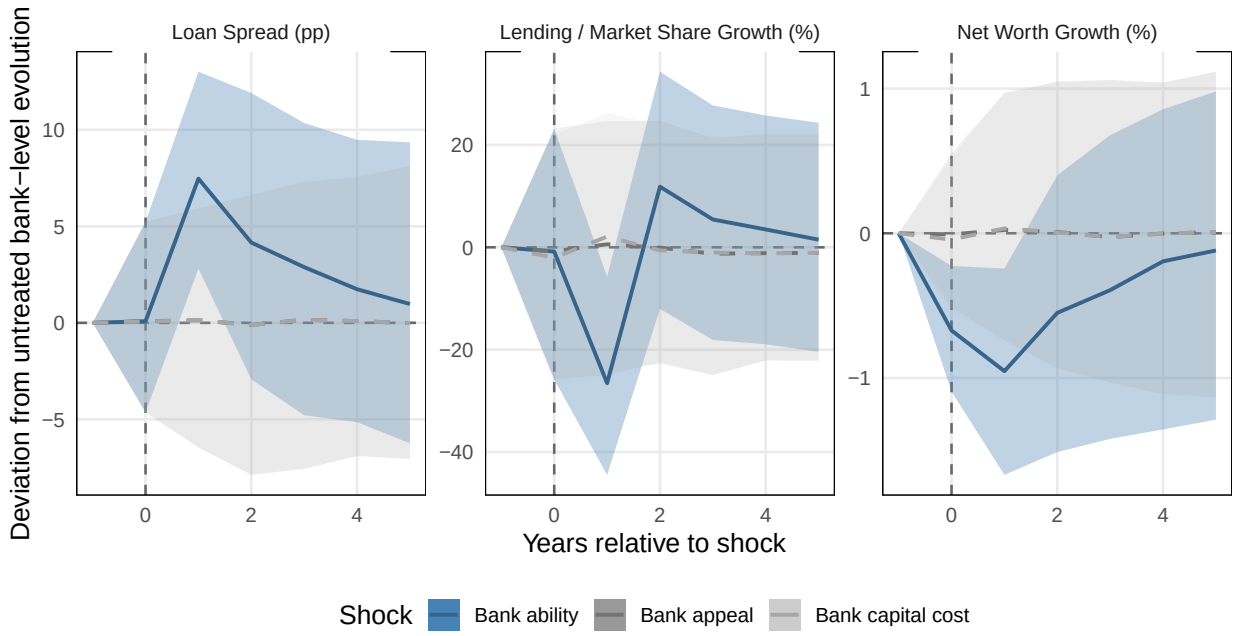
To then characterize the dynamic effects of shocks in this nonlinear environment, I compute Generalized Impulse Response Functions (GIRFs) following the simulation-based approach of Koop et al. (1996). For each structural shock j, the GIRF at horizon h is defined as the difference in conditional expectations of an endogenous variable y between two economies that share the same information set Ω_t and the same future sequence of stochastic innovations, except for a one standard deviation shock to fundamental or friction j at time t:

$$\text{GIRF}_h(\varepsilon_j) = \mathbb{E}\left[y_{t+h} \mid \varepsilon_j, \Omega_t\right] - \mathbb{E}\left[y_{t+h} \mid \Omega_t\right]. \tag{16}$$

This procedure isolates the causal effect of the exogenous fluctuation in fundamental or friction while preserving the full nonlinear state dependence of the system.

Lastly, to assess the relative importance of each structural shock for variation in endogenous bank dynamics over time, I construct a Forecast Error Variance Decomposition (FEVD) based on

Figure 5: Bank-level responses to fundamental shocks



Notes: This figure presents the response of a bank's loan rate, lending volume, and net worth to a one-standard-deviation deleterious shock to screening ability, the capital cost premium, and loan appeal. The solid and dashed lines mark the mean response of banks subject to the idiosyncratic shocks relative to the same bank absent the shock. Bands mark the 90 percent confidence interval. Lending and net worth are presented in percent deviation from the control and the spread in percentage points. Responses are based on sampling 50 banks in each of 500 simulations of the economy.

the simulated GIRFs, in the spirit of Pesaran and Shin (1998). For variable i and horizon t, the contribution of shock j is given by

$$\text{FEVD}_{i,t}^{(j)} = \frac{\left(\text{GIRF}_{i,t}^{(j)}\right)^2}{\sum_k \left(\text{GIRF}_{i,t}^{(k)}\right)^2}, \tag{17}$$

where the denominator aggregates the squared responses to all structural shocks. The FEVD is the decomposition of innovations in endogenous variables (e.g. loan rates and regional market shares) into the relative contributions of exogenous fluctuations in fundamentals and frictions.

5.2 How fundamentals shape dynamics

I begin my analysis by tracing how bank-specific shocks propagate through a bank's balance sheet. Figure 5 presents the response of a bank's loan rate, lending volume, and net worth to a one-standard-deviation deleterious shock to screening ability, the capital cost premium, and loan appeal. The screening ability shock decreases net worth by approximately 0.75 percent on impact,

and subsequent growth only slowly recovers over the next five years. At its peak, one year after impact, the screening-ability shock raises the loan spread by approximately 7 percentage points and slows lending growth, and thus market share growth, by almost one-third. In comparison, the response to a screening-ability shock exceeds the response to either of the other two shocks by several orders of magnitude, both of which are economically and statistically insignificant at any standard level across all horizons.

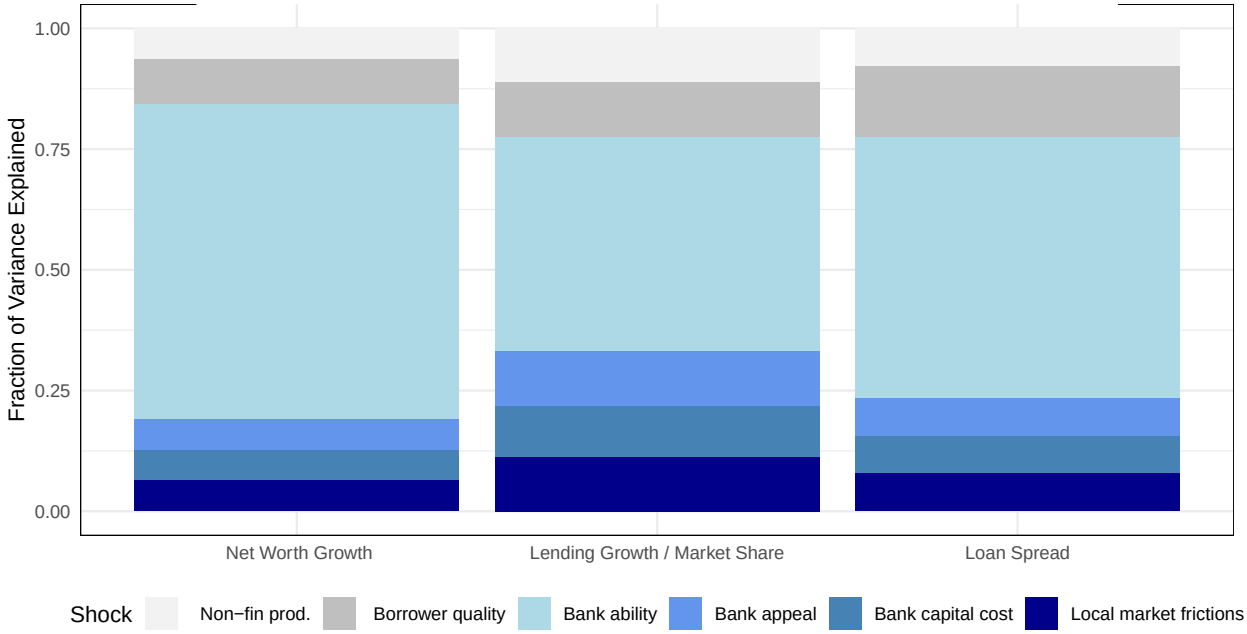
All three shocks theoretically set off the same self-reinforcing decline in a bank's pricing, lending, and size (as measured by net worth). A deterioration in the bank's fundamentals initiates an increase in loan rates and dampens lending volume. The decline in lending, paired with any increase in loan losses, erodes net worth. The shrinking balance sheet then further increases loan rates through an increasing regulatory cost, which exacerbates the decline in lending. This balance-sheet accelerator is common to all three fundamentals, but what differs is the magnitude and speed of the accelerator.

However, a deterioration in screening ability engages the accelerator immediately and generates by far the largest response. Because loan rates are posted before a bank's screening ability is realized each period, an unanticipated decline in ability is not priced into the current period's loan rate and the bank absorbs the resulting increase in loan losses before it has any opportunity to reprice. Net worth therefore falls on impact, and the resulting increase in regulatory costs compound with the downward revision to expected ability to push loan rates sharply higher the following period, further depressing lending and net worth growth in turn.

In contrast, the other two shocks are known and accounted for by the bank before lending occurs. A rise in the capital cost premium enters the loan spread directly as an increase in the marginal capital cost passed on to borrowers. A decline in loan appeal shifts loan demand and dampens quantities as fewer borrowers choose to work with the bank at the prevailing loan rate. Under either shock the accelerator engages only gradually, as the bank loses market share due to higher prices or lower demand, which in turn counterfactually slows net worth growth and elevates spreads while dampening profits.

The screening-ability shock is therefore the most consequential of the three not because its ultimate qualitative effect differs – all three fundamentals eventually generate the same self-reinforcing decline in a bank's pricing, lending, and size – but because it alone catches the bank's pricing off guard, exposing net worth to an unanticipated loss that immediately activates the balance-sheet accelerator. Capital cost and appeal shocks, in contrast, are either priced away on impact (capital cost) or never priced at all (appeal), sparing the bank's net worth in the near term and slowing the onset of the same downward spiral.

Figure 6: Decomposition of simulated *across*-bank dynamics



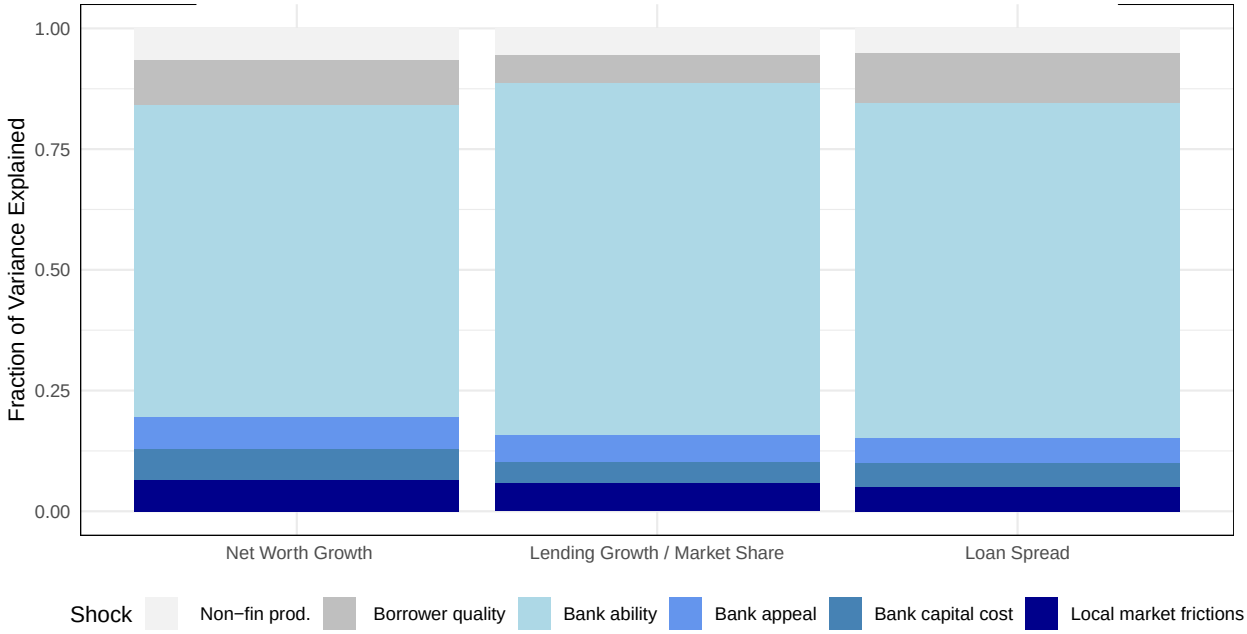
Notes: This figure presents the structural across-bank variance decomposition of bank dynamics into contributions from shocks to fundamentals and market frictions. The decomposition is the average contribution by shock to the endogenous outcome over five years after the realization of the structural shocks. The forecast error variance decompositions are based on GIRFs that are the average response of 500 simulated economies.

5.3 Decomposing bank dynamics

Realized shocks to bank fundamentals therefore determine both the trajectory of an individual bank as it grows over time. It also follows that as idiosyncratic realizations vary across banks, they too drive heterogeneity across the banking sector. I next decompose variation in lending outcomes within- and across-banks into contributions from the model's underlying fundamental shocks to understand their relative importance in driving bank dynamics. Figure 7 presents the within-bank FEVD while Figure 6 presents the across-bank FEVD.

Loan spreads. Changes in a bank's screening ability are the primary reason for changes in a bank's loan spread over time, accounting for 69.6 percent of the within-bank variation in loan rate pricing. The remaining 30.4 percent is largely described by fluctuations in external fundamentals—principally borrower quality (10.3 percent), along with non-financial productivity and regional market frictions—accounting for approximately 20.4 percent. Internal fundamentals—the capital cost premium and loan appeal—account for approximately 10.1 percent. Compensation for expected credit losses, captured jointly by screening ability and borrower quality, therefore accounts for approximately 79.9 percent of within-bank variation in loan pricing and is the primary

Figure 7: Decomposition of simulated *within*-bank dynamics



Notes: This figure presents the structural within-bank variance decomposition of bank dynamics into contributions from shocks to fundamentals and market frictions. The decomposition is the average contribution by shock to the endogenous outcome over five years after the realization of the structural shocks. The forecast error variance decompositions are based on GIRFs that are the average response of 500 simulated economies.

determinant of a bank's own loan-spread fluctuations over time.

Fluctuations in loan spreads *within* banks in turn generate variation in loan spreads *across* banks. Idiosyncratic screening ability shocks prompt banks to adjust their loan spreads, and as these shocks are realized differentially across banks within a period, dispersion in loan spreads offered across banks emerges as well. Screening ability accounts for 54 percent of the cross-sectional variation in loan spreads, borrower quality accounts for 14.8 percent, and the remaining 31.2 percent of the variation is evenly accounted for by fluctuations in the capital cost premium, loan appeal, non-financial productivity, and regional market frictions. Therefore, heterogeneity in bank-level loan pricing is primarily driven by differences in banks' required compensation for expected losses, manifesting from heterogeneity in banks' ability to assess risk and the inherent creditworthiness of borrowers across the markets banks operate in.

Lending. Variation in a bank's loan and market share growth over time is primarily driven by shocks to the bank's screening ability. Fluctuations in a bank's ability to assess risk account for 72.9 percent of medium-run fluctuations in its loan volume growth.[14] The remaining 27.1

[14]Recall that while the model gives screening ability the strict interpretation of a bank's ability to extract a signal

percent is not evenly split between external and internal factors: external fundamentals—non-financial productivity, borrower quality, and regional market frictions—account for approximately 17.0 percent, while internal fundamentals—loan appeal and the capital cost premium—account for approximately 10.2 percent.

Heterogeneity in banks' lending and market share growth across banks is likewise dominated by cross-sectional variation in screening ability, which accounts for 44.5 percent of the variation across banks' loan and market share growth. The remaining 55.5 percent of the differences across banks' lending growth rates is evenly distributed across the remaining shocks. This indicates that banks expand their lending volume and capture market share relative to their peers primarily on the basis of their ability to appropriately identify and price risk in their loan portfolios, rather than due to differential exposure to high-quality borrowers, deep local loan demand, or frictional market environments.

Net worth. Within banks, screening ability accounts for approximately 64.6 percent of medium-run fluctuations in net worth growth. The cross-sectional decomposition yields an almost identical share, with screening ability accounting for 65.1 percent of the heterogeneity in banks' balance sheet growth across the financial sector.

Summary. These results, taken together with the loan-spread and market-share decompositions above, describe a coherent mechanism: banks with improving screening ability first capture market share by reducing the compensation for expected losses they must charge, and as market share rises, so does the oligopolistic profit extracted from the regional loan market. Improving screening ability then protects these growing profits by reducing uncovered losses on the loan portfolio, raising retained earnings and, cumulatively, net worth. More simply put: as screening ability improves, the price of a bank's loans falls and the quantity of its lending rises; because banks compete oligopolistically, greater lending translates directly into greater market power and economic profit, fueling further growth. In comparison, banks with decreasing screening ability record larger uncovered losses, initiating the balance-sheet accelerator effect and a rapid deterioration in size. The speed and magnitude of the accelerator in response to a negative screening shock are what set a bank's screening ability apart from the other shocks as the most important driver of bank dynamics.

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of a borrower's creditworthiness from noisy information, the empirical measurement strategy requires the broader interpretation that screening ability captures a bank's ability to anticipate losses in its loan portfolio—which includes a bank's ability to identify loan applicants who will default or require a haircut, as well as its ability to recover valuable collateral in the case of default (analogous to a collateral recovery technology shock in Kiyotaki and Moore 1997 and Justiniano et al. 2011). A bank's ability to expand its lending volume and capture market share therefore varies with its ability to anticipate and mitigate losses on its balance sheet, or the level of recoverable collateral backing loans.

Empirical robustness. Appendix F.1 presents empirical evidence of the transmission channels, showing that fluctuations in screening ability induce empirical contractions in bank-level lending, growth and tightening loan rates, while capital cost and loan appeal fluctuations have mixed economic and statistical significance. Meanwhile, Appendix F.2 presents an empirical variance decomposition of observable loan spreads and market shares in the spirit of Hottman et al. (2016). Variation in screening ability accounts for the majority of both within- and across-bank variation in observable loan spreads and market share.

5.4 Comparing bank and non-financial firm dynamics

Bank dynamics are primarily driven by bank-level screening ability—an analog to a bank's lending productivity. This finding makes banks both like and unlike non-financial firms.

Banks' screening ability, rather than loan appeal, drives within-bank growth in balance sheet size and market share, as well as their heterogeneity across the banking sector. This is a sharp break from the non-financial firm dynamics literature, which typically finds product- or firm-specific demand to be the dominant driver of firm-level growth (see, for example, Foster et al. 2008; Hottman et al. 2016; Eslava et al. 2024). Loan appeal, by contrast, accounts for essentially none of the variance in spreads or market share in either the structural or empirical decomposition.

The most straightforward explanation is that borrowers treat loans as homogeneous good that only meaningfully differ on price. A borrower's loan is a claim to a fixed quantity of credit at a price, and unlike a consumer good, it carries no vertical or horizontal attribute (for example, style, brand, or convenience) that a borrower values independently of price and availability. If borrowers are close to indifferent between otherwise-identical loans, then screening ability and its effect on price, rather than appeal, determines which banks win business.

In contrast, bank dynamics mirror a key pattern in non-financial firm dynamics literature: Aggregate productivity growth is driven substantially by reallocation across heterogeneous producers rather than by within-producer technical change, even when average productivity is roughly flat (Foster et al., 2001). The same logic applies here: an aggregate fundamental, screening ability, can be stable on average while remaining the single most important source of which banks grow and capture market share.

6 Bank and Non-Financial Firm Dynamics

Bank dynamics determine which banks grow and capture market share and, because these dynamics are driven primarily by screening ability, they also shape the allocation of credit across borrowers. The following two sections examine how bank dynamics affect economic activity through two

channels. The first operates through non-financial firm dynamics, as differences in bank screening ability affect which firms receive credit and ultimately produce. The second operates through the transmission of monetary policy, as differences in bank fundamentals affect how regional economies respond to common policy shocks.

I now turn to the first channel. I first examine how bank dynamics shape non-financial firm dynamics, and then consider the implications for economic activity in the short and long run.

6.1 Two channels

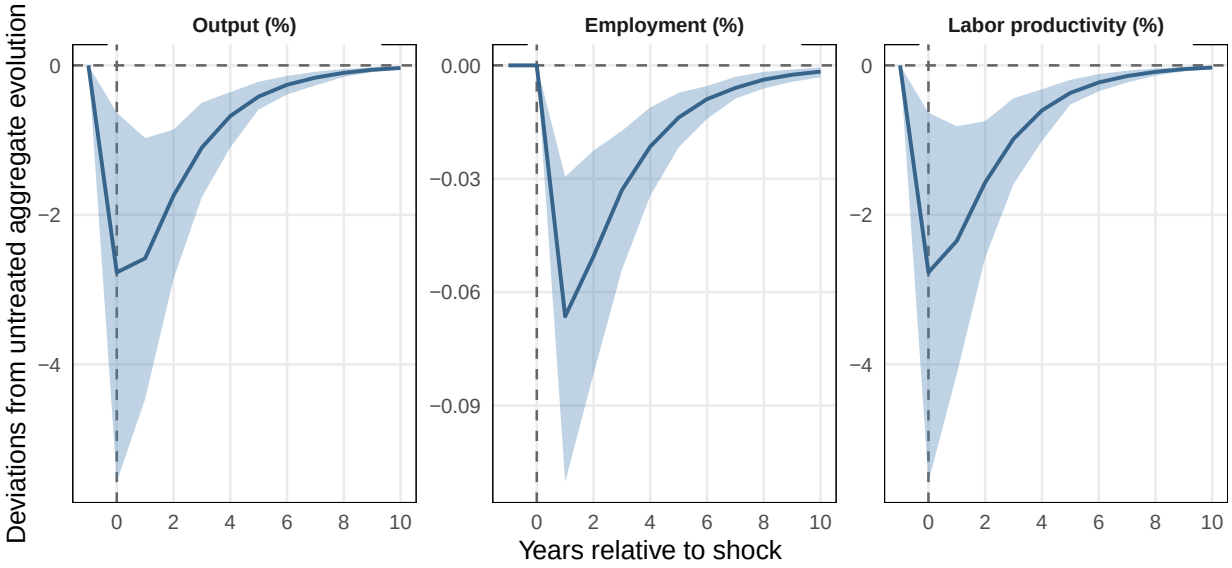
Bank dynamics shape non-financial firm dynamics through two channels: a selection channel and a factor pricing channel. The former determines the extensive margin of non-financial firm dynamics, how many and the quality of firms that enter the market. The latter determines the intensive margin of non-financial firm dynamics, how much each firm produces.

Figure 8 shows the response in aggregate output, employment, and labor productivity to a negative, one standard deviation, aggregate screening ability shock.[15] These three responses show the transmission channels at work. First, the selection channel is activated. On impact, screening ability falls, but banks do not yet observe the change. Holding all else equal, loan rates remain unchanged while a higher proportion of bank loans are directed toward firms that will exit before production. As a result employment is unaffected as loan rates, thus effective wage bills, remain constant. However, output falls by 2.64 percent. Labor productivity, measured as output divided by inputs (employment), therefore falls by 2.64 percent as well. Then one period after impact the factor pricing channel is activated. After banks realize an elevated degree of loan losses they revise down their expected screening ability and loan rates increase. As loan rates increase effective wage bills increase in lock-step, and employment falls approximately seven basis points. In the background monetary policy has become accommodative and output begins to recover, but not enough to return labor productivity to pre-shock levels. In fact, labor productivity does not fully recover before ten years after the screening shock was realized.

I next trace how these transmission channels shape short- and long-run productivity. Appendix G.1 presents an extended set of macroeconomic responses to all three bank fundamental shocks.

[15] I estimate the IRF of output, employment, and labor productivity as percent deviations from the economy's counterfactual evolution that would have been realized in the absence of the shock. Because these variables have a stationary ergodic mean in the simulated economy, averaging over a large number of simulations results in a response that marks the variable's deviation from its stationary steady state. This is in contrast to using growth rates for bank-level variables, because an individual bank's net worth can grow unbounded over time, so its lending and net worth outcomes are not stationary in level but instead must be presented in growth rates to achieve a stationary time series for analysis.

Figure 8: Macroeconomic response to a screening ability shock



Notes: This figure presents the response of aggregate output, employment, and labor productivity to a one standard deviation deleterious shock to screening ability. Solid blue lines are the average response and the blue bands mark the 90 percent confidence intervals. IRFs are based on the mean response of aggregate conditions after a common shock to screening ability relative to the counterfactual evolution of the same economy as if no shock had occurred. Outcomes are presented as percent deviation from the control. Responses are based on 500 simulations of the economy.

6.2 Business cycle fluctuations in labor productivity

I begin in the short run with business cycle fluctuations. The following formally decomposes regional and aggregate labor productivity and presents a quantitative exercise measuring the strength of either channel before comparing their effects to non-financial sources of fluctuations.

6.2.1 A framework for decomposing productivity

Given the effective regional loan rate $R_{s,t}$, the labor market clears at the wage $W_{s,t}$ and employment $M_{s,t}$ that solve the firm's problem. The *planned* output, defined as the output that would be realized if every funded borrower survived, is $Y^p_{s,t} = A_{s,t} M^{\alpha}_{s,t}$, with α the labor share. Borrower exit is realized after production is planned, labor is hired, and bank loans acquired. The realized regional exit rate coincides with the loan-weighted default rate:

$$\lambda_{s,t} = \chi^q_{s,t} \chi^w_{s,t}, \qquad \chi^w_{s,t} = \sum_{i \in \mathcal{B}_{s,t}} s_{i,s,t} \chi_{i,t} \qquad \sum_{\mathcal{B}_{s,t}} s_{i,s,t} = 1, \tag{18}$$

which is a function of both underlying borrower quality and banks' screening ability.[16] Realized regional output is therefore:

$$Y_{s,t} \;=\; (1-\lambda_{s,t})\, Y^{p}_{s,t}. \tag{19}$$

After combining Y^{p}_{st} and (19), the log-linearized regional labor productivity can be decomposed into three terms:

$$\log\!\left(\frac{Y_{st,}}{M_{s,t}}\right) \;=\; \underbrace{\log(1-\lambda_{s,t})}_{\text{selection}} \;+\; \underbrace{a_{s,t}}_{\text{technology}} \;+\; \underbrace{(\alpha-1)\log M_{s,t}}_{\text{input intensity}}. \tag{20}$$

The first term is determined entirely by the screening ability of the regional banking sector and the quality of local firms. The third term is determined by the equilibrium loan rate (as well as regional fundamentals and underlying production technologies), therefore also by the banking sector.

Aggregate labor productivity. Aggregate productivity follows as the activity-weighted average of regional productivity, which allows for a straightforward aggregation and subsequent decomposition. Define national aggregates $Y_t = \sum_s Y_{s,t}$, $Y^{p}_{t} = \sum_s Y^{p}_{s,t}$, and $M_t = \sum_s M_{s,t}$, and the three reported productivity objects:

$$\mathrm{LP}_t \equiv \frac{Y_t}{M_t} \qquad \mathrm{SEL}_t \equiv \frac{Y_t}{Y^{p}_{t}} \qquad \mathrm{LPP}_t \equiv \frac{Y^{p}_{t}}{M_t} \tag{21}$$

where LP_t is the aggregate labor productivity, SEL_t is the aggregate selection wedge, and LPP_t is the aggregate planned labor productivity. Note that the aggregate planned labor productivity subsumes the technology and input intensity productivity factors defined in Equation 20, and is by definition the labor productivity observed under perfect screening ability and borrower quality.

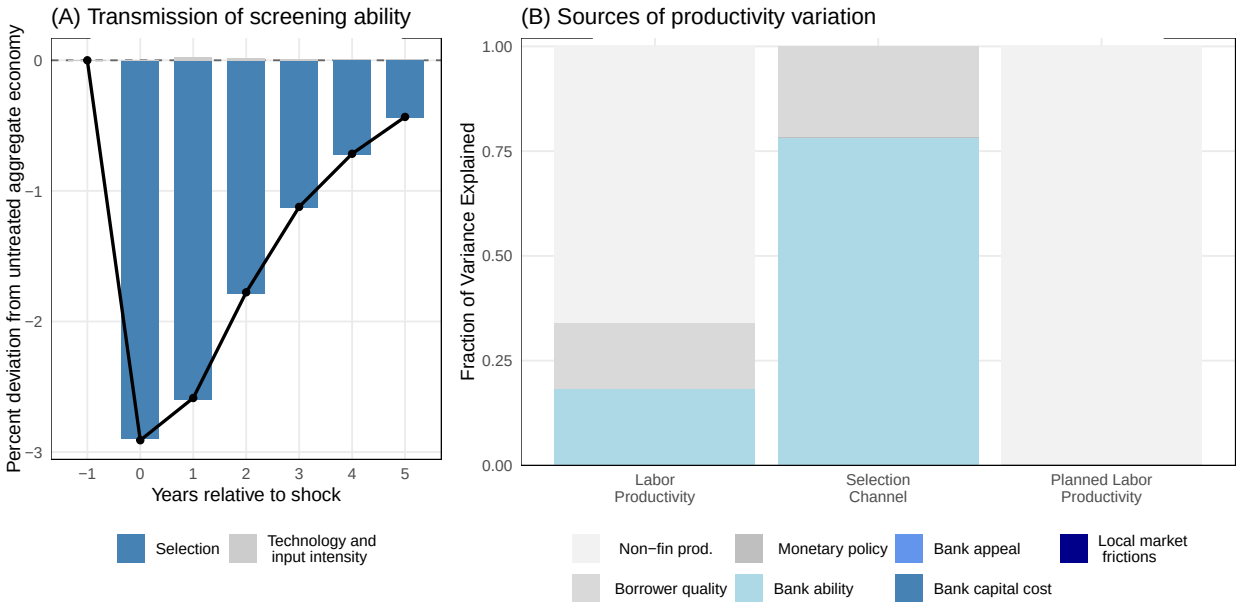
Aggregate labor productivity is then multiplicatively separable in an aggregate selection wedge and planned labor productivity:

$$\mathrm{LP}_t \;=\; \mathrm{SEL}_t \times \mathrm{LPP}_t, \qquad \mathrm{SEL}_t \;=\; \sum_s \frac{Y^{p}_{s,t}}{Y^{p}_{t}}\,(1-\lambda_{s,t}). \tag{22}$$

where the labor productivity identity is immediate from Equation 21, and the restatement of the selection wedge follows from substituting Equation 19. The aggregate selection wedge is thus the planned-output-weighted mean firm survival rate. Aggregate labor productivity factors exactly into a banking-sector component and a component that is determined as if under perfect screening.

[16] Only firms that are funded can plan to produce, so the exit rate is the proportion of funded borrowers that exit. Since all firms are ex ante identical, a bank's default rate is the exit rate of firms in its loan portfolio.

Figure 9: Decomposition of labor productivity by channels and structural shocks



Notes: The left panel plots the response of aggregate labor productivity to a one-standard-deviation negative shock to screening ability, together with its decomposition into the selection wedge and planned labor productivity components. The solid black line marks the mean response of labor productivity to the shock, while the blue bars indicate the selection channel contribution and the gray bars mark the input intensity channel contribution. The right panel reports the share of business cycle variance decomposition of labor productivity and its components. Responses and variance shares are based on 500 simulations of the economy.

6.2.2 Quantifying screening ability effects on productivity

I trace the effect of banks' screening ability on labor productivity by estimating the endogenous response of labor productivity to a one standard deviation negative screening ability shock. Then, using the decomposition in Equation 22, I decompose these fluctuations into their two underlying components. The left-hand panel of Figure 9 shows this GIRF and decomposition. The right-hand panel then places the importance of screening shocks in context with a full FEVD of labor productivity and its two components.

The selection channel accounts for almost the entirety of the change in labor productivity, with input intensity accounting for less than five percent of the transmission one year after the shock, at its maximum relevance. This muted role for input intensity reflects curvature in the decreasing-returns-to-scale production technology. Changes in labor input dampen, rather than proportionally translate into, changes in output. Meanwhile, Equation 20 implies that a change in default rates due to changes in banks' screening ability will have an almost one-to-one effect on labor productivity through the selection channel.

Fluctuations in screening ability account for the majority of business cycle fluctuations in the

screening wedge and approximately a quarter of overall labor productivity and TFP variation. Figure 9 shows that shocks to banks' ability account for 81.6 percent of fluctuations in the screening wedge, dominating the role of underlying borrower quality (the unconditional firm exit rate). Meanwhile, the input intensity and technology channels are dominated by fluctuations in the non-financial productivity shifter. Through these channels, fluctuations in banks' screening ability then account for 27 percent of business cycle frequency fluctuations in labor productivity.

6.3 Long-run productivity

As fluctuations in bank dynamics originating in screening ability shocks drive short-run fluctuations in productivity, the steady-state composition of screening ability across banks determines the long-run level of productivity. This is because both the level and dispersion in screening ability determine how effectively the financial sector can route resources to productive firms.

6.3.1 The cost of the level of screening ability

The average screening ability of banks determines how effectively they lend to firms that will produce. Therefore, any screening ability below the perfect-screening benchmark ($\psi = 1$) moves resources toward firms that will not produce, decreasing productivity in the economy.

The average screening ability estimated in the United States is $\bar{\psi} = 0.989$. This level of screening ability implies a 1.18 percentage point increase in the average annual default rate relative to an economy with perfect screening.[17] Then, because log labor productivity has a unit elasticity with the log firm survival rate $\log(1 - \lambda_{s,t})$, and λ is small so that $\Delta \log(1 - \lambda) \approx -\Delta\lambda$, this 1.18 percentage point increase in defaults translates to an approximately 1.18 percent decline in labor productivity. This is the direct effect through the selection channel, before any further adjustment in loan rates and employment through the factor-pricing channel.

To put this in further context, consider a simple back-of-the-envelope calculation. Take the average labor force, 151.6 million people (BLS), and real GDP, \$16.896 trillion (2017 dollars, BEA), from 1995 through 2023 as indicative of long-run steady-state values for the United States.Then an approximately 1.18 percent decline in labor productivity corresponds to a \$1.315 thousand per-worker economic cost.

However, it is not just the level of screening ability that shapes long-run labor productivity, but the entire distribution of screening ability across banks.

[17] This is based on the log-linear decomposition of loan default rates, which isolates the contribution of screening ability from the underlying borrower quality.

6.3.2 The cost of dispersion in screening ability

As bank dynamics generate heterogeneity across banks, the level of heterogeneity itself shapes labor productivity. The intuition is straightforward and works through two channels. First, there is a loss from realized dispersion in bank screening ability. A bank that screens badly funds projects that fail and destroys real output. Simultaneously, a bank that screens well can at best avoid making bad loans. The marginal damage from the left tail exceeds the marginal gain from the right tail, so heterogeneity destroys aggregate credit quality even holding the distribution of credit across banks fixed. Second, there is a gain from the heterogeneous allocation of lending across banks, that is, heterogeneity in market share. This is because the banks that grow large are those that are good at screening risk. Therefore, as the banking sector becomes more concentrated, and market share more dispersed, then a larger share of lending is accounted for by banks with high screening abilities, which may attenuate firm exit rates and improve labor productivity. This is a permanent, steady-state cost of cross-sectional heterogeneity, distinct from the transitory business-cycle response to a common screening ability shock. The following analytically characterizes these two forces and then quantifies their strength and cost in the U.S. economy.

A framework for decomposing the effects of bank heterogeneity. A region's exit rate is a function of heterogeneity in market share and screening ability. Therefore, to measure the marginal contribution of either, I need to construct counterfactual measures that isolate the change in defaults from heterogeneity in screening ability and market share. To do so, I turn to two alternative region-level aggregations of the log odds transform of screening ability, which I will refer to as the screening ability wedge, $\chi^{\psi} = (1 - \psi_{i,t})/\psi_{i,t}$:

$$\chi^{w}_{s,t} = \sum_{i} s_{i,s,t}\, \chi^{\psi}_{i,t}, \qquad \bar{\chi}_{s,t} = \frac{1}{n_{s,t}} \sum_{i} \chi^{\psi}_{i,t}, \qquad \chi^{g}_{s,t} = \exp\Big(\frac{1}{n_{s,t}} \sum_{i} \log \chi^{\psi}_{i,t}\Big) \tag{23}$$

where $n_{s,t}$ is the number of banks operating in region s and time t, χ^{w} is the loan-share weighted region-level selection wedge (the model's baseline measure), $\bar{\chi}$ is the equally weighted arithmetic mean, and χ^{g} is the equally weighted geometric mean.

The baseline model aggregation allows for heterogeneity in both market share and ability. The arithmetic mean, in comparison, measures the average bank-level screening ability under the realized dispersion in screening ability, as if there were no dispersion in market share. The geometric mean goes a step further and measures the average bank-level screening ability as if there were no dispersion in either banks' screening ability or market share. Comparing exit rates across the arithmetic and geometric means therefore isolates the contribution from dispersion in screening ability, while comparing exit rates across the model benchmark and the arithmetic mean isolates

the contribution from dispersion in market share.

These proposed aggregations follow the price index measurement literature which takes advantage of the properties of arithmetic versus geometric means for isolating the effects of dispersion in constructing aggregate indices (while assuming a lognormal random variable of interest). In this case, bank screening ability is estimated to follow the log-linear law of motion: $\log \chi_{i,t} = \mu_\psi + \rho_\psi \log \chi_{i,t-1} + \varepsilon^\psi_{i,t}$. Therefore ergodically $\log \chi \sim \mathcal{N}\left(\frac{\mu_\psi}{1-\rho_\psi}, \frac{\sigma^2_\psi}{1-\rho^2_\psi}\right)$ and the cross-sectional geometric mean is: $\chi^g = \exp\{\frac{\mu_\psi}{1-\rho_\psi}\}$. Note that the dispersion in bank-level screening ability, characterized by σ^2_ψ, does not enter the aggregate χ^g. In comparison, taking the arithmetic mean of the level of the screening ability wedge results in $\bar{\chi} = \exp\{\frac{\mu_\psi}{1-\rho_\psi} + \frac{1}{2}\frac{\sigma^2_\psi}{1-\rho^2_\psi}\}$. Thus the log difference between the arithmetic and geometric mean captures the marginal increase in the screening ability index due to the dispersion in bank-level screening ability. This mirrors the same identity that relates the Dutot and Jevons price indices (as shown in Silver and Heravi 2007).

Characterize dispersion cost. Guided by this framework, I derive two analytical characterizations of how heterogeneity in screening ability will impact productivity. Two results emerge. First, as dispersion in realized screening ability widens, labor productivity declines. Second, as dispersion in market share increases, labor productivity increases, if screening ability and market share are positively correlated. Proofs for both propositions are presented in Appendix G.2.

Cost of screening ability. First, it follows from the labor productivity decomposition, Equation 20, that log labor productivity changes approximately one-for-one with changes in the screening ability wedge. Therefore, analytically characterizing the relationship between heterogeneity in the screening ability wedge and log labor productivity simultaneously characterizes the relationship between heterogeneity in screening ability and the level of log labor productivity.

Proposition 6.1 (The cost of dispersion)**.** *For any region s and period t, $\chi^g_{s,t} \leq \bar{\chi}_{s,t}$, with equality if and only if χ_{it} is degenerate across i. If in addition $\log \chi_{i,t} \sim \mathcal{N}(\mu_{s,t}, \varsigma^2_{s,t})$ in the cross-section, then*

$$\bar{\chi}_{s,t} = \chi^g_{s,t} \exp(\varsigma^2_{s,t}/2), \qquad \bar{\chi}_{s,t} - \chi^g_{s,t} = \chi^g_{s,t}\left[\exp(\varsigma^2_{s,t}/2) - 1\right] \approx \tfrac{1}{2}\bar{\chi}_{s,t}\,\varsigma^2_{s,t}.$$

Because the exit rate is linear in χ while the law of motion governs $\log \chi$, a mean-preserving spread in log screening ability raises the arithmetic mean wedge and lowers realized output. That is, the average selection wedge $\bar{\chi}_{s,t}$ rises proportionally to the measured dispersion in banks' screening ability. This relationship is analogous to the mapping from wedge dispersion to aggregate productivity losses in the misallocation literature (Hsieh and Klenow 2009).

The gain from allocation. Second, it follows from the demonstrated relationship between banks' screening ability and size that heterogeneity in ability will drive heterogeneity in market concentration. This relationship is the basis of the second channel.

Proposition 6.2 (The gain from allocation)**.** *For all regions s and any period t with $\sum_i s_{i,s,t} = 1$,*

$$\chi^w_{s,t} - \bar{\chi}_{s,t} \;=\; \sum_{i \in \mathcal{B}_{s,t}} \left(s_{i,s,t} - \frac{1}{n_{s,t}} \right) \left(\chi_{i,t} - \bar{\chi}_{s,t} \right) \;=\; n_{s,t}\, \widehat{\mathrm{cov}}\left(s_{i,s,t},\, \chi_{i,t} \right),$$

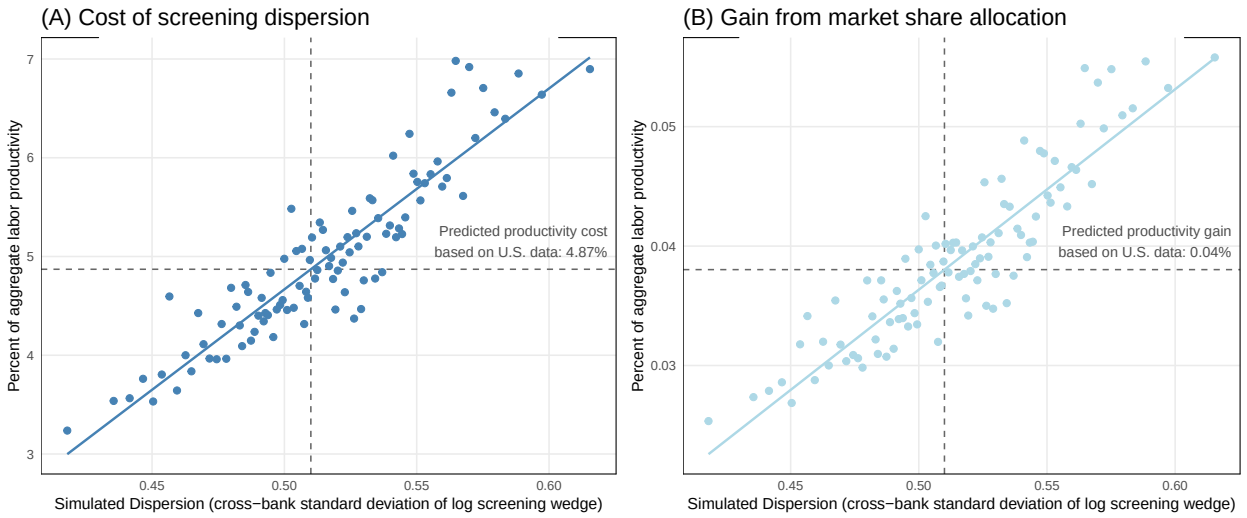
where $\widehat{\mathrm{cov}}$ denotes the unweighted cross-bank sample covariance.

The reported statistic is the covariance term normalized by the mean wedge, $\mathrm{cov}^{\mathrm{rel}}_{s,t} = 100\,(\chi^w_{s,t} - \bar{\chi}_{s,t})/\bar{\chi}_{s,t}$, which is scale free and directly interpretable as a percentage improvement in effective screening: because the screening ability wedge is decreasing in screening ability, a *negative* value of $\mathrm{cov}^{\mathrm{rel}}_{s,t}$ indicates that credit is disproportionately allocated toward better screeners, and therefore corresponds to a gain.

This proposition is the model's analogue of the Olley and Pakes (1996) productivity and market share relationship. It states that aggregate screening quality improves relative to the unweighted benchmark precisely when market share and the screening wedge covary negatively, that is, when credit is concentrated in better screeners —exactly as predicted in Section 5.

Quantify dispersion cost. To quantify the aggregate cost of heterogeneity in bank screening ability, I conduct a decomposition of measured labor productivity in the spirit of Hsieh and Klenow (2009), holding factor supplies fixed and varying only the aggregator applied to the realized distribution of bank-level wedges. Holding employment, the non-financial productivity shifter, and the loan rate at their simulated equilibrium values, I recompute the regional exit rate under each of the three aggregators applied directly to the realized, heterogeneous cross-bank distribution of screening wedges in each simulated economy's ergodic panel, so that each aggregator maps into a counterfactual level of aggregate labor productivity through realized output alone, via the selection channel. Two objects of interest follow, each expressed in log points of aggregate labor productivity: the difference between productivity under the geometric and the arithmetic mean, which captures the productivity cost of dispersion in bank screening ability, and the difference between productivity under the baseline loan-weighted mean and the arithmetic mean, which captures the productivity gain from heterogeneity in bank market shares. I compute both differences in labor productivity period-by-period across 500 simulated economies, and Figure 10 summarizes the relationship

Figure 10: Cost of screening ability heterogeneity on steady state labor productivity



Notes: This plot shows the simulated cost of screening ability dispersion and gain from market share allocation, and their predicted cost on U.S. labor productivity. The solid line is the line-of-best fit and dots reflect 100 representative points of the simulated values. The vertical dashed line in each panel marks the empirically estimated average annual screening ability dispersion, 0.51 in the U.S. from 1995 through 2023. The horizontal dashed line marks the implied predicted cost or gain to U.S. labor productivity. Data are based on 500 simulated economies.

between realized dispersion and each channel.[18]

Heterogeneity in banks' screening ability dampens steady-state U.S. labor productivity by 4.83 percent on average. However, this is a combination of two offsetting forces. The vertical dashed line in Figure 10 shows that the empirically estimated average of the within-year cross-sectional standard deviation of the log dispersion wedge is 0.51. The left panel then shows that the predicted cost to U.S. labor productivity is 4.87 percent relative to a baseline economy with no dispersion in screening ability. In contrast, the right panel show the gain from allocation boosts U.S. labor productivity by a limited 0.04 percent relative to a baseline economy with heterogeneity in screening ability but uniform market share. A simple back-of-the envelope accounting exercise similar to the one applied to the level of screening ability shows that the 4.83 percent decrease in labor productivity costs 5.38 thousand dollars per worker.

This estimate falls within the range of TFP losses the missalocation literature attributes to information and financial frictions. For example, David and Venkateswaran (2019) find that distortions correlated with firm-level productivity —which they argue includes sources such as financial frictions— generate a 3 percent loss in aggregate U.S. TFP. The same work also finds adjustment costs generate a 2 percent loss, uncertainty a 1 percent loss, and distortions uncorrelated with pro-

[18]Because the level cost above and the dispersion cost below are both constructed from the same aggregator, χ^g, they are additively separable and together capture the total cost of departing from perfect, homogeneous screening.

ductivity a 13 percent loss. Similarly, David et al. (2016) estimate information frictions to account for 4 percent of productivity losses in the United States. My 4.83 percent estimate is comparable in magnitude to the larger individual components of these decompositions, while arising from a single identified micro-foundation, cross-bank screening heterogeneity, rather than an unattributed residual wedge. Such a focus on more tightly identified micro-sources of distortions was the take-away in a recent synthesis of the macro-misallocation literature, Ghatak and Mookherjee (2025).

Robustness. The strength of the relationship between bank lending and real activity depends on the extent to which production is financed by bank lending. Appendix G.3 evaluates the sensitivity of the estimated productivity cost of bank dynamics to varying working capital financing intensities.

6.4 Implications for non-financial firm dynamics and competition policy

These results carry implications beyond the mechanisms already quantified above. Because the selection channel operates independently of loan pricing, reduced-form frameworks organized around a single factor-pricing channel (for example Gertler and Kiyotaki 2010 or Christiano et al. 2014) omit the key channel through which bank dynamics shape aggregate outcomes. Two further implications follow: one for how financial shocks reshape the firm distribution, and one for banking-sector competition policy.

Fewer and weaker. Ates and Saffie (2021) show that a tightening in financing conditions leads banks to raise underwriting standards at entry, producing a cohort of *fewer but more productive firms*. Financing frictions operate on the extensive margin through selection at birth, and the resulting cleansing raises average productivity. However, the bank dynamics characterized in this work show that when the financial shock is one that dampens banks' screening ability, then a sullying effect occurs. When screening ability decreases before banks realize it, the same number of firms will be funded, all else equal. However, a larger share of those funded firms will subsequently fail. The economy is therefore populated, in the near term, by *the same number, but weaker firms*. As screening ability remains impaired while shrinking balance sheets limit banks' ability to extend credit to replacements, the number of funded firms eventually falls, so the economy transitions from weaker to *fewer and weaker firms*.

Competition policy. The fact that screening ability is the primary driving force behind banks' growth has stark implications for banking-sector competition policy. Because market share flows to the banks best able to screen and price risk, restricting the growth or geographic expansion of large banks also restricts the growth of the banks best equipped to allocate credit productively.

Credit demand displaced from a constrained high-ability bank flows instead to banks with weaker screening ability, and, all else equal, more of that credit will end in default than if the constrained bank had been permitted to take the market share. This effect is not merely theoretical: Jayaratne and Strahan (1996); Bau and Matray (2023) document that removing geographic restrictions on bank and capital-market expansion accelerates growth by allowing credit to reallocate towards more competitive (i.e. more productive) banks. Therefore, size- and geography-based restrictions on bank growth (for example, the national deposit cap introduced by Riegle-Neal, merger-review standards applied by the Federal Reserve and Department of Justice) should be evaluated with an eye toward exactly which banks such restrictions are constraining. However, I leave a formal normative assessment of current policies to future work.

Two qualifications are worth noting. First, this argument implicitly holds other banks' behavior fixed: it does not follow that credit displaced from a single constrained bank is necessarily captured by weaker screeners rather than by other high-ability banks expanding to fill the gap. Second, and more fundamentally, there are financial-stability concerns that arise from concentration within the financial sector, and that would need to be weighed against the cost from misallocation (see for example Acharya and Yorulmazer 2008 and Farhi and Tirole 2012)

7 Bank Dynamics and Policy

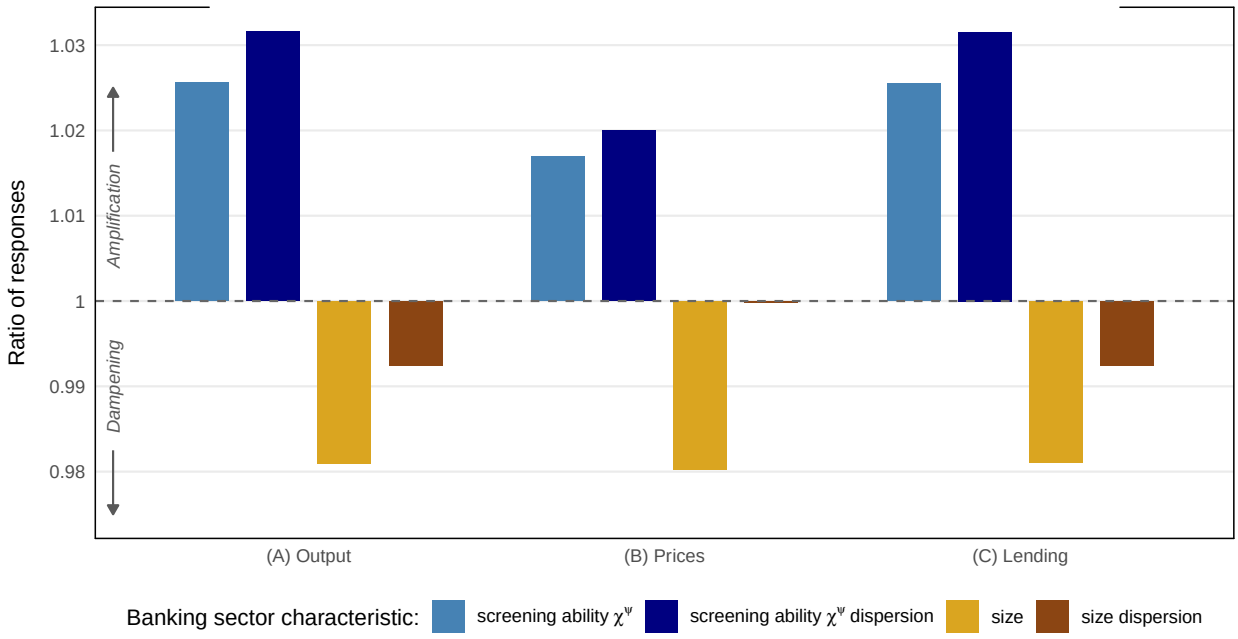
As bank dynamics shape non-financial firm dynamics through shifting the allocation of credit across (un)productive firms, they simultaneously determine the strength and shape of the monetary policy used to manage the resulting business cycle fluctuations.

7.1 The credit elasticity channel

The financial sector mediates the regional and aggregate elasticity to exogenous shocks through the *credit elasticity channel*. The worse (better) fundamentals or frictions are, the greater (smaller) the region's response to shocks, all else equal. This is because the bank-level loan rates that make up the regional cost of borrowing are more sensitive to shocks when bank fundamentals are weak or loan market frictions are high. For example, a deterioration in a bank's technology (screening ability or the capital cost premium) will steepen its marginal cost curve, amplifying movements along the curve in response to identically sized shocks pre- and post-technology shifts. Meanwhile, decreases in a bank's loan appeal will steepen its demand curve, likewise amplifying movements along the curve and the bank's response to external shocks.

The following formally characterizes this behavior, using changes in monetary policy shocks for concreteness of exposition. Taking the derivative of bank-level loan rates with respect to the

Figure 11: Heterogeneous effects of monetary policy by banking sector characteristics



Notes: This figure plots the ratio of the mean response to a monetary policy tightening shock in economies in the top quintile of a given banking-sector characteristic relative to economies in the bottom quintile, for lending, output, and prices. Sorting characteristics are the aggregate level of the screening wedge, the cross-sectional standard deviation of the screening wedge, aggregate bank net worth, and the standard deviation of bank net worth, each measured at the time of the shock. Screening wedge characteristics are oriented so that a higher value corresponds to worse average or more dispersed screening ability; size characteristics are oriented so that a higher value corresponds to larger or more dispersed net worth. A ratio above one indicates that the top-quintile economies' response exceeds the bottom-quintile economies' response. Ratios are based on 500 simulated economies.

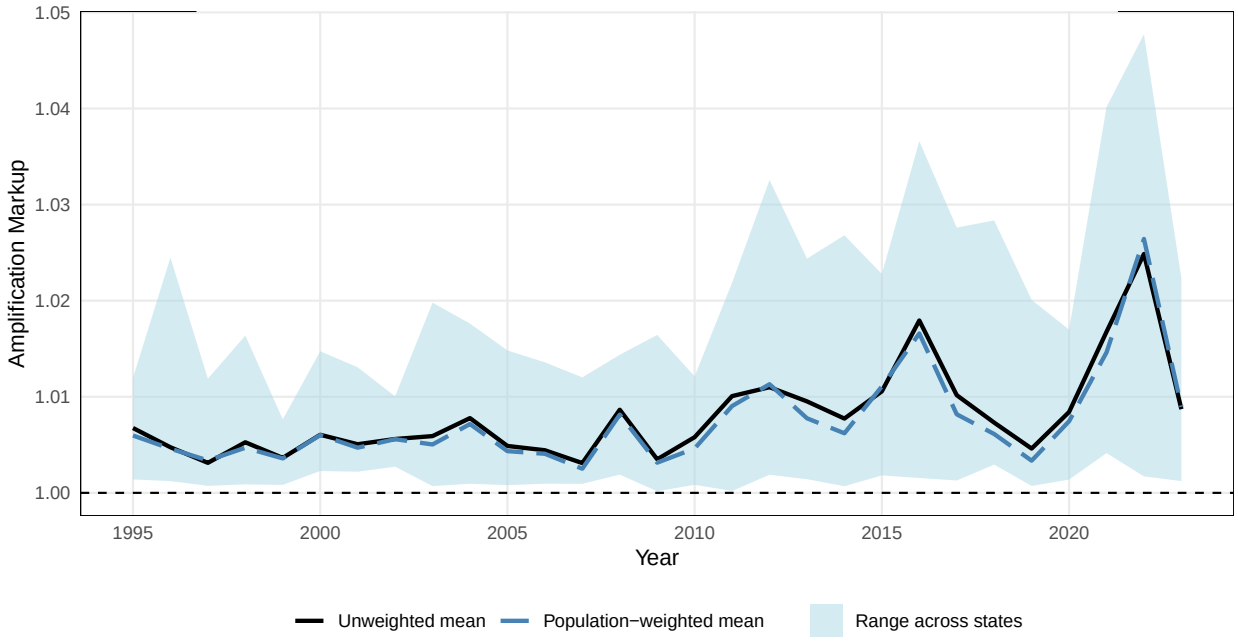
federal funds rate:

$$\frac{\partial R_{i,s,t}}{\partial I_t} = \left[1 + \mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right) \cdot \mathbb{E}_t\left(\frac{1-\psi_{i,t}}{\psi_{i,t}}\right)\right] \cdot \frac{\sigma_{s,t}}{\sigma_{s,t}-1} \frac{1}{1-s_{i,s,t}} \cdot \kappa_{i,t} \tag{24}$$

It then follows that the response of the loan rate to a change in monetary policy —systematic or otherwise— is a function of the bank's fundamentals and the region's underlying frictions, where stronger fundamentals, those that would decrease $R_{i,s,t}$, will dampen the response to a change in monetary policy. In contrast, weak fundamentals, those that would elevate $R_{i,s,t}$, will amplify the response to the change in monetary policy. Moreover, market share, a function of fundamentals, then further reinforces the amplification or mitigation as an accelerating force.

Output decreases as rising loan rates increase the effective wage bill faced by firms. Therefore, as fundamentals amplify or dampen the bank-level loan rate response to changes in monetary policy, they also amplify or dampen the response of regional output to the monetary policy.

Figure 12: Credit elasticity channel amplification over time and across regions



Notes: This figure plots the U.S. national average and regional amplification markups over time. The solid black line is the unweighted national average amplification markup, the dashed blue line marks the population-weighted state-level amplification markups, and the blue band marks the range of state-level amplification markups in a given year. Amplification markups are computed using observed bank sizes and estimated banking sector fundamentals and market frictions. A markup of one corresponds to no amplification of the policy rate through the credit elasticity channel. Data are annual from 1995 through 2023.

7.2 Quantifying the strength of the credit elasticity channel

I first quantify the strength of the credit elasticity channel in my simulated economic setting. To do so, I compare the economy's response to a tightening monetary policy shock when the economy has a relatively high value of screening ability compared to a low value. Equation 24 makes clear that as screening ability erodes, pushing up the expected level of defaults, the response of loan rates to a monetary policy shock will rise. Therefore, in the quantitative exercise I expect a larger response in lending, activity, and prices in economies with poor screening ability. Moreover, as described in Section 6.3.2, increasing dispersion in the screening ability wedge will increase the level of regional and aggregate screening ability wedges, also leading to an increase in the economy's sensitivity to rate shocks. While many pricing factors determine the overall level of amplification generated by the credit elasticity channel, Section 5 makes clear that the majority of within-bank fluctuations in the loan spread (a summary statistic for the credit elasticity channel) are driven by changes in banks' screening ability.

I additionally quantify the amplification (or dampening) due to differences in the aggregate size of the banking sector, which is the canonical macro-banking model mechanism for summarizing

amplification due to financial frictions (Gertler and Kiyotaki, 2010), as well as the dispersion in bank size within the banking industry, following the spirit of Bellifemine et al. (2026).

I simulate the economy 500 times and designate high-value (low-value) economies as those in the top (bottom) quintile of the realized sorting characteristic at the time of the shock.[19] Figure 11 shows the ratio of the mean high-value economy response relative to the mean low-value economy response. The sorting characteristics include: the aggregate level of the bank screening ability wedge χ_t^{ψ}, the cross-sectional standard deviation of the bank screening ability wedge, the net worth of the banking sector (size), and the standard deviation of bank net worth. Values are assigned such that the high value is the value that theoretically increases loan rates, all else equal. Both screening ability wedge variables are positively oriented, while the size variables are negatively oriented, such that large net worth and dispersion are counted as low-value economies.

Results. Poor screening ability amplifies the economy's response to monetary policy shocks. Figure 11 shows that economies in the top quintile of screening ability wedges had an approximately 2.55 percent larger decline in lending compared to economies in the corresponding bottom quintile. The amplified response is then transmitted through factor prices, increasing the decline in output by 2.56 percent and prices by 1.69 percent. A qualitatively similar effect is generated by differences in dispersion across realized screening ability. Lending falls by 3.15 percent more in economies with more heterogeneous banking sectors, which in turn amplifies the response of output and prices by 3.16 and 2 percent respectively.

In comparison, the net worth of the banking sector dampens the economic response to a monetary policy shock. As the size of the banking sector increases, banks are less sensitive in loan pricing, which attenuates the shock's effect on lending by 1.9 percent. The dampened response is then reflected in similarly sized reductions in output and prices. This is the standard financial accelerator outcome (Bernanke et al., 1999; Gertler and Kiyotaki, 2010; Christiano et al., 2014).

Lastly, dispersion across the size of banks similarly dampens the economic response to a monetary policy shock. However, the effect is not as large as with the aggregate size of the banking sector. Economies with greater dispersion within banking sectors experience a less than one percent smaller decline in lending compared to economies with low levels of dispersion. There is a 0.7 percent attenuation in the response of output, and no economically meaningful effect on prices. This mirrors the earlier result that the productivity gain from allocation is smaller than the cost of dispersion generated by heterogeneous bank-level screening ability.

Empirical robustness. Appendix H presents the empirical analog to these quantitative exercises and shows that the empirical transmission of monetary policy is similarly determined by the state

[19] Timing within the model is such that no conditioning states endogenously react to monetary policy on impact.

of the banking sector's aggregate screening ability.

7.3 Amplification in U.S. data and implications for policy

I lastly turn to asking how large the effect of the credit elasticity channel is in the U.S. economy. To answer this question, I calculate the amplification markup (Equation 24) using observed bank sizes and estimated region-level banking sector fundamentals and market frictions from Section 4. Figure 12 shows the average and range of regional amplification markups over time.

The aggregate amplification markup is minimal, only 1.007 on average over the entire sample. However, the degree of amplification has gone through two distinct regimes separated by the GFC.[20] Before the crisis (pre-2007), the average amplification markup was 1.005 and was stationary. After the crisis (post-2010), the average amplification markup rose to 1.011 and began positive secular growth. A Wald test rejects the null of a single, uniform, stationary regime across the full sample in favor of a level shift in the average markup across the pre- and post-crisis periods.

In comparison, dispersion in state-level amplification markups is significant and widening. The average range in state-level markups was less than 0.01 before the GFC. After the crisis, the range has steadily increased, reaching a maximum of nearly 0.045 in 2022. In other words, in 2022, some states were receiving monetary policy shocks amplified by approximately 2 percent relative to the national average and 4.5 percent relative to the lowest-markup states. This widening regional dispersion in the amplification markup is consistent with the model's own prediction that dispersion in bank characteristics, not merely their level, independently drives monetary policy amplification: if the well-documented post-crisis concentration of U.S. banking assets has widened the underlying cross-state dispersion in bank size and screening ability, the model implies exactly this pattern of rising regional divergence in transmission.

Policy consequence. Economies with a larger amplification markup receive larger monetary policy shocks because they are serviced by more policy-sensitive banks relative to states with lower amplification markups. The increase in loan rates leads to a decline in lending and is then transmitted to the real economy as larger declines in prices and output. This is true of both aggregate and regional markets. The asymmetric transmission of national monetary policy across regions therefore leads to heterogeneity in the real responses and outcomes of regional economies to a common policy shock. As a result, while monetary policy may be used to stabilize aggregate conditions, it is simultaneously raising regional income inequality through heterogeneity in the bank lending channel.

[20] Years spanning 2007–2010 are excluded from both regime averages, since these years span the acute phase of the crisis itself rather than either the pre-crisis or post-crisis steady state, and their inclusion in either regime would confound the level comparison.

8 Conclusion

Bank lending is central to both short-run fluctuations and long-run growth because it determines which borrowers receive credit and on what terms. However, aggregate outcomes depend not only on how much credit is supplied, but also on which banks supply it. This paper studies the firm dynamics of banks, identifying which banks grow large and capture market share, and in turn how this composition shapes the allocation of credit and policy. To do so, I develop a model of bank dynamics, estimate bank-level fundamentals that have not previously been measured, and embed them in a quantitative multi-region New Keynesian model. Screening ability emerges as the primary driver of bank dynamics, accounting for three-quarters of idiosyncratic loan growth within a bank and close to half of the variation across banks in the simulated economy. Because banks learn of a decline in screening ability only after realizing losses, the decline reduces net worth before any repricing takes place and triggers a self-reinforcing balance-sheet contraction. As a result, the banks that grow large and capture market share are those that are good at identifying and pricing risk, and this sorting shapes the macroeconomy through two channels. First, it governs non-financial firm dynamics: a one standard deviation negative screening shock lowers labor productivity by 2.74 percent on impact through selection, while heterogeneity in screening ability dampens long-run labor productivity by 4.83 percent relative to a benchmark without it. Second, it shapes the transmission of monetary policy: aggregate amplification is under one percent through the GFC, but regional dispersion spans 4.5 percentage points across states by 2022. Monetary policy can therefore stabilize aggregate conditions, but only at the cost of raising regional income inequality.

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A Theory

This appendix completes the description of the regional NK DSGE model, analytically characterizes non-financial outcomes in the model, and provides proofs for the financial outcomes discussed in Section 3.

A.1 Regional economies

This section describes the regional economic agents (firms, retailers, and households). These agents are standard but presented here for completeness. The Central Bank and its policies are described in the following section after a brief discussion of regional activity, prices, and aggregation.

Firms. Firms produce a homogeneous consumption good to be sold in a perfectly competitive market by combining bank-specific credit products into a composite input (taken as a stand-in for working capital constraints on a variety of factor inputs, including labor, durable capital, and inventory) and expanding this input with a decreasing returns to scale technology. Firms are potentially subject to an exogenous exit shock each period. The following formalizes the individual and representative firm's problem.

There is a continuum of individual firms. Each firm applies to a local bank for a loan to fund its labor bill before starting production. However, each firm receives an exogenous exit shock with probability $1 - q_{s,t}$. If a firm is forced to exit, then it does not produce that period and defaults on its loan. If a firm is denied a loan by the bank then it cannot produce that period.[21]

The probability of receiving an exit shock is time-varying and region specific, following the law of motion:

$$q_{s,t} = \rho^q q_{s,t-1} + \epsilon^q_{s,t}$$

where ρ^q is the regional persistence of exit rates and $\epsilon^q_{s,t}$ is the stochastic exit shock, which may be correlated across regions.

The decisions and outcomes of the individual firms can be studied as a single regionally representative firm. The representative firm aggregates bank-specific loans into a composite credit

[21] Because firms only match with and submit loan applications to a single bank each period, and there is no explicit persistence in borrower-lender relationships, there is no multiple-equilibria or "winners curse" present in this economy. These outcomes can arise in the presence of sticky borrower-lender relationships and asymmetric screening ability, as analyzed by works recently reviewed by Blickle et al. (2024).

product $L_{s,t}$ using CES aggregator technology:

$$L_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} \right)^{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}}, \qquad \sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t} = 1,$$

where $\sigma_{s,t} > 1$ is the elasticity of substitution across bank loans, and $\eta_{ist} \geq 0$ is the firm's idiosyncratic preference weight for bank i. Note that the firms preference weights may change period-by-period as the set of banks operating in region s, $\mathcal{B}_{s,t}$, changes over time. The firm-specific preference weights over bank-specific loans represent the firm's borrowing requirements across different types of loans (for example commercial and industrial loans versus real estate), industrial specialization of the bank, or geographic locations of bank branches.

The firm uses the composite loan $\bar{L}_{s,t}$ to hire labor as a composite input to produce output with standard decreasing returns to scale technology and time-varying total factor productivity:

$$Y_{s,t} = A_{s,t} \cdot M_{s,t}^{\alpha}, \quad \text{with } \alpha \in (0,1),$$

$$a_{s,t} = \rho^{a} a_{s,t-1} + \epsilon_{s,t}^{a} \qquad a_{s,t} = \log A_{s,t}$$

where $\epsilon_{s,t}^{a}$ is the firm's productivity shock.

Combing the loan aggregation and output production technology, the firm's problem is to maximize profits by choosing labor inputs M_t, taking wages as given, while minimizing financing costs by choosing loans $\{L_{ist}\}_{i \in \mathcal{B}_{s,t}}$, taking interest rates $\{R_{i,s,t}\}_{i \in \mathcal{B}_{s,t}}$ as given:

$$\max_{M_{s,t},\ \{L_{is,t}\}_{i \in \mathcal{B}_{s,t}}} A_{s,t} M_{s,t}^{\alpha} - R_{s,t} L_{s,t} \quad \text{s.t} \tag{25}$$

$$L_{s,t} = W_{s,t} M_{s,t} \qquad L_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t}^{1/\sigma_{s,t}} \cdot L_{i,s,t}^{\frac{\sigma_{s,t}-1}{\sigma_{s,t}}} \right)^{\frac{\sigma_{s,t}}{\sigma_{s,t}-1}}$$

where $L_{s,t} \geq W_{s,t} M_{s,t}$ is the condition that the Firm must pay its labor bill with working capital before production can begin.

Retailers. Each region is home to a continuum of retailers indexed by $j \in [0,1]$, matching the index over consumption varieties in the household's problem. Retailer j purchases the homogeneous

good produced by the region's firms in a perfectly competitive market and uses a linear technology,

$$\bar{Y}_{j,s,t} = Y_{j,s,t}, \qquad \int_0^1 Y_{j,s,t}\, dj = Y_{s,t},$$

to differentiate it, where $Y_{j,s,t}$ denotes retailer j's purchases of the homogeneous good and $\bar{Y}_{j,s,t}$ its supply of variety j. Retailers sell directly to households as a set of competitive monopolists, facing the inverse demand schedule for variety j derived in the household's problem, and are subject to menu cost nominal rigidities. The retailer's problem is:

$$\max_{P_{j,s,t}} \; P_{j,s,t}\bar{Y}_{j,s,t} - Y_{j,s,t} - \frac{\theta}{2}\left(\ln P_{j,s,t} - \ln P_{j,s,t-1}\right)^2 Y_{j,s,t},$$

where the second term is the cost of acquiring the homogeneous input, priced at unity because the firms' intermediate good is the numeraire within each island, and θ is the adjustment cost of changing the retailer's nominal price, which captures the degree of nominal rigidity in the region.

Households. The region's continuum of households maximizes expected lifetime utility by supplying labor to firms for wages, purchasing a continuum of consumption goods indexed by $j \in [0, 1]$, and saving in an intra-island bond market. Households in region s can be written as a representative agent, whose problem is:

$$V_H(B_{s,t-1}) = \max_{B_{s,t}, M_{s,t}, C_{s,t}} \{U(C_{s,t}, M_{s,t}) + \beta \mathbb{E}[V_H(B_{s,t})]\} \quad \text{s.t.} \tag{26}$$

$$C_{s,t} \equiv \left(\int_0^1 C_{j,s,t}^{\frac{\epsilon-1}{\epsilon}}\, dj\right)^{\frac{\epsilon}{\epsilon-1}} \qquad U_{s,t} = \left(\frac{C_{s,t}^{1-\varrho}}{1-\varrho} - \frac{M_{s,t}^{1+\phi}}{1+\phi}\right)$$

$$P_{s,t}C_{s,t} + Q_{s,t}B_{s,t} \leq B_{s,t-1} + W_{s,t}M_{s,t} + D_{s,t} \qquad 0 \leq M_{s,t} \leq 1$$

where $C_{s,t}$ is the aggregate consumption in region s at time t, created by combining varieties $C_{j,s,t}$ via CES aggregator, $P_{s,t}$ is the nominal aggregate price of the household's consumption bundle, $Q_{s,t}$ is the bond market interest rate, $B_{s,t}$ is the level of household savings in time t, $W_{s,t}$ are wages in time t, $M_{s,t}$ is the household's hours worked or its supply of labor, and $D_{s,t}$ are the dividends from firms, ϱ is the households coefficient of relative risk aversion ($1/\varrho$ is the elasticity of intertemporal substitution), ϵ is the elasticity of substitution between consumption goods, and ϕ is the curvature of the labor disutility (i.e. inverse Frisch elasticity). The firms intermediate good is the numeraire within each island.

The household's demand for total consumption, savings, and its labor supply follow from the standard conditions:

$$Q_{s,t} = \beta \mathbb{E}\left(\frac{U_{c,t+1}}{U_{c,t}}\right) \qquad M_{s,t}^{\phi} C_{s,t}^{\varrho} = W_{s,t}$$

and the intra-island bond market is in zero net supply, so $B_{s,t} = 0$ for all t. The household's demand for individual consumption varieties is standard and follows the inverse demand function:

$$C_{j,s,t} = \left(\frac{P_{j,s,t}}{P_{s,t}}\right)^{-\epsilon} C_{s,t} \qquad P_{s,t} = \left(\int_0^1 P_{j,s,t}^{1-\epsilon}\, dj\right)^{\frac{1}{1-\epsilon}}$$

where the regional price index $P_{s,t}$ is also standard.

By having firms pay wage bills with working capital loans, the household wage income is guaranteed despite some individual firms receiving an exit shock and not producing, thus not needing the labor. The structure of the economy effectively shifts the income risk from regional households to the banking sector.

A.2 Activity, prices, and policy

Regional loan demand, employment, and GDP. Solving the regional representative firm's dual problem leads to the region's demand for bank i's loans and composite region-specific loan rate:

$$L_{i,s,t} = \eta_{ist} \cdot \left(\frac{R_{s,t}}{R_{i,s,t}}\right)^{\sigma_{s,t}} \cdot L_{s,t}, \qquad R_{s,t} = \left(\sum_{i \in \mathcal{B}_{s,t}} \eta_{i,s,t} \cdot R_{i,s,t}^{1-\sigma_{s,t}}\right)^{\frac{1}{1-\sigma_{s,t}}}$$

while the primal profit maximization problem leads to the standard factor input demand:

$$M_{s,t} = \left(\frac{\alpha A_{s,t}}{R_{s,t} W_{s,t}}\right)^{\frac{1}{1-\alpha}}$$

where $M_{s,t}$ is in turn the regional labor demand and when markets clear, the region's employment level.
Total region production is then a function of the firm's technology, labor inputs, and firm exits.

$$Y_{s,t} = (1 - \lambda_{s,t}) \cdot A_{s,t} \cdot \left(\frac{\alpha A_{s,t}}{R_{s,t} W_{s,t}}\right)^{\frac{\alpha}{1-\alpha}}$$

where is the $\lambda_{s,t}$ is the firm exit rate in region s at time t. Note that regional production is increasing in regional productivity $A_{s,t}$, but decreasing in regional loan rates $R_{s,t}$, wages $W_{s,t}$, and firm exits

$\lambda_{i,t}$ which are equivalently the region's loan default rate.
Given the structure of the economy and market clearing conditions, the regional equilibrium wages, labor, loan demand, and planned production can all be characterized analytically in terms of the regional loan rate $R_{s,t}$, economic fundamentals, and model parameters:

$$W_{s,t} = \left[A_{s,t}^{\varrho} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{\alpha\varrho+1}{1-\alpha}} \right]^{\frac{1}{\frac{\alpha\varrho+1}{1-\alpha}+1}} \tag{27}$$

$$M_{s,t} = \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{1}{1-\alpha}} \left[A_{s,t}^{\varrho} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{\alpha\varrho+1}{1-\alpha}} \right]^{\frac{1}{\alpha\varrho+2-\alpha}} \tag{28}$$

$$L_{s,t} = A_{s,t}^{\frac{\varrho}{1+\varrho}} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{\alpha\varrho+1}{(1-\alpha)(1+\varrho)}+\frac{1}{1-\alpha}} \tag{29}$$

$$Y_{s,t} = A_{s,t} N_{s,t}^{\alpha} = A_{s,t} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{1}{1-\alpha}} \left[A_{s,t}^{\varrho} \left(\frac{\alpha A_{s,t}}{R_{s,t}} \right)^{\frac{\alpha\varrho+1}{1-\alpha}} \right]^{\frac{1}{\alpha\varrho+2-\alpha}} \tag{30}$$

which all reveal that as loan rates increase, economic activity, labor, and wages all decline in turn. This further highlights the transmission of monetary policy through loan rates. As the monetary authority tightens (eases) policy and increases (decreases) loan rates through the banks' cost of borrower capital funds, real economic activity and employment will slow.

Regional prices and inflation. Given two conditions (along with regional goods and financial market clearing conditions) one can analytically derive the the regional nominal price level and inflation: first $\epsilon = 1$, second, the intermediate goods firms produce with a decreasing returns to scale technology. Meeting these conditions, the the log nominal price level and inflation are:

$$\Pi_{s,t} = \ln(P_{s,t}/P_{s,t-1}) = \theta Y_{s,t} \implies \ln(P_{s,t}) = \theta Y_{s,t} + \ln P_{s,t-1} \tag{31}$$

where $Y_{s,t}$ is the level of regional intermediate output in terms of model primitives, as described in Equation 30. This implies that as the Central Bank tightens policy, inflation declines.

National activity and prices. The aggregate output, Y_t, and inflation rate Π_t are the equally-weighted average of region-level output and inflation. I abstract from differences in population, otherwise national activity and inflation would be population-weighted.

Central Bank. The Central Bank sets the nominal interest rate following an inertial Taylor

Rule:

$$I_t = I_{t-1}^{\rho_i} \left(\hat{Y}_t^{\psi_y} \hat{\Pi}_t^{\psi_\pi} \right)^{(1-\rho_i)} U_t \tag{32}$$

where $\hat{Y}_t$ is the aggregate output gap (with respect to steady state $\bar{Y}$) and $\hat{\Pi}_t$ is the inflation gap (with respect to an exogenously determined inflation target).

The Central Bank interacts with the economy primarily through lending capital funds directly to banks at the nominal policy rate. I assume the Central Bank is a deep-pocketed investor and can lend as much capital as is demanded by banks at the policy rate it sets.

A.3 Timeline and equilibrium

I conclude the description of the economic environment by summarizing the within-period timing and formally defining the economy's equilibrium. The model's equilibrium concept has two components, corresponding to two distinct margins on which banks interact. Within a period, banks compete for market share in an aggregative game: because loans enter the representative firm's problem only through the CES aggregator in Equation 1, a bank's profits depends on its rivals only through a single regional index, and each bank internalizes its own effect on that index (statements formalized in Section 3). Across time, a bank's decision to enter or exit a region requires forecasting how this same regional aggregate will evolve, which depends on the entry, exit, and idiosyncratic fundamental draws of every other bank potentially active in that market. However, because regional participation costs are per-period fixed costs and there are no sunk costs of entry, prior activity in a region confers no advantage, and the participation decision reduces to a static comparison of expected current profit against the current fixed cost. I resolve this second, dynamic problem using a standard large-market simplification, described below.

Timeline within a Period. Each period t is divided into a financing morning and a real-activity afternoon. The sequence of events is as follows:

- *Partial states are realized.* Regional productivity $A_{s,t}$, regional loan market frictions $\sigma_{s,t}$, and banks' capital cost premiums $\kappa_{i,t}$ are observed.
- *Morning: participation, funding, and lending.* Given the policy rate and the observed states, banks choose which regions to serve, pay the associated fixed costs, and regional appeal $\{\eta_{i,s,t}\}$ is realized. Firms submit loan applications. Banks form expectations $\mathbb{E}_t \psi_{i,t}$ and $\mathbb{E}_t q_{s,t}$, screen applicants, and choose regional loan quantities $\{L_{i,s,t}\}_{s \in S}$, funding the resulting portfolio from net worth and Central Bank borrowing. Loan rates $\{R_{i,s,t}\}$ are the rates at which the CES demand system clears at those quantities, and are offered to applicants on a take-it-or-leave-it basis.

- *Remaining states are realized.* Banks' screening abilities $\psi_{i,t}$ and firms' survival rates $q_{s,t}$ are realized.

- *Afternoon: production and repayment.* Financed firms hire labor and produce output. Exit shocks are realized, defaults occur, and surviving firms repay their loans. Retailers set prices subject to nominal rigidities and sell to households. Banks realize profits, net worth is updated, and exogenous bank exit and replacement occur.

This timing ensures that banks' loan pricing and market entry decisions are forward-looking, embedding expectations over borrower risk and screening ability.

Within-period equilibrium. Let $\mathcal{B}_{s,t} \subseteq \mathcal{B}_t$ denote the set of banks active in region s at time t and let $\mathcal{I}_{i,t}$ denote bank i's morning information set.

Definition A.1 (Within-period equilibrium)**.** Given the set of active bank–region pairs and expected fundamentals $\{\mathbb{E}[\psi_{i,t} \mid \mathcal{I}_{i,t}],\ \mathbb{E}[q_{s,t} \mid \mathcal{I}_{i,t}],\ \kappa_{i,t},\ \eta_{i,s,t},\ N_{i,t},\ \sigma_{s,t}\}$, a within-period equilibrium is a vector of loan quantities $\{L_{i,s,t}\}_{i\in\mathcal{B}_t,\, s\in S}$ such that each active bank's quantity in each region solves Equation 3, taking as given the quantities supplied by all other banks, and such that regional loan rates $\{R_{i,s,t}\}$ and the regional indices $\{R_{s,t}, L_{s,t}\}$ are those implied by the CES demand system in Equation 1 at those quantities.

The equilibrium is defined over the full vector of bank–region quantities rather than region by region because, under a strictly convex regulatory cost function, the marginal regulatory cost $\Gamma'(\sigma^R L_{i,t}/N_{i,t})$ depends on the bank's total portfolio $L_{i,t} = \sum_s L_{i,s,t}$, linking a multi-rgional bank's optimal supply across the regions it serves. When Γ is linear, marginal regulatory cost is $\zeta_t \sigma^R / N_{i,t}$ and is independent of loan volume. Therefore, under linearity, the problem separates into $|S|$ independent regional Cournot games linked only through the predetermined state $N_{i,t}$. The general definition is retained because the weak convexity assumption of Section 2 admits both cases. I will use a linear functional form in subsequent analysis.

Recursive equilibrium. Let the aggregate state at time t be $(\mathcal{M}_t, \Omega_t, \Upsilon_t)$,, where $\mathcal{M}_t$ denotes the cross-sectional distribution of banks over net worth, capital costs, screening ability, and regional loan appeal; Ω_t denotes the cross-sectional distribution of regions over productivity, firm exit rates, and loan market frictions; Υ_t denotes the monetary policy states, including the lagged policy rate I_{t-1}, and policy shock ν_t.

Definition A.2 (Recursive equilibrium)**.** A recursive equilibrium consists of policy functions for banks, firms, households, and retailers, a pricing system, and a transition kernel Φ such that, given $(\mathcal{M}_t, \Omega_t, \Upsilon_t)$:

- Banks choose loan supply and regional entry to maximize expected discounted net worth, taking the evolution of $(\mathcal{M}_t, \Omega_t, \Upsilon_t)$ as given by Φ.
- Firms choose loan demand and labor inputs to maximize profits, taking loan rates and wages as given.
- Households and retailers solve their respective optimization problems.
- The Central Bank sets the policy rate according to its monetary policy rule.
- All regional loan, labor, and goods markets clear. Local bond markets clear with households in zero net supply.
- The banking sector distribution and regional fundamentals evolve consistently with individual policies, realized loan outcomes, exogenous processes, and bank exit and replacement, according to the transition kernel

$$(\mathcal{M}_{t+1}, \Omega_{t+1}, , \Upsilon_{t+1}) \sim \Phi\big(\mathcal{M}_t, \Omega_t, , \Upsilon_t, \varepsilon_{t+1}\big),$$

  where ε_{t+1} collects the aggregate, regional, and idiosyncratic innovations realized at $t+1$.

Because $\mathcal{M}_t$ is a cross-sectional distribution, a fully general specification of Φ would require each bank to condition its entry and lending decisions on the entire joint distribution of every other bank's state in every region it might operate in (a dynamic game in the tradition of Ericson and Pakes (1995). This becomes intractable for an economy with many banks and many regions. I instead assume that regional loan markets are "large enough" that no single bank's entry, exit, or realized fundamentals materially alter the average fundamentals of the regional banking sector, in the spirit of Weintraub et al. (2008). Under this assumption, each bank treats the future evolution of a regional loan market as an exogenous, correctly-anticipated, stochastic process. This reduces the bank's dynamic problem to a standard single-agent optimization against an aggregate state it does not itself influence (i.e. Krusell and Smith (1998) for the dynamic entry games). In smaller regions, where an individual bank's entry could materially alter the regional aggregate, this simplification would fail and a full dynamic oligopoly equilibrium concept would be required instead. I will assume that regions are equivalent to U.S. states and satisfy the properties of "large" markets for the duration of this paper.[22]

[22] I present empirical evidence in Section 4.3 that supports this assumption.

A.4 Additional financial outcomes

I next turn to analytically characterizing additional bank-level outcomes that complement those discussed more extensively in Section 3, including bank's loan demand, the sources of banks' portfolio return risk, profits, and comparative loan rates and quantities.

A.4.1 Bank-specific demand

The demand for a bank's loans within a region is necessarily the aggregation of demand for loans from each firm operating in the region. If we take the representative firm view of the region's manufacturing sector, then the bank's loan demand follows from a CES demand for working capital.

Loan demand for bank i in region s at time t is therefore standard and given as:

$$L_{i,t} = \eta_{i,t}\left(\frac{R_t}{R_{i,t}}\right)^{\sigma_{s,t}} L_{s,t} \tag{33}$$

where $L_{i,t}$ is bank i's loans, $L_{s,t}$ are total regional loans, $R_{i,t}$ is bank i's portfolio average loan rate, $R_{s,t}$ is the effective regional loan rate, $\eta_{i,t}$ is the bank-specific appeal (i.e. the CES preference weight), and $\sigma_{s,t} \geq 1$ is the time-varying elasticity of intermediate loan substitution.[23]

A.4.2 Portfolio risk

The portfolio risk σ^R is the variance of portfolio returns rate, which are a function of the bank's borrower quality and screening ability given all other variables of Equation 8 are known at the time loan offers are made. As a result, the portfolio risk is simply the variance of sum of these two stochastic fundamentals:

$$\lambda_{i,t} = \underbrace{\frac{1-\psi_{i,t}}{\psi_{i,t}}}_{\equiv \chi^{\psi}_{i,t}} + \underbrace{\frac{1-q_{i,t}}{q_{i,t}}}_{\equiv \chi^{q}_{i,t}} \quad \Longrightarrow \quad \sigma^R = \sigma^{\psi} + \sigma^{q} + cov(\sigma^{\psi}, \sigma^{q})$$

where σ^{ψ}_t is the variance of $\chi^{\psi}_{i,t}$ and σ^{q}_t is the variance of $\chi^{q}_{i,t}$. A further transformation of variables may solve the exact mapping from the variance of ψ and q to the portfolio risk, but the log-odds terms will be used later in the quantitative estimation.

[23] Note that demand is homothetic across the size of total loans demanded, $L_{s,t}$. The CES demand structure is key in maintaining the tractability of the empirical measurement equation. Non-homothetic preferences over bank-specific loans may allow for the demand ratios for loans to vary across the total demand for loans, leading to a non-linear model of loan demand and potential identification issues measuring bank and regional fundamentals.

A.4.3 Loan returns and total profits

Realized loan return and net interest margin. The bank's realized return in region s at time t is:

$$R^*_{i,s,t} = \frac{\lambda_{i,s,t}}{1 - E\lambda_{i,s,t}} \cdot \frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}} \cdot \left(C_{i,t} + \zeta_t \frac{\sigma_{R^*}}{N_{i,t}} \right)$$

while assuming a linear functional form for the regulatory cost function, $\Gamma(\cdot) = \zeta_t \frac{\sigma_{R^*} L_{i,t}}{N_{i,t}}$. Then rewriting the risk assessment in terms of deviations from realized defaults, $E\lambda_{i,s,t} = \lambda_{i,s,t} - \epsilon_{i,s,t}$, as well as assuming a small default rate and marginal capital cost, $c_{i,t}$, implies the net profit margin:

$$\pi_{i,s,t} = r^*_{i,s,t} - c_{i,t} = \mu + \rho_{i,t} - \epsilon_{i,s,t}$$

The bank's profit margin is thus its observable net interest margin ($r^*_{i,s,t} - c_{i,t}$), which is equivalent to a linear combination of log markups $\mu = log(\frac{\sigma_{s,t}}{\sigma_{s,t} - t}) + s_{i,s,t}$, approximate residual unobservable costs $\rho_{it} = log(C_{i,t} + \frac{\zeta_t \sigma_{R^*}}{N_{it}}) - log(C_{i,t})$, and default assessment error $\epsilon_{i,s,t}$.

Total profits. The realized profits of a bank's regional loan portfolio in region s at time t are:

$$\begin{aligned} \Pi_{i,s,t} &= R^*_{i,s,t} \, L_{i,s,t} \\ &= \underbrace{(1 - \lambda_{i,s,t})}_{\text{performing loans}} \cdot \underbrace{\eta_{i,s,t} R^{\sigma_{s,t}}_{s,t} L_{s,t}}_{\substack{\text{demand adjusted} \\ \text{aggregate conditions}}} \cdot \underbrace{\left[\frac{1}{1 - E\lambda_{i,t}} \frac{\theta}{\theta - 1} \left(C_{i,t} + \zeta_t \cdot \frac{\sigma_{R^*}}{N_{i,t}} \right) \right]^{1-\theta}}_{\text{expected return per loan}} \end{aligned}$$

which is a function of the realized vs expected loan losses, the aggregate lending conditions of the local market scaled by the bank-specific demand within the market, and the loan rate charged for each loan, while continuing to assume a linear functional form for the regulatory cost function, $\Gamma(\cdot) = \zeta_t \frac{\sigma_{R^*} L_{i,t}}{N_{i,t}}$. The banks total profits are then the sum of regional loan portfolio profits less fixed operating costs:

$$\Pi_{i,t} = \sum_{s \in S} \left\{ R^*_{i,s,t} L_{i,s,t} - f_{i,s,t} \right\} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma(\sigma^R \frac{L_{i,t}}{N_{i,t}}) - f_{i,t}$$

recalling that the bank does not pay an operating cost $f_{i,s,t}$ if it does not issue loans in a region s in time t.

A.4.4 The sources of differences in bank lending volume and loan rates

Understanding the banks' regional market share Equation 10 begins with comparing the relative lending between any two banks in market s:

$$\frac{L_{i,s,t}}{L_{j,s,t}} = \frac{\eta_{i,s,t}\left(\frac{R_{s,t}}{R_{i,s,t}}\right)^{\sigma_{s,t}} L_{s,t}}{\eta_{j,s,t}\left(\frac{R_{s,t}}{R_{j,s,t}}\right)^{\sigma_{s,t}} L_{s,t}} = \frac{\eta_{i,s,t} R_{i,s,t}^{-\sigma_{s,t}}}{\eta_{j,s,t} R_{j,s,t}^{-\sigma_{s,t}}} \implies$$

$$\frac{s_{i,s,t}(1-s_{i,s,t})^{-\sigma_{s,t}}}{s_{j,s,t}(1-s_{j,s,t})^{-\sigma_{s,t}}} = \left(\frac{1/L_{s,t}}{1/L_{s,t}}\right)\frac{\eta_{j,s,t}}{\eta_{i,s,t}}\left[\frac{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\psi_{j,t}}{\psi_{j,t}}\right)\right]\cdot\left(C_{j,t}+\zeta_t\frac{1}{N_{j,t}}\right)}{\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\psi_{i,t}}{\psi_{i,t}}\right)\right]\cdot\left(C_{i,t}+\zeta_t\frac{1}{N_{i,t}}\right)}\right]^{\sigma_{s,t}}$$

One can further take the log of both sides to show that the difference in lending is a function of log differences in their respective pricing factors and bank specific characteristic. In other words, the difference in market share, thus relative lending, between any two banks is a function of their relative appeal, borrower quality, ability, capital costs, and regulatory cost.

A similar analysis can be applied to comparing the loan rates charged by any two banks operating in the same region. In fact, that analysis is subsumed the comparison of lending quantities above, and the comparative loan rate quotient:

$$\frac{R_{j,s,t}}{R_{i,s,t}} = \left[\frac{\frac{1}{1-s_{j,st}}\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\psi_{j,t}}{\psi_{j,t}}\right)\right]\cdot\left(C_{j,t}+\zeta_t\frac{1}{N_{j,t}}\right)}{\frac{1}{1-s_{i,st}}\left[1+\mathbb{E}_t\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\left(\frac{1-\psi_{i,t}}{\psi_{i,t}}\right)\right]\cdot\left(C_{i,t}+\zeta_t\frac{1}{N_{i,t}}\right)}\right]$$

shows a similar result. The log-difference between any two banks' loan rates in the same region will be due to the comparative difference in their screening ability, capital cost, loan appeal (through differences in market share), and their pre-existing size (measured by net worth).

A.5 Proofs

The following presents the proofs of Propositions 3.1, 3.2, and 3.3. Throughout, $\mathcal{S}$ denotes the set of regions, $\mathcal{B}_{s,t}$ denotes the set of banks operating in region s at date t, and $B_{s,t} = |\mathcal{B}_{s,t}| \geq 1$. For bank i, $L_{i,s,t}$ denotes lending in region s, $L_{s,t}$ denotes the CES composite of regional lending, $R_{s,t}$ denotes the regional loan-rate index, and $s_{i,s,t} = R_{i,s,t}L_{i,s,t}/(R_{s,t}L_{s,t})$ denotes the regional loan mar-

ket share, measured as the bank's share of regional loan billings; $L_{i,t} = \sum_{s \in \mathcal{S}} L_{i,s,t}$ and $N_{i,t}$ denote total lending and net worth. The parameter $\sigma_{s,t} > 1$ is the region-specific elasticity of substitution across intermediate loans, $\eta_{i,s,t}$ is bank appeal, $C_{i,t}$ is the capital cost, $\lambda_{i,s,t}$ is the loan default rate, and $\Gamma(X)$ is the regulatory cost of the risk-adjusted leverage ratio $X = \sigma^R L_{i,t}/N_{i,t}$, where σ^R is the portfolio return variance and, as in Section 2, $\Gamma'(X) \geq 0$ and $\Gamma''(X) \geq 0$. Screening ability enters through $\chi^{\psi}_{i,t} = (1 - \psi_{i,t})/\psi_{i,t}$. For any bank-level variable x, $\tilde{x}_{s,t} = B^{-1}_{s,t} \sum_{j \in \mathcal{B}_{s,t}} \log x_{j,s,t}$ denotes the cross-sectional mean of $\log x$ across banks operating in region s at date t, that is, the logarithm of the geometric mean.

Proposition 3.1 (Loan rate)**.** *Bank i's equilibrium loan rate in region s at time t is*

$$R_{i,s,t} = \underbrace{\frac{1}{1 - \mathbb{E}\lambda_{i,s,t}}}_{\text{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}}}_{\text{markup}} \cdot \underbrace{\left(C_{i,t} + \frac{\sigma^R}{N_{i,t}} \Gamma'\left(\sigma^R \frac{L_{i,t}}{N_{i,t}} \right) \right)}_{\text{marginal cost}}. \tag{7}$$

Expanding the expected default rate, regulatory cost, and capital cost terms gives the equivalent expression

$$R_{i,s,t} = \underbrace{\left[1 + \mathbb{E}_t\left(\frac{1 - q_{s,t}}{q_{s,t}} \right) \cdot \mathbb{E}_t\left(\frac{1 - \psi_{i,t}}{\psi_{i,t}} \right) \right]}_{\text{risk premium}} \cdot \underbrace{\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \frac{1}{1 - s_{i,s,t}}}_{\text{markup}} \cdot \underbrace{\left(I_t \kappa_{i,t} + \zeta_t \frac{\sigma^R}{N_{i,t}} \right)}_{\text{marginal cost}}, \tag{8}$$

where the regulatory cost function is assumed to be linear in a time-varying cost coefficient ζ_t, so that $\Gamma(\cdot) = \zeta_t\left(\sigma^R L_{i,t}/N_{i,t}\right)$.

The proof relies on two features of the environment that the main text uses implicitly; I state them explicitly here. First, beliefs are certainty equivalent. In the morning, bank i's beliefs over the current-period states $(\psi_{i,t}, q_{s,t})$ place unit mass on its conditional forecasts $\mathbb{E}[\psi_{i,t} \mid \mathcal{I}_{i,t}]$ and $\mathbb{E}[q_{s,t} \mid \mathcal{I}_{i,t}]$, and $\mathbb{E}_t$ denotes expectations under these beliefs. Hence $\mathbb{E}_t\, g(\psi_{i,t}, q_{s,t}) = g(\mathbb{E}_t\psi_{i,t}, \mathbb{E}_t q_{s,t})$ for any function g. This is the convention under which the expected default rate is defined in Section 3. Second, conduct is Cournot with unit-elastic regional credit expenditure. Bank i chooses its regional lending taking rivals' quantities as given, internalizes its effect on the regional indices $(R_{s,t}, L_{s,t})$, and takes regional output $Y_{s,t}$ as given. As Step 3 shows, the last condition fixes regional credit expenditure $R_{s,t}L_{s,t}$ from the bank's perspective.

Proof. The proof proceeds in five steps. First, I write the bank's expected profit under certainty-equivalent beliefs. Second, I show that the bank's dynamic problem reduces to maximizing current expected profit. Third, I derive the marginal revenue from lending implied by the CES demand system under Cournot competition. Fourth, I combine the first-order condition with marginal

revenue to obtain Equation (7). Fifth, I expand the risk premium, capital cost, and regulatory cost terms to obtain Equation (8).

Step 1: Expected profit. Realized profit in the bank's problem (Equation (3)) is

$$\Pi_{i,t} \;=\; \sum_{s\in\mathcal{S}} \left\{(1-\lambda_{i,s,t})\, R_{i,s,t} L_{i,s,t} - f(d_{i,s})\right\} - (L_{i,t} - N_{i,t})\, C_{i,t} - \Gamma\!\left(\sigma^R \tfrac{L_{i,t}}{N_{i,t}}\right),$$

where $L_{i,t} = \sum_{s\in\mathcal{S}} L_{i,s,t}$. Under certainty-equivalent beliefs, the bank's expected repayment rate in region s is the expected probability that an applicant with an h-type signal is an h-type borrower:

$$1 - \mathbb{E}_t \lambda_{i,s,t} \;=\; \frac{\mathbb{E}_t \psi_{i,t}\, \mathbb{E}_t q_{s,t}}{\mathbb{E}_t \psi_{i,t}\, \mathbb{E}_t q_{s,t} + (1-\mathbb{E}_t \psi_{i,t})(1-\mathbb{E}_t q_{s,t})},$$

where $\mathbb{E}_t \lambda_{i,s,t}$ is the expected default rate, written $\mathbb{E}\lambda_{i,s,t}$ in the proposition. The bank's expected profit is therefore

$$\mathbb{E}_t \Pi_{i,t} \;=\; \sum_{s\in\mathcal{S}} \left\{(1-\mathbb{E}_t\lambda_{i,s,t})\, R_{i,s,t} L_{i,s,t} - f(d_{i,s})\right\} - (L_{i,t} - N_{i,t})\, C_{i,t} - \Gamma\!\left(\sigma^R \tfrac{L_{i,t}}{N_{i,t}}\right). \tag{34}$$

Because beliefs are degenerate, realized profit coincides with $\mathbb{E}_t \Pi_{i,t}$ from the bank's perspective at the lending stage.[24]

Step 2: Reduction to a static problem. Lending affects the bank's continuation value only through next period's net worth, $N_{i,t+1} = N_{i,t} + \Pi_{i,t}$, because the bank takes the evolution of the aggregate state $(\mathcal{M}_t, \Omega_t)$ as given. Under certainty-equivalent beliefs, the objective in Equation (3) is therefore $\Phi(\mathbb{E}_t \Pi_{i,t})$, where

$$\Phi(x) \;\equiv\; x + (1-m)\beta\, \mathbb{E}_t \mathbf{V}\big(N_{i,t} + x,\; \mathcal{M}_{t+1},\; \Omega_{t+1}\big).$$

I next show that $\mathbf{V}$ is nondecreasing in net worth. For any lending choice, profit is nondecreasing in net worth, since

$$\frac{\partial \Pi_{i,t}}{\partial N_{i,t}} \;=\; C_{i,t} + \Gamma'\!\left(\sigma^R \tfrac{L_{i,t}}{N_{i,t}}\right) \frac{\sigma^R L_{i,t}}{N_{i,t}^2} \;\geq\; 0,$$

where the inequality follows from $C_{i,t} = I_t \kappa_{i,t} > 0$ and $\Gamma' \geq 0$. Regulation enters as a cost rather than a constraint, so the set of feasible lending plans does not depend on net worth. Fix any feasible plan and consider two levels of current net worth, $N' > N$. Current profit is weakly higher under N', so next period's net worth satisfies $N'_{i,t+1} - N_{i,t+1} \geq N' - N > 0$. By induction, net worth and profit are weakly higher under N' in every subsequent period and state. Because exit occurs with

[24] Under the objective conditional distribution of $(\psi_{i,t}, q_{s,t})$ given $\mathcal{I}_{i,t}$, the expected repayment rate is $\mathbb{E}[1-\lambda_{i,s,t} \mid \mathcal{I}_{i,t}]$. This differs from $1-\mathbb{E}_t\lambda_{i,s,t}$ by a Jensen term that is second order in forecast uncertainty. Moreover, $N_{i,t+1}$ would then be random at the lending stage, so the first-order condition would acquire a covariance between the marginal value of net worth and realized repayment. Both terms vanish under certainty-equivalent beliefs.

the exogenous probability m, the value of every feasible plan is weakly higher under N'; taking the supremum over plans gives $\mathbf{V}(N', \cdot) \geq \mathbf{V}(N, \cdot)$. Hence Φ is strictly increasing. Maximizing $\Phi(\mathbb{E}_t \Pi_{i,t})$ over lending therefore has the same solution set as maximizing $\mathbb{E}_t \Pi_{i,t}$, and the bank chooses lending to maximize current expected profit period by period.

Step 3: Marginal revenue under Cournot conduct. Fix region s and suppress the subscripts (s, t) where no confusion arises. The representative firm minimizes the cost of the composite loan L_s subject to the CES aggregator, which yields the demand system

$$L_i \;=\; \eta_i \Big(\frac{R_i}{R_s}\Big)^{-\sigma} L_s, \qquad R_s \;=\; \Big(\sum_{j \in \mathcal{B}_{s,t}} \eta_j R_j^{1-\sigma}\Big)^{\frac{1}{1-\sigma}}.$$

Multiplying each demand by R_j, summing over j, and using the definition of R_s gives $\sum_j R_j L_j = R_s L_s$. Market shares $s_i = R_i L_i / (R_s L_s)$ therefore sum to one. Substituting the inverse demand $R_i = R_s (\eta_i L_s / L_i)^{1/\sigma}$ gives

$$s_i \;=\; \frac{x_i}{\sum_{j \in \mathcal{B}_{s,t}} x_j}, \qquad x_j \;\equiv\; \eta_j^{1/\sigma} L_j^{(\sigma-1)/\sigma}.$$

Next, the firm's first-order condition for labor, $\alpha A_s M_s^{\alpha-1} = R_s W_s$, multiplied by M_s and combined with the working-capital constraint $L_s = W_s M_s$, gives $R_s L_s = \alpha Y_s$. Under Cournot conduct, bank i takes $\{L_j\}_{j \neq i}$ and Y_s as given, so its regional revenue is

$$R_i L_i \;=\; \alpha Y_s \, \frac{x_i}{x_i + X_{-i}}, \qquad X_{-i} \;\equiv\; \sum_{j \neq i} x_j.$$

Differentiating with respect to L_i and using $\mathrm{d}x_i / \mathrm{d}L_i = \frac{\sigma-1}{\sigma} \, x_i / L_i$ gives

$$\frac{\partial (R_i L_i)}{\partial L_i} \;=\; \alpha Y_s \, \frac{X_{-i}}{(x_i + X_{-i})^2} \cdot \frac{\sigma - 1}{\sigma} \, \frac{x_i}{L_i} \;=\; \frac{\sigma - 1}{\sigma} \, (1 - s_i) \, \frac{\alpha Y_s \, s_i}{L_i} \;=\; \frac{\sigma - 1}{\sigma} \, (1 - s_i) \, R_i.$$

Marginal revenue therefore equals the loan rate scaled by $(\sigma - 1)(1 - s_i)/\sigma$. It is positive whenever rivals lend ($X_{-i} > 0$, so that $s_i < 1$).[25]

Step 4: The loan rate. Fixed costs do not depend on lending, and $\partial L_{i,t} / \partial L_{i,s,t} = 1$. Differentiating Equation (34) and using Step 3 therefore gives

$$\frac{\partial \, \mathbb{E}_t \Pi_{i,t}}{\partial L_{i,s,t}} \;=\; (1 - \mathbb{E}_t \lambda_{i,s,t}) \, \frac{\sigma_{s,t} - 1}{\sigma_{s,t}} \, (1 - s_{i,s,t}) \, R_{i,s,t} - C_{i,t} - \frac{\sigma^R}{N_{i,t}} \, \Gamma'\Big(\sigma^R \tfrac{L_{i,t}}{N_{i,t}}\Big).$$

[25] If the bank also internalized the response of regional output to its lending, taking A_s and W_s rather than Y_s as given, the firm's labor demand would imply $R_s = \alpha A_s W_s^{-\alpha} L_s^{\alpha-1}$. Marginal revenue would then be $R_i \big[(\sigma-1)(1-s_i)/\sigma + \alpha s_i\big]$, and the markup would be $\sigma / \big[(\sigma - 1)(1 - s_i) + \sigma \alpha s_i\big]$.

Expected profit is strictly concave in the lending vector $\{L_{i,s,t}\}_s$, for three reasons. Each regional revenue term is strictly concave in own lending, because $x \mapsto x/(x + X_{-i})$ is increasing and strictly concave for $X_{-i} > 0$ and x_i is increasing and concave in $L_{i,s,t}$ for $\sigma_{s,t} > 1$. Funding cost is linear. The regulatory cost is convex, because $\Gamma'' \geq 0$ and its argument is linear in $L_{i,t}$. Moreover, marginal revenue diverges as $L_{i,s,t} \to 0$, since $R_{i,s,t} \propto L_{i,s,t}^{-1/\sigma_{s,t}}$ as $s_{i,s,t} \to 0$. It converges to zero as $L_{i,s,t} \to \infty$, since $s_{i,s,t} \to 1$. Marginal cost, in contrast, is bounded below by $C_{i,t} > 0$. The optimum is therefore interior and unique, and the first-order condition characterizes it. Setting the derivative to zero and solving for the loan rate gives

$$R_{i,s,t} \;=\; \frac{1}{1-\mathbb{E}_t\lambda_{i,s,t}} \cdot \frac{\sigma_{s,t}}{\sigma_{s,t}-1}\,\frac{1}{1-s_{i,s,t}} \cdot \left(C_{i,t} + \frac{\sigma^R}{N_{i,t}}\,\Gamma'\!\left(\sigma^R\,\frac{L_{i,t}}{N_{i,t}}\right)\right),$$

which is Equation (7).

Step 5: The expanded expression. By Step 1, the risk premium is

$$\frac{1}{1-\mathbb{E}_t\lambda_{i,s,t}} \;=\; 1 + \frac{(1-\mathbb{E}_t\psi_{i,t})(1-\mathbb{E}_t q_{s,t})}{\mathbb{E}_t\psi_{i,t}\,\mathbb{E}_t q_{s,t}} \;=\; 1 + \mathbb{E}_t\!\left(\frac{1-q_{s,t}}{q_{s,t}}\right)\cdot\mathbb{E}_t\!\left(\frac{1-\psi_{i,t}}{\psi_{i,t}}\right),$$

where the second equality applies certainty-equivalent beliefs to each odds ratio. The capital cost is $C_{i,t} = I_t\kappa_{i,t}$. With the linear regulatory cost $\Gamma(X) = \zeta_t X$, $\Gamma'(X) = \zeta_t$, so the marginal regulatory cost of lending is $\zeta_t\sigma^R/N_{i,t}$. Substituting these three expressions into Equation (7) gives Equation (8). ■

Proposition 3.2 (Market share)**.** *A bank's log regional loan market share (i.e., relative regional lending volume) is*

$$\log(s_{i,s,t}) \;=\; \frac{\sigma_{s,t}\log\bar{s}_{s,t}}{\sigma_{s,t}\log\bar{s}_{s,t}+1}\Big[(1/\sigma_{s,t})\big(\log(\eta_{i,t}) - \tilde{\eta}_{s,t}\big) + \big(\tilde{C}_{s,t} - \log(C_{i,t})\big) + \big(\tilde{\Gamma}'_{s,t} - \log(\Gamma'_{i,t})\big) + \big(\mathbb{E}\tilde{\chi}^{\psi}_{s,t} - \mathbb{E}\log(\chi^{\psi}_{i,t})\big)\Big], \quad (10)$$

where $s_{i,s,t} = L_{i,s,t}/L_{s,t}$, $\bar{s}_{s,t} = (1/B_{s,t})\sum_{i\in\mathcal{B}_{s,t}} s_{i,s,t}$, $\tilde{x}_{s,t} = (1/B_{s,t})\sum_{i\in\mathcal{B}_{s,t}}\log x_{i,s,t}$, *and* $\chi^{\psi}_{i,t} = (1-\psi_{i,t})/\psi_{i,t}$.

Proof. The proof proceeds in three steps. First, I derive the relative lending of any two banks operating in the same region. Second, I log-linearize this relationship and aggregate it across all competitors in the region. Third, I rewrite the aggregated condition as a closed-form expression for log market share.

Step 1: Relative lending. Fix (s,t) and consider any two banks $i, j \in \mathcal{B}_{s,t}$. The bank-region-specific

demand curve implies

$$\frac{L_{i,s,t}}{L_{j,s,t}} \;=\; \frac{\eta_{i,s,t}\big(R_{s,t}/R_{i,s,t}\big)^{\sigma_{s,t}} L_{s,t}}{\eta_{j,s,t}\big(R_{s,t}/R_{j,s,t}\big)^{\sigma_{s,t}} L_{s,t}} \;=\; \frac{\eta_{i,s,t}\, R_{i,s,t}^{-\sigma_{s,t}}}{\eta_{j,s,t}\, R_{j,s,t}^{-\sigma_{s,t}}}.$$

Substituting the loan rate from Proposition 3.1 and expressing lending in terms of market shares gives

$$\frac{s_{i,s,t}(1-s_{i,s,t})^{-\sigma_{s,t}}}{s_{j,s,t}(1-s_{j,s,t})^{-\sigma_{s,t}}} \;=\; \Big(\frac{1/L_{s,t}}{1/L_{s,t}}\Big)\frac{\eta_{j,s,t}}{\eta_{i,s,t}}\left[\frac{1+\mathbb{E}_t\big(\frac{1-q_{s,t}}{q_{s,t}}\big)\,\mathbb{E}_t\big(\frac{1-\psi_{j,t}}{\psi_{j,t}}\big)}{1+\mathbb{E}_t\big(\frac{1-q_{s,t}}{q_{s,t}}\big)\,\mathbb{E}_t\big(\frac{1-\psi_{i,t}}{\psi_{i,t}}\big)}\cdot\frac{C_{j,t}+\zeta_t/N_{j,t}}{C_{i,t}+\zeta_t/N_{i,t}}\right]^{\sigma_{s,t}}.$$

Taking logarithms of both sides shows that the log difference in lending between any two banks is a function of the log differences in their pricing factors and bank-specific characteristics. That is, the difference in market share, and hence in relative lending, between any two banks is a function of their relative appeal, borrower quality, ability, capital costs, and regulatory costs (i.e., risk aversion).

Step 2: Aggregation across competitors. Next, I log-linearize the relative market shares of bank i and sum them over all banks $j \in \mathcal{B}_{s,t}$:

$$\sum_{j\in\mathcal{B}_{s,t}}\Big[\log s_{i,s,t}-\sigma_{s,t}s_{i,s,t}-\log s_{j,s,t}+\sigma_{s,t}s_{j,s,t}\Big] \;=\; -B_{s,t}\,\sigma_{s,t}\,s_{i,s,t}+\frac{\log s_{i,s,t}}{\sum_{j\in\mathcal{B}_{s,t}}\log s_{j,s,t}}+\sigma_{s,t}.$$

Using the tilde operator defined above, so that $\tilde{s}_{s,t} = B_{s,t}^{-1}\sum_{j\in\mathcal{B}_{s,t}}\log s_{j,s,t}$, both sides of the sum can be written as

$$s_{i,s,t}+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \bar{s}_{s,t}}+1 \approx (1/\sigma_{s,t})\big(\log(\eta_{i,s,t})-\tilde{\eta}_{s,t}\big)+\big(\tilde{C}_{s,t}-\log(C_{i,t})\big) + \big(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t})\big)+\big(\tilde{\chi}^{\psi}_{s,t}-\log(\chi^{\psi}_{i,t})\big).$$

where $\log \bar{s}_{s,t}$ replaces the undefined $\tilde{s}_{s,t}$ as an approximating device.[26] This expresses bank i's market share in terms of its fundamentals relative to the regional averages.

Step 3: The closed form. Using the properties of logarithms and the definition of market share, I

[26] The coefficient would exactly require $\log \tilde{s}_{s,t}$. However, $\tilde{s}_{s,t}$ is already a logarithm, the log of the geometric-mean share $s^g_{s,t} = \exp(\tilde{s}_{s,t})$, and it is negative whenever $s^g_{s,t} < 1$, so $\log \tilde{s}_{s,t}$ is undefined. The coefficient therefore uses the arithmetic mean $\bar{s}_{s,t}$. By the inequality of arithmetic and geometric means, $s^g_{s,t} \le \bar{s}_{s,t}$, with equality if and only if all banks in the region have equal shares, so $\log \bar{s}_{s,t} \ge \tilde{s}_{s,t}$.

rewrite the left-hand side as

$$\frac{L_{i,s,t}-L_{s,t}}{L_{s,t}}+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}} \;\approx\; \log\left(L_{i,s,t}/L_{s,t}\right)+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}}$$
$$= \log(s_{i,s,t})+\frac{1}{\sigma_{s,t}}\frac{\log s_{i,s,t}}{\log \tilde{s}_{s,t}} \;=\; \log(s_{i,s,t})\,\frac{1+\sigma_{s,t}\log \tilde{s}_{s,t}}{\sigma_{s,t}\log \tilde{s}_{s,t}}.$$

Solving for $\log(s_{i,s,t})$ yields the closed-form expression

$$\log(s_{i,s,t}) \approx \frac{\sigma_{s,t}\log \tilde{s}_{s,t}}{\sigma_{s,t}\log \tilde{s}_{s,t}+1}\Big[(1/\sigma_{s,t})\big(\log(\eta_{i,t})-\tilde{\eta}_{s,t}\big)+\big(\tilde{C}_{s,t}-\log(C_{i,t})\big)$$
$$+\big(\tilde{\Gamma}'_{s,t}-\log(\Gamma'_{i,t})\big)+\big(\mathbb{E}\tilde{\chi}^{\psi}_{s,t}-\mathbb{E}\log(\chi^{\psi}_{i,t})\big)\Big],$$

which is the expression in Proposition 3.2. The bank's market share is therefore increasing in its appeal relative to the average appeal, borrower quality relative to the average borrower quality, and ability relative to the average ability, and decreasing in its capital and regulatory costs relative to the average capital and regulatory costs. ■

Proposition 3.3 (Regional market entry)**.** *Under the linear regulatory cost of Proposition 3.1, bank i is indifferent between entering and not entering region s at time t when*

$$\mathbb{E}_t \log \Pi_{i,s,t} \;=\; \mathbb{E}_t\Big\{\log\frac{1+(\sigma_{s,t}-1)\,s_{i,s,t}}{\sigma_{s,t}}+\log s_{i,s,t}+\log(1-\lambda_{i,s,t})+\log\alpha+y_{s,t}\Big\} \;=\; \log f_{i,s,t}, \quad (11)$$

where $\Pi_{i,s,t}$ is bank i's operating profit in region s, gross of the fixed cost $f_{i,s,t}$, and $s_{i,s,t}$ is the market share the bank obtains upon entry. The bank enters if and only if the left-hand side weakly exceeds $\log f_{i,s,t}$.

The proof maintains the certainty-equivalent beliefs and Cournot conduct stated before the proof of Proposition 3.1, together with the large-market assumption of Section 2.4, under which a potential entrant takes rivals' regional lending as given. Following Equation (3), $f_{i,s,t} = f(d_{i,s})$ denotes the per-period fixed cost of operating in region s, which is increasing in the distance $d_{i,s}$ from the bank's headquarters, and $y_{s,t} = \log Y_{s,t}$.

Proof. The proof proceeds in three steps. First, I show that the participation decision for each region reduces to a static comparison of that region's expected operating profit with its fixed cost. Second, I evaluate operating profit at the optimal lending choice characterized in Proposition 3.1. Third, I take logarithms.

Step 1: The entry criterion. Under the linear regulatory cost $\Gamma(X) = \zeta_t X$, the marginal cost of lending, $MC_{i,t} \equiv C_{i,t} + \zeta_t \sigma^R / N_{i,t}$, does not depend on lending. Expected profit in Equation (34) is

therefore additively separable across the set of regions the bank serves, $\mathcal{S}_{i,t}$:

$$\mathbb{E}_t \Pi_{i,t} \;=\; \sum_{s \in \mathcal{S}_{i,t}} \left\{ \mathbb{E}_t \Pi_{i,s,t} - f_{i,s,t} \right\} + C_{i,t} N_{i,t}, \qquad \mathbb{E}_t \Pi_{i,s,t} \;\equiv\; (1 - \mathbb{E}_t \lambda_{i,s,t})\, R_{i,s,t} L_{i,s,t} - MC_{i,t}\, L_{i,s,t}.$$

Participation costs are per-period fixed costs with no sunk component, so participation affects the continuation value only through next period's net worth. The argument in Step 2 of the proof of Proposition 3.1 therefore applies to participation as well as to lending: the bank chooses the set of regions it serves to maximize current expected profit. Because profit is separable across regions and the bank takes rivals' regional lending as given, optimal lending in region s, and hence $\mathbb{E}_t \Pi_{i,s,t}$, does not depend on participation elsewhere. The bank therefore enters region s if and only if $\mathbb{E}_t \Pi_{i,s,t} \geq f_{i,s,t}$, and it is indifferent at equality.[27]

Step 2: Operating profit at the optimum. Fix region s and suppress the subscripts (s,t) where no confusion arises. If bank i enters, its lending satisfies the first-order condition in Step 4 of the proof of Proposition 3.1, $(1 - \mathbb{E}_t \lambda_i) R_i = \mu_i\, MC_{i,t}$, where $\mu_i \equiv \sigma / [(\sigma - 1)(1 - s_i)]$ is the markup. Substituting into the definition of operating profit gives

$$\mathbb{E}_t \Pi_i \;=\; (1 - \mathbb{E}_t \lambda_i)\, R_i L_i \left(1 - \frac{1}{\mu_i}\right) \;=\; (1 - \mathbb{E}_t \lambda_i)\, R_i L_i\, \frac{1 + (\sigma - 1) s_i}{\sigma}.$$

By Step 3 of the same proof, $R_i L_i = s_i R_s L_s = s_i\, \alpha Y_s$. Therefore

$$\mathbb{E}_t \Pi_i \;=\; (1 - \mathbb{E}_t \lambda_i) \cdot \alpha Y_s \cdot s_i \cdot \frac{1 + (\sigma - 1) s_i}{\sigma}.$$

Operating profit is strictly positive, because the optimum is interior with $s_i \in (0,1)$ and $\mathbb{E}_t \lambda_i < 1$ whenever $\mathbb{E}_t q_s > 0$.

Step 3: The log condition. Taking logarithms gives

$$\log \mathbb{E}_t \Pi_{i,s,t} \;=\; \log \frac{1 + (\sigma_{s,t} - 1)\, s_{i,s,t}}{\sigma_{s,t}} + \log s_{i,s,t} + \log(1 - \mathbb{E}_t \lambda_{i,s,t}) + \log \alpha + y_{s,t}.$$

Under certainty-equivalent beliefs, $\log \mathbb{E}_t \Pi_{i,s,t} = \mathbb{E}_t \log \Pi_{i,s,t}$ and $\log(1 - \mathbb{E}_t \lambda_{i,s,t}) = \mathbb{E}_t \log(1 - \lambda_{i,s,t})$, so the right-hand side equals the expectation in Equation (11). By Step 1, the bank is indifferent between entering and not entering when this expression equals $\log f_{i,s,t}$, and it enters if and only if the expression weakly exceeds $\log f_{i,s,t}$.

■

[27] A bank-level fixed cost of operating, paid irrespective of the regions served, does not depend on participation and drops out of this comparison. In the headquarters region, $f(d_{i,s}) = 0$ and operating profit is strictly positive (Step 2), so the bank always serves that region. The condition in the proposition is therefore relevant for regions with $f_{i,s,t} > 0$.

B Data

This appendix provides additional details, discussion, and figures for the data used in the estimation of the bank and region fundamentals, market frictions, and regulatory costs.

B.1 Call Reports: bank balance sheet and income data

Bank-level balance sheet and income statement data are from the Consolidated Report of Condition and Income, also known as Call Reports. A form of Call Report data was first collected in 1933 by the Federal Deposit Insurance Corporation (FDIC) as a part of the regulatory reform responding to the Great Depression. However, there was a substantial update to the required reporting frequency, the number of variables collected, and variable definitions in 1984, resulting in a bifurcation between the pre- and post-1984 Call Report sample history. Because of the incongruity between variable definitions across the two sample periods modern Call Report data effectively begins in 1984. I use the historical Call Report collected by Correia et al. (2026).

The bank-level sample is representative of the U.S. banking sector. Figure 15 shows that the sum of deposits and loans reported in the micro-data (approximately) match the level of deposits, loans, and net worth reported by the Federal Reserve Board at the national-level. Figure 15 likewise shows that the loan-weighted charge-offs ratio reported in the micro-data matches the national charge-off ratio reported by the Federal Reserve. The sample extends from 1984 through 2024, and includes 13.9 thousand unique banks, although banks enter and exit due to failures. The maximum number of banks any given period is close to 8 thousand (see Figure 13 for further description of the number and entry/exit of banks).

Most variables of interest in my study are not directly observable from bank balance sheets and income statements. Table 5 reports the variable mnemonics used to construct these variables. Figure 15 presents net charge offs (along side deposits, loans, and net worth), Figure 16 presents loan rates, and Figure 17 shows the quarterly median interest income and costs. The difference between these two loan characteristics is the observable net interest margin, which is used for estimating the regulatory costs and markups described in Section 4.3. Note that similar to the inferred loan rates, both interest income and costs have been in secular decline since the 1980s, but unlike loan rates, have a pro-cyclical component increasing before recessions.

The panel of banks display substantial heterogeneity in both balance sheet size, income, and cost behavior. Table 1 summarizes the pooled distribution of key observable characteristics in the bank sample. The median bank holds 80 million dollars in loans and 10 million dollars in net

worth, while the right tail is pronounced —banks at the 95th percentile hold 1.24 billion dollars in loans and 190 million dollars in net worth— and the extreme upper tail skews sharply even further —at the 99th percentile, loans reach 11.78 billion dollars and net worth reaches 1.98 billion dollars, nearly an order of magnitude above the 95th percentile in both cases. On the pricing side, the median loan rate is 6.74 percent, with an interquartile range from 5.43 to 8.58 percent, whereas average capital costs are lower and more dispersed, reflecting meaningful cross-sectional variation in funding conditions. Credit risk is modest for the typical bank, the median default rate is 0.10 percent, but rises sharply in the upper tail, reaching 1.46 percent at the 95th percentile and 3.99 percent at the 99th percentile.

B.2 Summary of Deposits: spatial distribution of banking activity

County-level deposit data are from the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits (SOD). The SOD is an annual survey that collects branch-level information on deposit balances for all FDIC-insured institutions as of June 30th each year. The data provide the geographic distribution of deposits across counties and include identifiers that allow deposits to be linked to bank holding companies and Call Reports. Because the survey is conducted annually and reflects deposits at a point in time, it is best suited for measuring cross-sectional exposure and geographic deposit shares rather than within-year flows. Consistent and comprehensive coverage begins in 1994 following improvements in reporting standardization and electronic submission procedures. I will specifically use the SOD information on the branch state-level locations of each bank and the level of deposit holdings at each of those branches. These information are required to calculate regional loan market shares.

B.3 Calculating regional market shares

A bank's regional market share is calculated by the bank's loan share of all loans in a U.S. state. The algorithm for calculating a bank's regional market share is as follows: For banks that only have branch locations in one state, then the bank's entire loan portfolio is assumed to be made to borrowers in that state. For banks with branch locations across multiple states, the bank's loan portfolio is allocated across states based on the bank's deposit holdings by state, following Nguyen (2019). That is, if a bank has raised 3/4 of its deposits in Virginia and 1/4 in Maryland, then 3/4 of its loan portfolio is counted as its Virginia loan portfolio and the other 1/4 as its Maryland loan portfolio. These regional loan portfolios are then summed across all banks operating in a given region to determine the total region loan market size. Dividing the bank's regional loan portfolio by the total regional loan market size gives the bank's regional loan market.

Testing the spatial distribution of loans assumption. The key assumption motivating the the regional loan market calculation is that banks lend where they raise deposits. This is statutorily mandated by the Community Reinvestment Act of 1977. However, recent work also shows that banks sort into localities with excess deposits to lending ratios so they can use the additional deposits to fund loans in higher margin markets, see for example Oberfield et al. (2024). Therefore I test the validity of the spatial sorting assumption by calculating the within-bank correlation of a bank's county-level deposits and lending activity (either mortgage lending using HMDA data or small business lending using CRA data). Figure 18 shows a high correlation between where banks raise deposits and where they lend. The median bank has a correlation coefficient greater than 0.9 on average over the sample for either type of lending. I take this as sufficient evidence to allocate a bank's loan portfolio across states by using their relative deposit holdings, and in turn calculate its state-level loan market share.

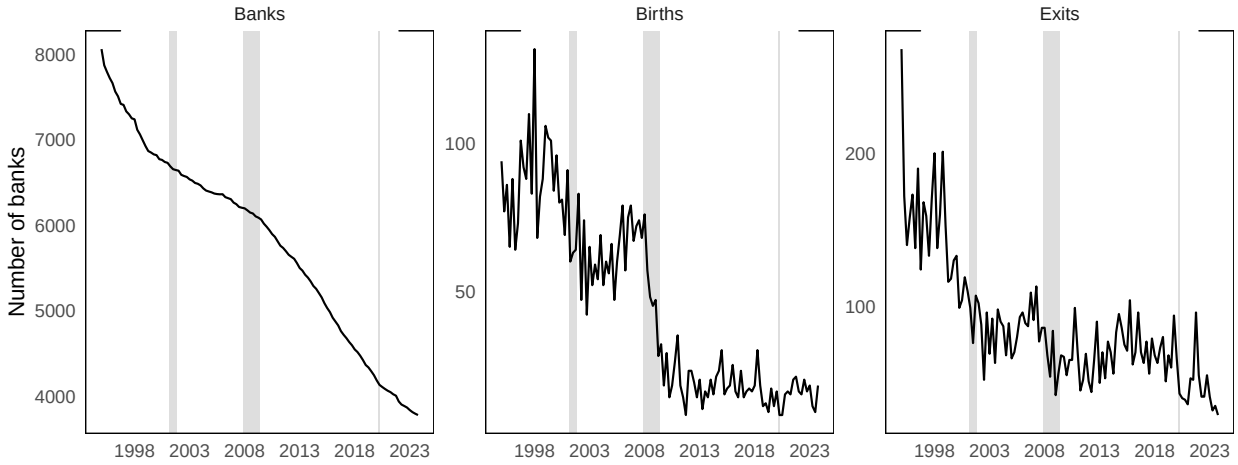
Estimated market shares. Figure 19 shows the average bank-state-level market share from 1995 through 2023, while Figure 2 shows the maximum bank-state-level market share across the sample per state.

B.4 Non-financial data

The key non-financial data used to estimate fundamentals and frictions are the state-level establishment exit rate, which acts an observable proxy for the firm-level probability of receiving an exogenous exit shock, and the real median household income, which is a stand-in for household labor income. Both are collected from the U.S. Census Bureau, the latter from the Business Dynamism Statistics dataset and the latter from the report on Income and Poverty in the Unites States.

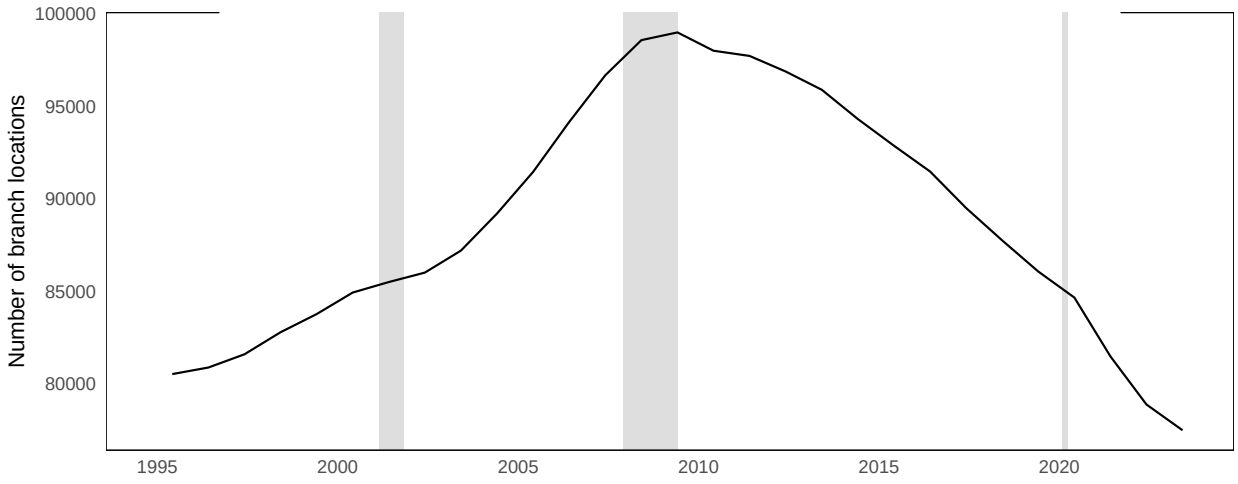
Figure 20 shows the mean, median, and interquartile range of either non-financial data series of interest. The data is a balanced panel of the 50 U.S. states. Establishment exit rates are countercyclical, with their largest spike during the global financial crisis. Real median household income has been was generally stable from 1995 through 2010 and then slowly increased through 2023.

Figure 13: Banks, entry, and exit



Notes: These plots present the number, entry, and exits of banks from the FFIEC Call Reports that operate a physical branch location. A bank is born when its unique RSSD ID is observed for the first time in the sample, while a bank is considered to have exited when its RSSD ID has been observed for the last time. Data are annual from 1995 through 20124.

Figure 14: Number of FDIC Insured Bank Branches in Sample



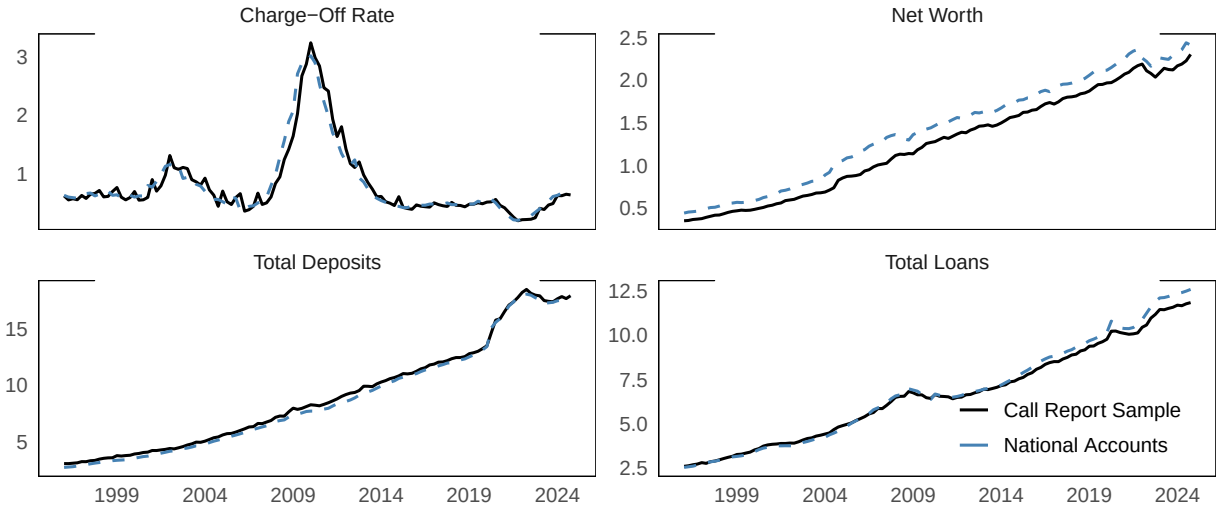
Notes: This plot presents the number of branch locations operated by banks in the estimation sample. Branch locations are identified by bank owner RSSID in the FDIC's Summary of Deposits. Data are annual from 1995 through 2023.

Table 5: Bank-level variable details

Variable	Variable	Formula	Call Report variables
Net worth	N		RCON3210
Loans	L		RCON2122, RCFD2122
Capital costs	C	total interest expense / total loans	RIAD4073, RCON2122, RCFD2122
Loan return	R^*	loan interest income / total loans	RIAD4010, RCON2122, RCFD2122
Charge-off ratio	λ	net charge offs / total loans	RIAD4635, RIAD4605
			RCON2122, RCFD2122

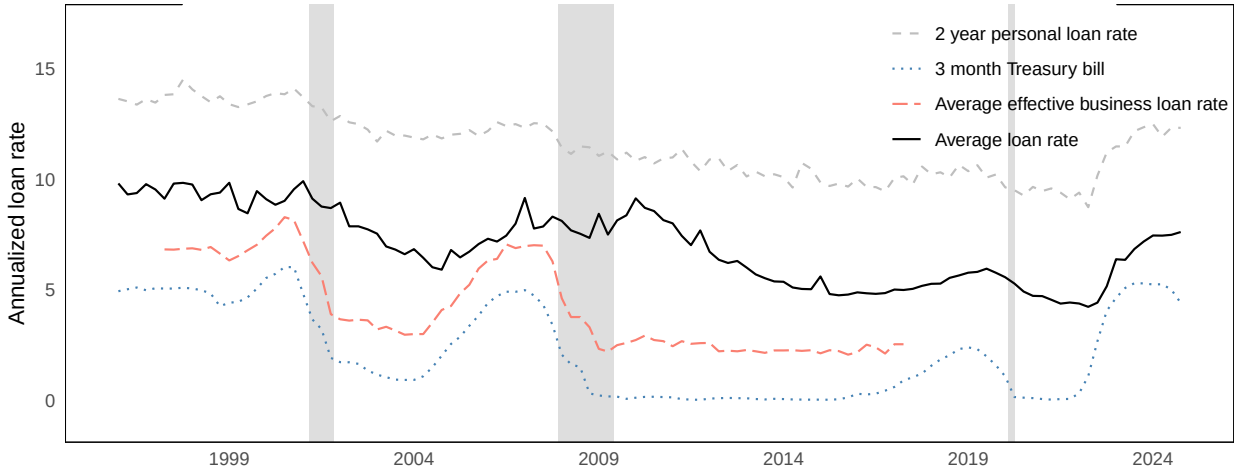
Notes: This table presents the Call Report filing variables used to create observable proxies for banks' capital cost premium, loan return, and charge off ratios. Variable codes are based on the Federal Reserve Bank of Chicago managed set of Call Report data.

Figure 15: Banking industry size and loan performance, sample vs national accounts



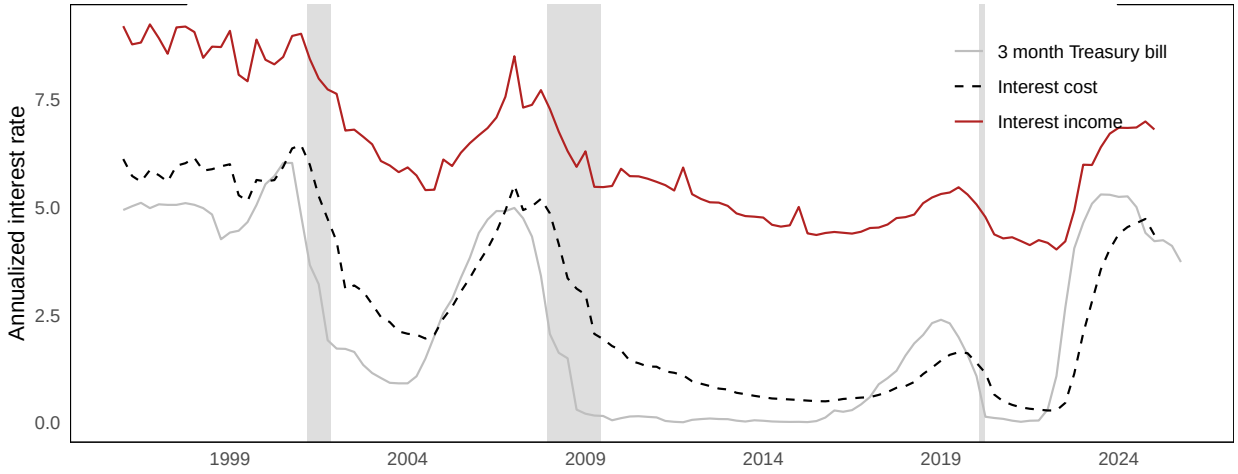
Notes: These plots compare the charge-off rate, total loans, total deposits, and banking sector net worth in the sample to those reported by National Accounts. The charge-off rate is reported in percent while net worth, total deposits, and loans are reported in trillions of nominal U.S. dollars. The sample covers the set of U.S. commercial banks that report FFIEC Call Reports and operate physical branch locations. The National Account values are from the H.8 Assets and Liabilities of Commercial Banks in the United States, reported by the Federal Reserve Board. Charge-offs are net of recoveries. Data are quarterly from 1995 through 2024.

Figure 16: Bank loan rates



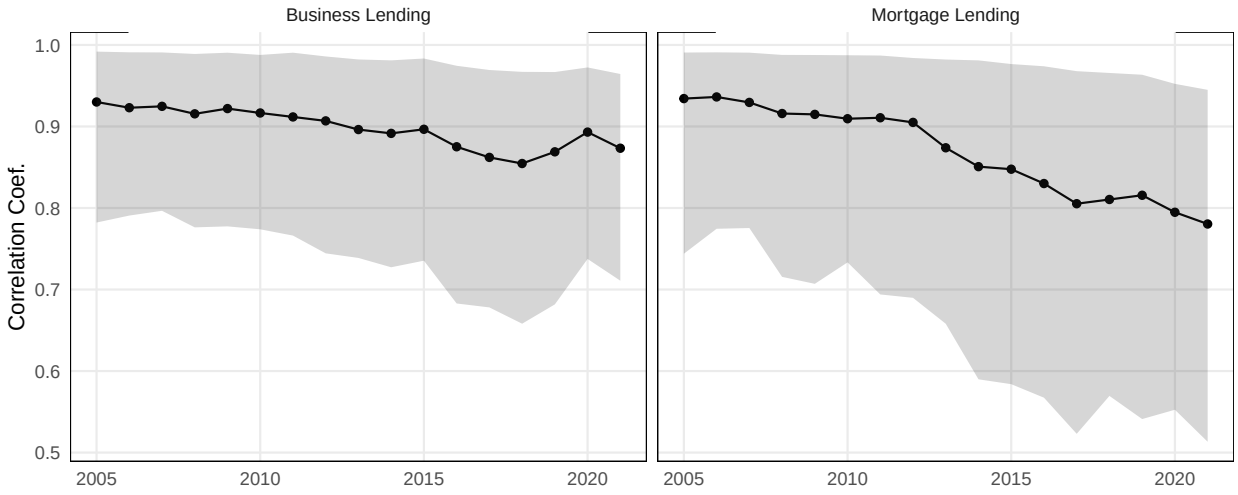
Notes: This plot compares the sample implied loan-weighted loan rate to published risk free and consumer loan rates. The bank-level sample loan rate is calculated as the reported quarterly loan income divided by total loans, scaled by the inverse one minus net charge-off ratio. The average loan rate is then the loan-weighted average of bank-level loan rates over the cross-section of banks in the given quarter. The risk-free rate is proxied by the US 3-month T-bill. The consumer loan rate is the Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan, reported by the Federal Reserve Board's G.19 Consumer Credit. The average business loan rate is the weighted-average effective loan rate for all Commercial and Industrial Loans reported in the Federal Reserve Board's E.2 Survey of Terms of Business Lending. Gray shaded regions mark NBER dated recessions. Data are annual from 1995 through 2024.

Figure 17: Loan income and costs



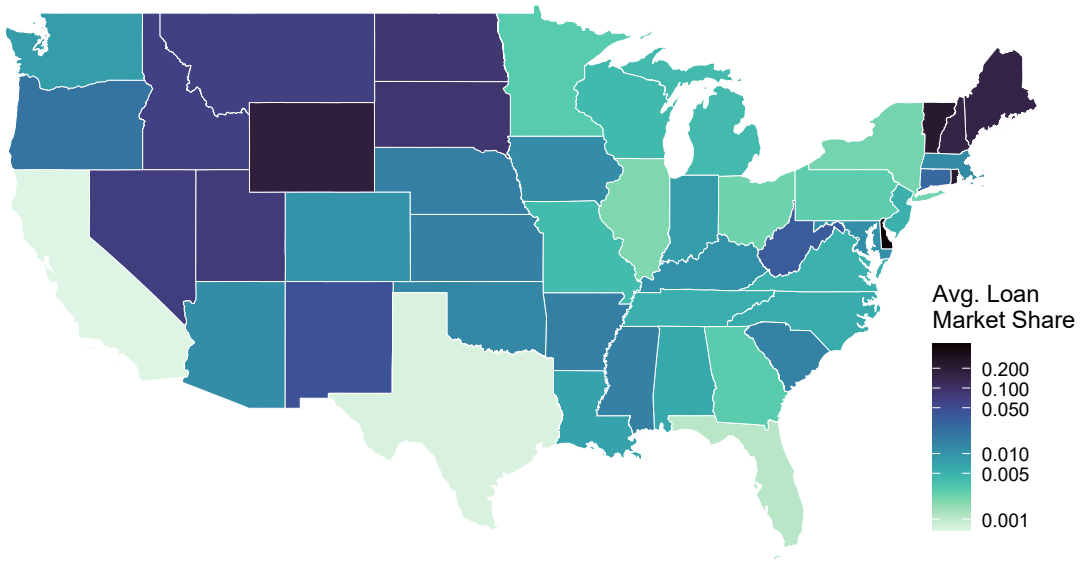
Notes: This plot presents loan interest income, costs, and the 3-month Treasury rate for comparison. Bank loan interest income and costs are loan-weighted. Gray shaded regions mark NBER dated recessions. Data are annual from 1995 through 2024.

Figure 18: Correlation between county lending and deposits shares within banks



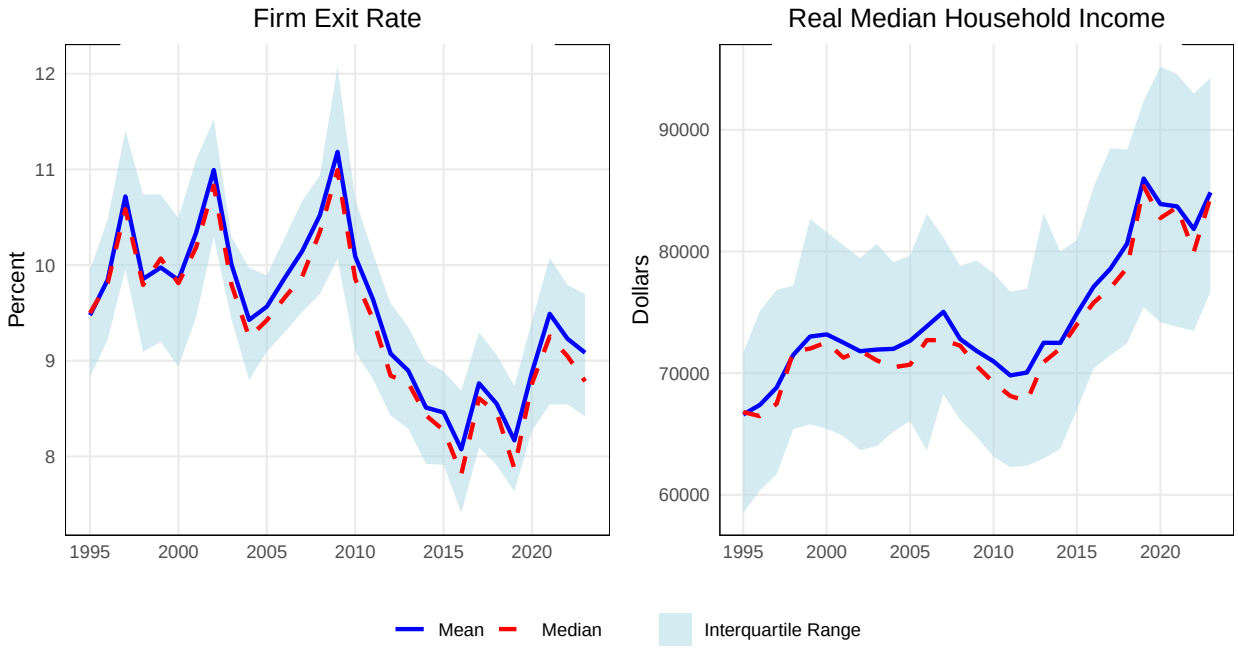
Notes: This plot shows the within-bank correlation between branch deposit holding and lending originations. The left panel shows the annual correlation between a bank's county-level small business lending (gathered from CRA filings) and branch deposit holdings (gathered from the FDIC's Summary of Deposits report). The right panel shows the annual correlation between a bank's county-level mortgage lending origination and branch deposit holdings. The gray bands mark the inter-quartile range of bank-level correlation coefficients, while the black line marks the median bank-level correlation. Data are annual from 2005 through 2021.

Figure 19: Average bank market share by state



Notes: This plot shows the average loan market share of banks by U.S. state. Data are annual from 1995 through 2023.

Figure 20: State-level non-financial data



Notes: This plot summarizes the distribution of the state-level firm exit rates and real median household income. The solid blue line is the unweighted mean, the red dashed line is the median. The blue bands mark the inter-quartile range. Data is annual from 1995 through 2023.

C Details on measuring fundamentals and market frictions

This appendix presents a detailed discussion of the empirical measurement model, estimation, identification, robustness, and accuracy.

C.1 Empirical Model

I use a state-space model to jointly estimate bank fundamentals, $\psi_{i,t}$, $\kappa_{i,t}$, region fundamentals $q_{s,t}$, $A_{s,t}$, bank-region fundamental $\eta_{i,s,t}$, along with parameters ζ_t and $\sigma_{s,t}$.

State equation. The state transition equation is a standard AR(1) process with *i.i.d.* Gaussian errors:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \qquad \epsilon_t \sim N(0, Q), \tag{35}$$

where the state vector is the stacked partitioned vector $X_t = \{\log \chi_t^\phi \ \log \kappa_t \ \log \chi_t^q \ \log \eta_t\}$ in which $\chi_t^\phi = \{\chi_{i,t}^\phi\}_{i\in\mathcal{B}}$, $\kappa_t = \{\kappa_{i,t}\}_{i\in\mathcal{B}}$, $\chi_t^q = \{\chi_{i,t}^q\}_{s\in S}$, $\eta_t = \{\eta_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; Γ is the conformable matrix of auto-regressive coefficients, and ϵ_t is the analogously stacked vector of fundamental shocks. The shocks are *i.i.d.* across and within banks and regions, thus Q is a diagonal matrix. Note that I use the log-odds transformation of both screening ability $\chi^\phi = \frac{1-\phi}{\phi}$ and borrower quality $\chi^q = \frac{1-q}{q}$ rather than the direct objects of interest $\phi_{i,t}$ and $q_{s,t}$. This transformation of the latent states is necessary to maintain the linear structure of the measurement equations.

Consistent with the model structure, estimation proceeds in two steps: an aggregate component that identifies the laws of motion for economy-wide states, and an idiosyncratic component that disciplines bank- and region-level deviations from these aggregates. The bifurcated estimation is discussed in detail in Appendix C.2.

Measurement equation. The unobservable fundamentals are then related to a set of observable bank and region characteristics: the regional firm survival odds, $E_{s,t} = \frac{1-q_{s,t}}{q_{s,t}}$, bank loan losses $\lambda_{i,t}$, bank capital costs $\log \tilde{C}_{i,t}$, and bank-region market share $s_{i,s,t}$.[28] The measurement equation relating the latent states (i.e. unobservable fundamentals) to the observable data then follows:

$$Y_t = \gamma + \underbrace{H_t X_t}_{\substack{\text{bank/region}\\ \text{fundamentals}}} + \underbrace{\tilde{H}_t X_t}_{\substack{\text{regional}\\ \text{banking sector}\\ \text{characteristics}}} + \underbrace{B_t Z_t}_{\substack{\text{exogenous}\\ \text{terms}}} + \underbrace{\nu_t}_{\substack{\text{meas.}\\ \text{error}}} \qquad \nu_t \sim N(0, R) \tag{36}$$

[28] Note that the bank's capital cost markup $C_{i,t}$ may be approximate observed using the bank's income statement. Therefore I will use $\tilde{\ }_t$ to denote the observable approximation of the latent state $C_{i,t}$ to not confuse the observable proxy and true fundamental in exposition.

where the stacked vector of observables is $Y = \{\log E_t \ \log \lambda_t \ \log \tilde{C}_t \ \log s_t\}$ in which $\lambda_t = \{\lambda_{i,t}\}_{i\in\mathcal{B}}$, $\tilde{C}_t = \{\tilde{C}_{i,t}\}_{i\in\mathcal{B}}$, $E_t = \{E_{s,t}\}_{s\in S}$, $s_t = \{s_{i,s,t}\}_{i\in\mathcal{B},s\in S}$; the matrix H_t relates the latent fundamentals to the observable outcomes; $\tilde{H}_t$ is the conformable matrix such that $\tilde{H}_t X_t$ creates a stacked vector of region-specific banking industry (geometric) average fundamentals for calculating the bank-region market share following Equation 10; the vector Z_t is the stacked vector of exogenous variables (the bank-level marginal regulatory costs and nominal federal funds rate), while ν_t captures the possible measurement error and γ the vector of constants.[29] The measurement matrices are time-varying because the parameters in the market share equation are time-varying functions of regional frictions, $\sigma_{s,t}$.

Estimating region-specific non-financial productivity wedges. After estimating the bank fundamentals, regulatory costs, and market frictions, I need to estimate one more object, the regional non-financial productivity shifter, $A_{s,t}$. The goal of this estimation is not to precisely measure local industry TFP, but rather to measure a reduced-form wedge that characterizes fluctuations in regional activity for reasons other than shocks originating in banks.

I construct the regional productivity shifter starting from the model-implied structural wage equation,

$$W_{s,t} = \left[A_{s,t}^{\varrho}\left(\frac{\alpha A_{s,t}}{R_{s,t}}\right)^{\upsilon}\right]^{(\upsilon+1)^{-1}}, \qquad \upsilon = \frac{\alpha\varrho+1}{1-\alpha},$$

where α denotes the labor share of output, governing the curvature of decreasing returns to scale, and ϱ is the elasticity of labor supply. Solving this expression for $A_{s,t}$ delivers a structural measure of local productivity that can be expressed as a function of observable real income, regional loan rates, and calibrated deep parameters. Therefore I jointly estimate the productivity shifter alongside the full set of bank- and region-level fundamentals as an ancillary step in the EM algorithm (after constructing a measure of regional loan rates with the estimated fundamentals and frictions).

Empirically, I identify $A_{s,t}$ with the residual variation in local real income after netting out the effects of local financial conditions, summarized by regional loan rates. This residual captures regional heterogeneity in real economic fundamentals not driven by fluctuations originating in the financial sector, and serves as the empirical proxy for the local productivity shifter.

C.2 State-space model details

A more detailed discussion of the formal state-space model and its aggregate versus idiosyncratic decomposition are as follows.

[29] The regional banking sector matrix $\tilde{H}_t$ additionally introduces the possibility of regional spillovers, and resembles the spatial autocorrelation term present in spatial vector-autoregression models, Beenstock and Felsenstein (2019).

Table 6: Notation for Bank Fundamentals and Regional Frictions

Unobservable Fundamentals, Frictions, and Distortions		*Observable Bank and Regional Outcomes*	
$\psi_{i,t}$	Bank i's screening ability	$E_{s,t}$	Regional firm survival rate
$\chi^{\psi}_{i,t}$	Log-odds transformation of $\psi_{i,t}$	$\lambda_{i,t}$	Bank i's loan default rate
$\kappa_{i,t}$	Bank i's capital cost premium	$s_{i,s,t}$	Bank i's market share in region s
$q_{s,t}$	Regional borrower quality	i_t	Net nominal federal funds rate
$\chi^{q}_{s,t}$	Log-odds transformation of $q_{s,t}$	$\tilde{C}i,t$	Bank i's capital cost
$\eta_{i,s,t}$	Bank-region loan appeal		
$\sigma_{s,t}$	Regional elasticity of loan substitution		
ζ_t	Time-varying regulatory cost coefficient		
$A_{s,t}$	Regional non-financial productivity shifter		

State vector and transition equation. Let $i \in \mathcal{B}$ index banks, $s \in \mathcal{S}$ index regions, and $t = 1, \ldots, T$ index time. The latent state vector is stacked as

$$X_t = \left[\log \chi^{\psi}_{1,t} \cdots \quad \log \chi^{\psi}_{B,t} \quad \log \kappa_{1,t} \cdots \quad \log \kappa_{B,t} \quad \log \chi^{q}_{1,t} \cdots \quad \log \chi^{q}_{S,t} \quad \log \eta_{1,1,t} \cdots \log \eta_{B,S,t}\right]'$$

where:

- $\chi^{\psi}_{i,t} = \frac{1-\psi_{i,t}}{\psi_{i,t}}$ is the odds transformation of screening ability.
- $\kappa_{i,t}$ is the bank capital cost premium.
- $\chi^{q}_{s,t} = \frac{1-q_{s,t}}{q_{s,t}}$ is the odds transformation of borrower quality.
- $\eta_{i,s,t}$ is bank-region loan appeal.

The law of motion is linear and Gaussian:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q).$$

The matrix Γ is comformable block diagonal, with AR(1) persistence parameters for each latent component and their constant. The shock covariance matrix Q is diagonal, implying conditional independence across banks and regions given past states.

Measurement Equation and observables. Let the stacked observable vector be

$$Y_t = \left[\log \tilde{C}_{1,t} \cdots \quad \log \tilde{C}_{B,t} \quad \log \lambda_{1,t} \cdots \quad \log \lambda_{B,t} \quad \log E_{1,t} \cdots \quad \log E_{S,t} \quad \log s_{1,1,t} \cdots \quad \log s_{B,S,t}\right]'.$$

where

- $\tilde{C}_{i,t}$ is the observable approximation of a bank's capital cost
- $\lambda_{i,t}$ is a bank's loan portfolio default rate
- $E_{1,t} = \frac{1-q_{s,t}}{q_{s,t}}$ is the observable log-odds firm exit rate
- $s_{i,s,t}$ is a bank's regional loan market share

The measurement equation is

$$Y_t = \gamma + H_t X_t + \tilde{H}_t X_t + D Z_t + \nu_t, \quad \nu_t \sim \mathcal{N}(0, R).$$

The matrix H_t maps bank- and region-specific fundamentals into observables implied by equilibrium conditions. The matrix $\tilde{H}_t$ constructs region-level geometric averages of bank fundamentals that enter the market share equation. The measurement error covariance R is diagonal. The vector Z_t includes exogenous terms, bank-level regulatory costs and the nominal federal funds rate, to the measurement equation. The vector Z_t is a stacked analogously to X_t, with bank-specific regulatory costs in order and then the nominal federal funds rate repeated for each bank.

To make the measurement matrix concrete, consider the model with only one bank and one region so that that the system would follow as:

$$\mathbf{X}_{i,s,t} = \begin{bmatrix} \log \kappa_{i,t} \\ \log \eta_{i,s,t} \\ \log \chi^{\psi}_{i,t} \\ \log \chi^{q}_{s,t} \end{bmatrix}, \quad \mathbf{Y}_{i,s,t} = \begin{bmatrix} \log \tilde{C}_{i,t} \\ \log \lambda_{i,t} \\ \log E_{s,t} \\ \log s_{i,s,t} \end{bmatrix} = \gamma_t + \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ \bar{\sigma}_{s,t} & 0 & \frac{\bar{\sigma}_{s,t}}{\sigma_{s,t}} & \bar{\rho}^{\psi}_{s,t} & 0 \end{bmatrix}}_{\equiv \mathbf{H}_t} \mathbf{X}_{i,s,t} + \tilde{H}_t X_t + B_t Z_t + \nu_t$$

$$\bar{\sigma}_{s,t} \equiv \frac{\sigma_{s,t} \log \bar{s}_{s,t}}{\sigma_{s,t} \log \bar{s}_{s,t} + 1} \qquad \rho^{\psi}_{s,t} \equiv \rho^{\psi} \bar{\sigma}_{s,t}$$

where both s and i are equal to one. The model with multiple banks and regions then expands this simplified matrix to be a blocked matrix with scalar ones becoming identity matrices and parameters being sorted accordingly. The blocks of the full measurement matrix are determined by the stacking of the latent state vector X_t.

The regional matrix $\tilde{H}_t$ collects the fundamentals of all banks operating in a region at time t and produces the unweighted mean of their log fundamentals. This results in a stacked vector

of region-specific geometric means of bank fundamentals, which are necessary for computing the bank-region-specific market share. This additionally introduces the possibility of regional spillovers, and resembles the spatial autocorrelation term present in spatial vector-autoregression and panel models.

Aggregate-Idiosyncratic Decomposition. The system admits an orthogonal decomposition that separates economy-wide common variation from bank- and region-specific deviations. This decomposition achieves computational efficiency while preserving the full covariance structure. The following decomposition has been used for these reasons to estimate modern regional NK DSGE models, including Beraja et al. (2019) and Jones et al. (2022). The following will describe the decomposition.

For any latent state $x_{i,t}$, define:

$$\bar{x}_t \equiv \frac{1}{N}\sum_{i=1}^{N} x_{i,t}, \quad \tilde{x}_{i,t} \equiv x_{i,t} - \bar{x}_t$$

The linearity of the state and measurement equations ensures this decomposition preserves the Gaussian structure:

$$\begin{aligned}
\text{Aggregate system:} \quad & \bar{X}_t = \bar{\Gamma}\bar{X}_{t-1} + \bar{\varepsilon}_t, && \bar{\varepsilon}_t \sim \mathcal{N}(0, \bar{Q}) \\
& \bar{Y}_t = \bar{H}_t\bar{X}_t + \tilde{\bar{H}}_t\bar{X}_t + \bar{B}_t\bar{z}_t + \bar{\nu}_t, && \bar{\nu}_t \sim \mathcal{N}(0, \bar{R}) \\
\text{Idiosyncratic system:} \quad & \tilde{X}_t = \tilde{\Gamma}\tilde{X}_{t-1} + \tilde{\varepsilon}_t, && \tilde{\varepsilon}_{i,t} \sim \mathcal{N}(0, \tilde{Q}) \\
& \tilde{Y}_t = \tilde{H}_t\tilde{X}_t + \tilde{\tilde{H}}_t\tilde{X}_t + \tilde{B}_t\tilde{Z}_t + \tilde{\nu}_{i,t}, && \tilde{\nu}_t \sim \mathcal{N}(0, \tilde{R})
\end{aligned}$$

where matrices are now divided by mapping idiosyncratic or aggregate laws of motion.

By construction, $\mathbb{E}[\bar{x}_t\tilde{x}_{i,t}] = 0$ for all i, t, and the likelihood function factors:

$$\mathcal{L}(Y_1, \ldots, Y_T \mid \theta) = \mathcal{L}^{agg}(\bar{Y}_1, \ldots, \bar{Y}_T \mid \theta) \times \prod_{i=1}^{N} \mathcal{L}^{idio}(\tilde{Y}_{i,1}, \ldots, \tilde{Y}_{i,T} \mid \theta)$$

This factorization permits separate estimation of the two subsystems, dramatically reducing computational burden when N is large, as is the case in my setting.

Resulting estimates. The resulting estimation yields a set of aggregate and idiosyncratic shocks and laws of motions for bank-level fundamentals (screening ability and capital cost premiums), and

region-level fundamentals (firm exit rates and non-financial productivity). However, bank-region loan appeal is only constructed at the idiosyncratic level, because its normalization keeps the aggregate level of appeal constant.

C.3 Estimating regional frictions and regulatory costs

The previously described state-space model tracks bank- and region-level fundamentals, and takes the regulatory cost and regional market frictions as parameters. I next describe how these parameters of interest are estimated.

Method. Bank-level net interest income can be partitioned into observable and unobservable components:

$$\underbrace{r^*_{i,t} - \sum_{s \in S_{i,t}} \frac{L_{i,s,t}}{L_{i,t}} \left[s_{i,s,t} + c_{i,t} - (\lambda_{i,s,t} - \mathbb{E}\lambda_{i,s,t}) \right]}_{\equiv \xi_{i,t}} = \zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right) + \log \left(\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \right)$$

where $\xi_{i,t}$ is the portfolio loan-weighted average interest income adjusted for capital costs, endogenous markups, and the net compensation for loan defaults, ζ_t is the time-varying regulatory cost coefficient from assuming a linear functional form for $\Gamma'(\mathbb{E}\sigma^{R^*} \frac{L_{i,t}}{N_{i,t}})$, and $\sigma_{s,t}$ is the regional elasticity of loan substitution. The left hand side, $\xi_{i,t}$, is observable or can be estimated with bank fundamentals measured in the E-step. Therefore, I can jointly estimate the regulatory cost function and elasticity of substitution in two steps.

I first estimate the following bank-level interest income process:

$$\xi_{i,t} = \zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right) + \log \left(\frac{\sigma_{s,t}}{\sigma_{s,t} - 1} \right) + \epsilon^{\xi}_{i,t}$$

where $\epsilon^{\xi}_{i,t}$ is the idiosyncratic measurement error. the regulatory cost coefficient ζ_t can be consistently estimated by projecting observable $\xi_{i,t}$ onto the bank's inverse log net worth, maintaining the identifying assumption that the regional elasticity of substitution and idiosyncratic measurement error is not correlated with the firms time t net worth. The resulting term $\zeta_t \left(\frac{\sigma^R}{N_{i,t}} \right)$ then gives the estimated regulatory cost of bank i at time t.

Second, I recover the elasticity of substitution $\sigma_{s,t}$ by taking the mean residual of banks only operating in region s at time t. Assuming mean zero measurement error, the average residual then represents the exogenous markup from regional frictions. This approach is equivalent to estimating

$\sigma_{s,t}$ by matching the first moment of the observed realized return distribution with the return equation implied by the banks' FOC conditions, and follows the "demand function approach" widely employed by the industrial organization literature.[30]

Identification. The regulatory cost function is identified by the covariance between the bank's net worth and its net interest income residualized against fundamentals. The intuition is that distortions in the banks interest income, after accounting for its own and borrower fundamentals, must be coming from either regulatory distortions, which realistically vary with bank size, or from market frictions. The former is therefore identified by the cross-sectional correlation between bank-level residualized income and size, while the latter is identified by the common bias in residualized income of banks operating within the same region.

Threats to identification. The threats to identification come from a failure of the conditional independence assumption in the weighted least squares estimating the regulatory cost function. The residual term from the estimation is made up of the region-specific friction term and possible bank-level measurement error. The region friction is common across all banks operating in the region, thus its covariance with the regressor is zero. The estimation of the regulatory cost function is then unbiased as long as measurement error in the residualized interest income is uncorrelated with bank size. In an ex post analysis we can observe the measurement error in fundamentals estimated by the state-space model and can confirm this assumption.

C.4 Identifying bank-level fundamentals

There is a key threat to identifying bank-level fundamentals in the data and a two-step approach to overcoming this threat.

Identification. The identification of bank and region fundamentals come from the theoretically disciplined measurement equation, which maps observable data to unobservable latent states. First, the bank-level capital cost premium is approximately observed on the banks balance sheet, thus the fundamental is actually well proxied by observable data. Second, from public U.S. Census data one can construct an observable proxy for the state-level firm survival rate, $E_{s,t}$. From the model, the log-odds survival rate is solely a function of the exogenous regional fundamental $q_{s,t}$, thus the observable proxy for the odds of survival uniquely identifies $\chi^q_{s,t}$ (after explicitly accounting

[30] See De Loecker et al. (2020) for a practical overview of the three primary strategies for estimating markups. I take the "demand function approach" in contrast to Corbae and D'Erasmo (2021) or Jamilov and Monacelli (2025) who use the "production function approach" because the former subsumes all unexplained variation in loan rates, over and above the marginal cost, and defines it as a bank's markup, leaving no room for risk adjustments to play a role.

for measurement error in the state-space model). Third, the structural equation for a bank's loan default rate, Equation 6, shows that log loan default rate is a function of both the regional exit rate, $1 - q_{s,t}$, and the bank's screening ability $\psi_{i,t}$. However, with the regional exit rate pinned down by the observable firm survival odds, the residual loan default rate is accounted for by the bank's screening rate. Fourth, with all three other fundamentals already uniquely identified, the residual regional market share is determined by, and thus measures, the bank-region loan appeal.

Threat to identification. The primary threat to identification is within-region sorting of borrower and lenders. This is because the variation used to identify and measure banks' screening ability follows from the equation for log loan losses:

$$\log \lambda_{i,s,t} = \log \frac{1 - q_{s,t}}{q_{s,t}} + \log \frac{1 - \psi_{i,t}}{\psi_{i,t}}$$

In other words, if the bank's loan portfolio is not a function of $q_{s,t}$ then the resulting estimate of $\psi_{i,t}$, conditional on $q_{s,t}$, may be biased.

The structural model assumes that banks face common borrower quality within a state-level loan market. However, this assumption can fail for two primary reasons. First, banks operate in a sub-region of the state that is subject to its own idiosyncratic exit rate shocks. For example, banks operating in rural upstate New York may operate in communities with very different local economic conditions, and exit rate fluctuations, than in New York city and the aggregate state economy. Second, banks may serve a specific industry or type of borrower that has a systematically elevated or dampened risk profile than the aggregate state-level economy. For example, banks serving the oil and gas industry in Pennsylvania may serve borrowers with a systematically amplified boom-bust cycle relative to the well-diversified aggregate state economy. If either of these cases are true, then the use of $q_{s,t}$ in the measurement equation for log loan losses will not reflect the true borrower quality the bank faces and the resulting estimate of the banks screening ability will be biased.

Solution. To remedy this problem, I will estimate a bank-specific borrower quality, $q_{i,t}$, and use it in the subsequent estimation of the state-space model. To do so, I will proceed in two steps.

Step 1: Bank-specific borrower quality. I create a proxy for the firm survival rate in a tight geographic region around a bank's branch footprints. That is, I assume banks lend to borrowers residing or operating close to the bank's physical branch locations and in turn create a proxy for the borrower quality of firms operating in that narrowly defined banking market. I calculate the narrowly defined $\bar{q}_{i,t}$ as the (deposit-weighted) average county-level firm survival rate for counties

with active bank branches. Formally, the proxy is calculated as:

$$\bar{q}_{i,t} = \sum_{c \in C_{i,t}} \left(\frac{d_{i,c,t}}{\sum_{j \in C_{i,t}} d_{i,j,t}} \right) q_{c,t}$$

where $C_{i,t}$ is the set of counties a bank is operating in at time t, $d_{i,c,t}$ is the number of deposits a bank holds in branches located in county c, and $q_{c,t}$ is the firm survival rate of county c. This first step ameliorates concerns that banks are exposed to credit markets that differ in their regional exit rate shocks compared to the broader state-wide economy.

Figure 21 shows the distribution of bank-specific borrower quality proxy follows closely the state-level distribution of log-odds borrower quality proxies shown in Figure 30. This shows that while adopting a proxy for borrower quality based on bank-specific geographic footprints, the aggregate dynamics of the borrower quality represented in the sample are unaffected by this transformation.

Step 2: Bank-specific loan losses elasticity. I allow banks to have an amplified or attenuated response to the regional exit rate fluctuations to ameliorate concerns about exposure to scaled risk profiles. To do so, I estimate a bank-specific elasticity of log loan default rate to local borrower quality. However, I cannot simply project the proxy for local borrower quality onto the bank's log loan default rate because of the potential presence of shocks that impact both the exit rate and screening ability simultaneously, that is, the threat of omitted variable bias. For example, drones have been used by both farmers and banks to monitor crops, making their development a type of positive technology shock boosting both farmers productivity and banks ability to assess risk in real time (Blickle et al. 2024).

I therefore turn to an instrumental variable approach and estimate the the following model:

$$\log \lambda_{i,t} = \alpha_i \left[\log \widehat{\frac{1-\bar{q}_{i,t}}{\bar{q}_{i,t}}} \right] + \epsilon_{i,t}, \qquad \epsilon_{i,t} \equiv \log \frac{1-\psi_{i,t}}{\psi_{i,t}} + e_{i,t}$$

$$\log \frac{1-\hat{q}_{i,t}}{\hat{q}_{i,t}} = a + b x_t + \nu_{i,t} \qquad \nu_{i,t} \perp e_{i,t} \text{ and } \nu_{i,t} \perp \psi_{i,t}$$

where the elasticity of the default rate to local borrower quality, α_i, is the statistic of interest, $\hat{q}_{i,t}$ is the observable proxy for borrower quality $q_{i,t}$ estimated in the previous step, $\widehat{\frac{1-\bar{q}_{i,t}}{\bar{q}_{i,t}}}$ is the predicted borrower quality term, x_t is the instrument for borrower quality shocks, and $\epsilon_{i,t}$ is the residual that contains the unobservable log odds screening ability and measurement error $e_{i,t}$. The instrument

must satisfy the standard relevance condition, as well as a two-faced exclusion restriction:

$$\mathbb{E}[x_t e_{i,t}] = 0, \quad \mathbb{E}[x_t \psi_{i,t}] = 0.$$

That is, the instrument must be exogenous (statistically independent) to both the measurement error $e_{i,t}$ and the bank-specific productivity $\psi_{i,t}$. Estimating the model will result in a consistent estimate of α_i and isolate the unobservable bank productivity in the residual loan default rate.

I use Känzig (2021) oil news shocks as the instrument x_t.[31] These shocks easily satisfy the relevance condition both theoretically (oil price shocks are effectively a marginal cost shock for non-financial firms that induce exits in standard Hopenhayn (1992) style models of firm dynamics), as well empirically (the first stage F-stat of the pooled model is 18.48). The oil news shocks additionally satisfy the exclusion restriction —bank's screening technology is reasonably not impacted by oil price shocks.[32] See Appendix C for a detailed discussion of IV diagnostics and results.

As a result, I create an observable proxy for the bank's log borrower quality term and residualized loan losses. The former is the fitted value while the latter is the residual of the second stage regression. With an observable proxy for loan losses that is only a function of the bank's log productivity term and measurement error, I can update the state-space model of the bank-level fundamentals such that the residualized loan losses is used as the observable in place of the total loan losses. The bank's log borrower quality term then no longer loads onto the observable, removing the potential identification problem and allowing for the consistent estimation of the latent bank screening ability.

The remainder of this section will discuss the estimation robustness to alternative bank-specific borrower quality specifications and standard IV diagnostics.

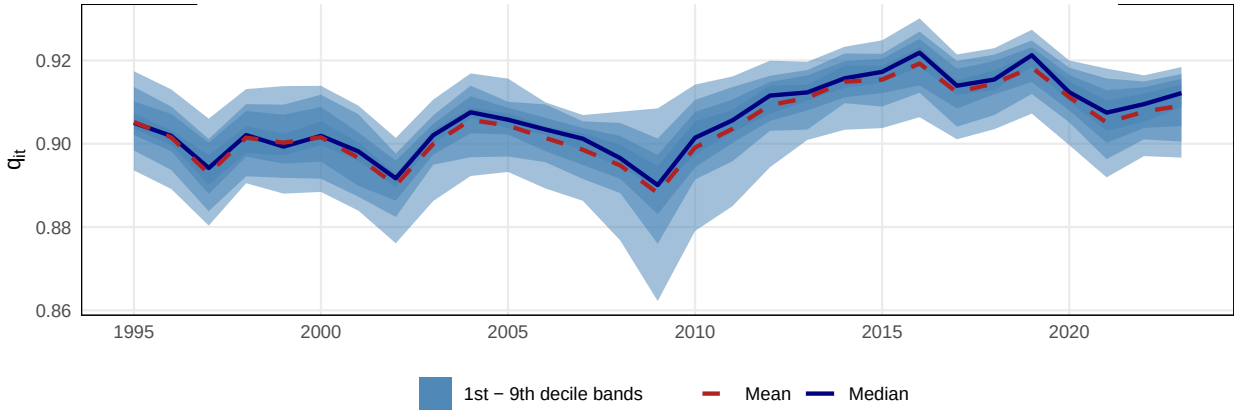
C.4.1 Robustness: activity weighted exit rates

A concern that may emerge regarding the estimated bank-level fundamentals (and the ensuing macroeconomic analysis) is that they do not represent the activity-weighted dynamics of the economy. For example, the majority of employment in the U.S. is concentrated in large firms (those with 500+ employees), while at the same, the exit rate of these firms is far lower than the exit rate of small firms. Therefore, by taking the unweighted exit rate of establishments within a region, I may be overstating the degree of exits as they matter for economic outcomes like employment and

[31] Känzig (2021) identifies oil supply news shocks by exploiting high-frequency changes in oil futures prices around OPEC announcements. By focusing on asset price fluctuations during a tight window around the macroeconomic news announcement, he isolates variation in oil prices to supply shocks.

[32] This condition holds in the context of the theoretical framework guiding the empirical estimation. However, it may fail under deviations from the full-information rational expectations paradigm. For example, if banks exhibit bounded attention, large oil price shocks may divert resources away from screening and toward processing aggregate information, violating the exclusion condition.

Figure 21: Observable proxy for local borrower quality



Notes: This plot shows the observable proxies for the bank-location specific market odds-transformation of borrower quality. The solid blue line marks the median and the dashed red line the unweighted mean. The blue banks mark the first through ninth deciles in each period. Data are annual from 1996 through 2019.

lending. I address these concerns with an additional robustness exercise: I re-estimate the bank-specific borrower quality and screening ability using (employment) activity-weighted regional exit rates. These bank-specific borrower quality are measured as:

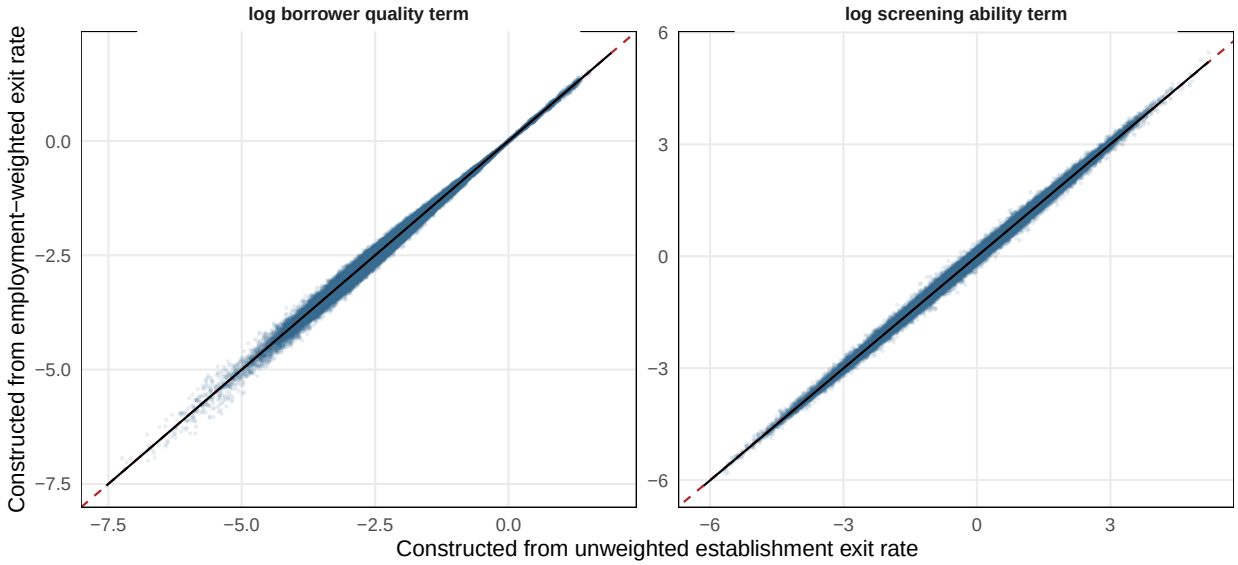
$$\bar{q}^{w}_{i,t} = \sum_{c \in C_{i,t}} \sum_{e \in E} \left(\frac{d_{i,c,t}}{\sum_{j \in C_{i,t}} d_{i,j,t}} \right) \left(\frac{M_{e,c,t}}{\sum_{e \in E} M_{c,t}} \right) q_{e,c,t}$$

where $C_{i,t}$ is the set of counties a bank is operating in at time t, $d_{i,c,t}$ is the number of deposits a bank holds in branches located in county c, $M_{e,c,t}$ is the total employment of firms in region c, at time t, in firm-size bucket bucket e, and $q_{e,c,t}$ is the firm survival rate of firms in firm-size bucket e in county c. Firms are categorized into the following 10 discrete buckets reported in the Census Business Dynamics Statistics dataset: (a) 1–4, (b) 5–9, (c) 10–19, (d) 20–99, (e) 100–499, (f) 500–999, (g) 1,000–2,499, (h) 2,500–4,999, (i) 5,000–9,999, and (j) 10,000+.

The alternative produces both bank-level borrower quality and screening ability that are quantitatively similar to their baseline, unweighted, counterparts. Figure 22 shows the bank-year-level comparison of screening ability and borrower quality constructed under unweighted vs activity-weighted establishment exit rates. The observations tightly cluster around the forty-five degree line, which is overlapped by the estimated line-of-best fit —indicating little variation between the two construction methods. A corresponding t-test of differences cannot reject the null hypothesis that their is no difference in observations between the two procedures (with a p-value of approximately 0.78 while clustering standard errors at the bank and state level).

The reason for the consistency across estimation procedures is due to the IV-powered decomposition. I first estimate banks' elasticity of losses with respect to local exit rates. I then use this

Figure 22: Comparing activity-weighted borrower quality and screening ability



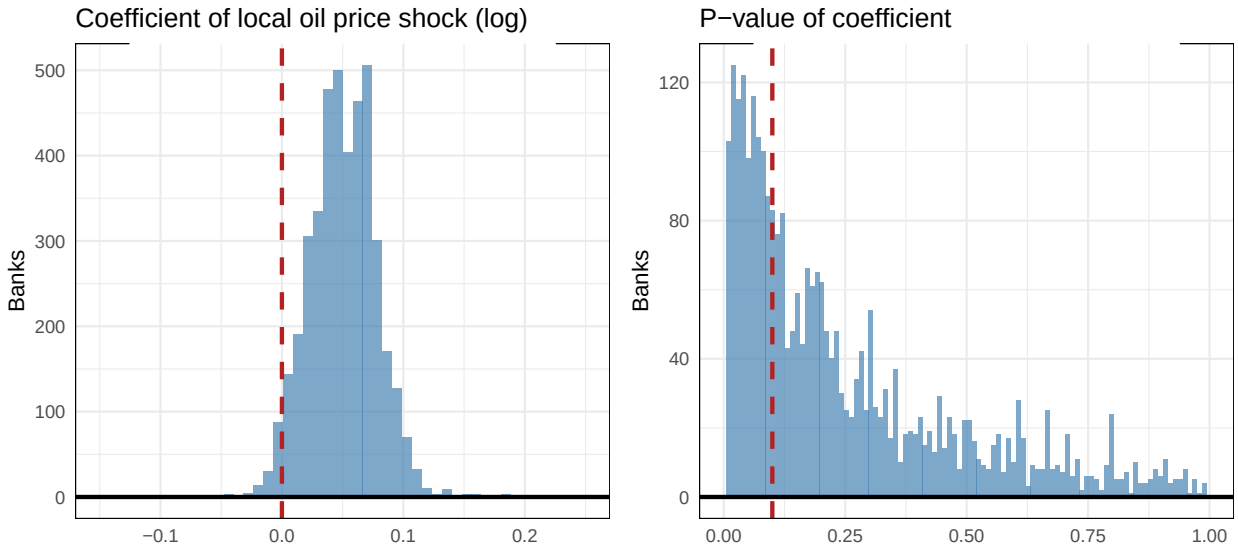
Notes: This plot compares the bank-specific borrower quality and screening ability constructed with activity-weighted and unweighted local establishment exit rates. Blue dots represent matched bank-year level observations, the red dashed line is the forty-five degree line, and the black line is the line-of-best fit. Both borrower quality and screening ability are compared in their log χ^x form, which is how they enter the structural loan rate equation. Data are at the bank-year level from 1995 through 2023.

elasticity and a bank-specific proxy for exit rates in a narrowly defined geographic region around its branch locations to estimate the level of a bank's loan defaults predicted by local borrower quality. The residual realized loan defaults captures variation due to the bank's screening ability. Therefore, if local activity weighted and equally weighted exit rates are correlated (which they are, with a correlation coefficient of 0.764), and the loan default rate is a function of the activity weighted rather than equally weighted exit rate, then the estimated elasticity of losses with respect to exits will scale the losses proportional to the correlation between the weighted and unweighted exits.

C.4.2 IV diagnostics

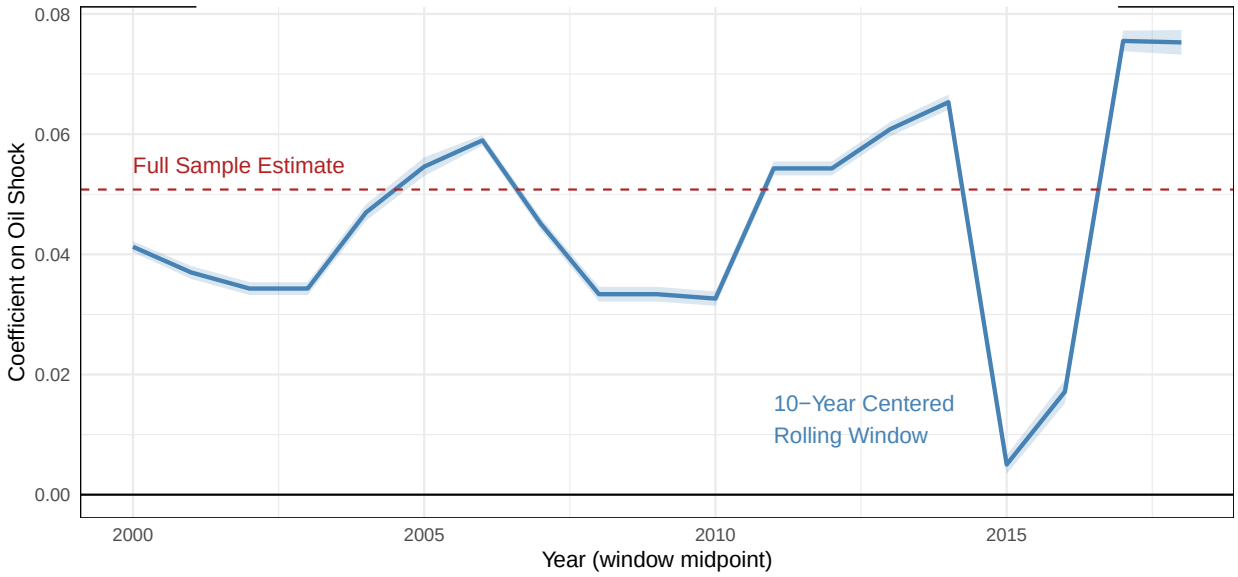
Relevance criterion. The IV must satisfy the standard relevance condition. The Känzig (2021) oil price new shocks satisfy the relevance condition both on theoretical and empirical grounds. First, oil price shocks are effectively a marginal cost shock for non-financial firms that induce exits, holding all else equal, in standard Hopenhayn (1992) style models of firm dynamics. While my model of the economy does not explicitly model these non-financial entry/exit forces and ultimately represents regional production as a representative firm, the regional firm exit shocks can be considered a reduced-form stand in for the individual firm-level entry and exit problems that are abstracted by

Figure 23: Distribution of bank-level first-stage regressions



Notes: These plots show the distribution of bank-level first-stage elasticities of borrower quality to oil price shocks and their probability values of being statistically different than zero. The left panel shows the elasticities estimated by bank-level univariate models with a constant. The red dashed line marks zero, where positive values are the theoretically expected results. The left panel shows the p-values of the elasticities and the red line marks the 10 percent confidence level.

Figure 24: Pooled first-stage regression



Notes: This plot shows the first-stage elasticity of borrower quality to oil price shocks estimated with the pooled sample or with 10-year rolling windows. Each regression includes bank fixed effects. The rolling window estimated elasticity is marked its 99-percent confidence interval. Standard errors are clustered at the bank-level.

Table 7: Testing the effects of oil price shocks on loan losses

Specification	Coefficient	Std. Error	p-value
only oil shocks	-0.371	0.168	0.0478
adding borrower quality...	-0.058	0.101	0.574
adding capital cost, net worth...	-0.058	0.101	0.575
adding loan market shares...	-0.058	0.101	0.579

Notes: This table shows the estimated coefficient, standard error, and implied p-values for regressions mapping bank-level loan losses to oil price news shocks. The first reports the univariate regression with only oil price shocks, while the following rows successively add controls. Each regression has bank- and year-level fixed effects. Standard errors are clustered at the year-level. Each regression is estimated on a sample of 73.5 thousand bank-year observations.

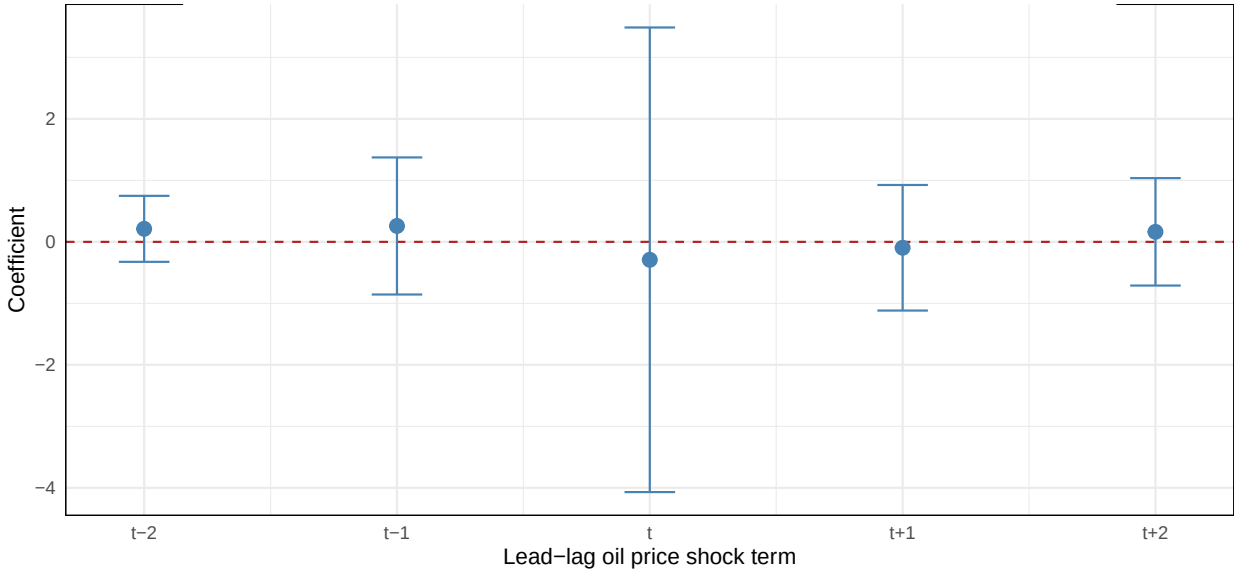
the representative agent framework.

Second, the oil price news shocks are empirically relevant. Figure 23 shows the distribution of bank-level first-stage coefficients and corresponding probability values that the coefficient is different than zero. The coefficient maps the log oil price news shock to the log odds of the observable proxy for local firm exits, meaning they can be interpreted as elasticities. The mean coefficient is 0.05 and the median 0.04, indicating that a one percent increasing oil price shock will increase the log-odds of firm exits by approximately five basis points on average. Only 3.25 percent of bank-level elasticities fall below zero. That is, approximately 97 percent of banks have the "correctly" signed first-stage, consistent with the theoretical framework. In a just-identified IV, the correctly signed coefficient is a sufficient first stage condition to then rely on inference tests of the second stage estimates, following Angrist and Kolesár (2024). However, only, 36.9 percent of bank-level regressions report first-stage coefficient p-values below the standard 0.1 threshold.

Since the low p-values may reflect noise from a small sample problem (banks can only have up to 24 observations), I additionally estimate an analogous pooled model that includes bank-level fixed effects and clustered errors at the bank-level. Figure 24 shows the full sample estimated coefficient is approximately 0.507, in line with the average of the bank-level regression distribution. The pooled sample has a robust F-statistic of 18.48 (while double clustering standard errors at the bank and year level), exceeding the standard relevance thresholds (see Stock and Yogo 2002). The sign and significance of the pooled sample elasticity are also stable over time. Figure 24 likewise shows the centered 10-year rolling window estimated coefficient of interest, which remains centered around 0.051 and is always positive, consistent with the theoretically implied effect of oil price shocks on the log-odds of firm exits.

Exclusion restriction. The oil news shocks must additionally satisfy the exclusion restriction —bank's screening technology is not impacted by oil price shocks. This condition holds in the

Figure 25: Lead-Lag Structure: Oil Shocks and Loan Losses



Notes: This plot shows the estimated coefficients and 95 percent confidence intervals for a regression mapping bank-level loan losses to leads and lags oil price news shocks. The x-axis marks the coefficients by the number of period leads or lags on the oil price shock regressor. The regression also include the observable proxy for borrower quality. The regression has bank- and year-level fixed effects. Standard errors are clustered at the year-level. The regression is estimated on a sample of 73.5 thousand bank-year observations.

context of the theoretical framework guiding the empirical estimation. There is no modeled channel for oil price shocks to influence the exogenous shocks to the bank-level screening ability. However, it may fail if we relax the assumption of banks' operating with full-information rational expectations. For example, if banks exhibit bounded attention, large oil price shocks may divert resources away from screening and toward processing aggregate information, violating the exclusion condition.

I empirically test the exclusion restriction with two exercises. I first test that oil price shocks are uncorrelated with loan losses while controlling for borrower quality, showing that if the shocks have an effect on loan losses, it is through the borrower quality channel as assumed. Second, I test that loan losses do not anticipate future or lag oil price new shocks, disqualifying the possibility that the oil price shocks violate the exclusion restriction by working through non-borrower quality channels with a lead or delay that would keep them hidden from the first test.

Test 1: Independence from screening ability. Table 7 reports a progressive controls test. I regress bank loan losses on oil shocks, sequentially adding controls for predicted borrower quality, capital costs, net worth, and market share. In Specification (1), without controls, oil shocks predict loan losses and are statistically significant at the five percent level. However, once I control for the observable borrower quality proxy in Specification (2), the oil shock coefficient shrinks considerably and becomes statistically insignificant at any standard level. Adding capital costs and

market share controls in Specifications (3) and (4) does not meaningfully change this result. The attenuation of the oil shock coefficient to zero once borrower quality is controlled for supports the exclusion restriction: oil shocks affect loan losses only through borrower quality, not through screening ability.

Test 2: Pre-trends and lead-lag structure. A violation of the exclusion restriction might manifest as oil shocks predicting future loan losses, even after controlling for current borrower quality. Figure 25 report the results of a lead-lag test. I include two leads and two lags of oil shocks in the regression, controlling for the observable borrower quality proxy. The lead coefficients are small and statistically insignificant ($p > 0.75$ for both $t + 1$ and $t + 2$), indicating no anticipation effects or omitted variable bias. The contemporaneous and lag effects are also insignificant once borrower quality is controlled for.

Collectively, these tests provide support for the exclusion restriction. Oil supply news shocks predict loan losses only through their effect on borrower quality, with no direct effect on screening ability.

C.5 Estimation algorithm

I jointly estimate the latent bank- and region-level fundamentals, time-varying loan market frictions, and regulatory costs using the following full-information EM algorithm.

1. **Residualize loan default rates.** Use bank-specific localized firm exit rates and oil price shocks as instruments to estimate the borrower quality component of observed loan losses. Then residualize the observable portfolio-level loan default rate to the borrower quality proxy. Use the residualized loan default rate it in place of the bank-region-specific loan default rate in the observables vector and zero out its loading on the regional borrower quality latent state in the measurement equation.

2. **E-step (Kalman Smoother).** For the aggregate and idiosyncratic system, separately use the Kalman filter and Rauch–Tung–Striebel smoother to recover smoothed latent states $\hat{x}_{i,t}$, given the current measurement equation parameters.

3. **M-step.**

    - *Update State Transition Parameters.* Estimate the autoregressive coefficients $\rho^{\psi}, \rho^{q}, \rho^{g}, \rho^{\eta}$ by regressing smoothed latent states on their own lags at the bank- and region-level for both idiosyncratic and aggregate systems.
    - *Estimate Structural Loan Rate Equation.* After combining the idiosyncratic and aggregate states:

- Construct expected loan default risk $\mathbb{E}[\lambda_{i,s,t+1}]$ from estimated bank ability and borrower quality states and their updated laws of motion to then construct bank-level portfolio average loan rates.
- Estimate the regulatory cost term via a weighted regression of residualized loan rates on the risk-adjusted inverse bank net worth.
- Invert the banks' first-order conditions to recover the elasticity of substitution $\sigma_{s,t}$

- *Update Measurement Equation Coefficients.* Use the estimated $\sigma_{s,t}$ and average market shares to construct:

$$\bar{\sigma}_{s,t} = \frac{\sigma_{s,t} \cdot \log \tilde{s}_t}{\sigma_{s,t} \cdot \log \tilde{s}_t + 1}, \quad \bar{\rho}_t^{\psi} = \bar{\sigma}_{s,t} \cdot \rho^{\psi}, \quad \bar{\rho}_t^{q} = \bar{\sigma}_{s,t} \cdot \rho^{q}$$

  These coefficients define the time-varying measurement matrix for the next E-step.

- *Update Exogenous States*. Use the estimated bank-level regulatory costs to update the vector of exogenous states in the measurement equation, Z_t.

4. **Convergence Check.** Terminate the algorithm if the maximum absolute change in all parameters falls below a pre-specified tolerance level.

5. **Repeat.** Iterate steps 2–4 until convergence.

6. **Non-financial fundamentals.** Using the estimated latent states, market frictions, and regulatory costs, estimate the region-specific non-financial productivity level, shocks, and laws of motion.

This procedure efficiently combines the state-space estimation of the bank-level fundamentals and structural parameter estimation of the regional market frictions and regulatory costs into a into a single full-information approach. The result is a set of consistently estimated fundamentals $\{\psi_{i,t},\ q_{s,t},\ \eta_{i,s,t},\ c_{i,t},\ A_{s,t}\}$, market frictions $\{\sigma_{s,t}\}$, and regulatory cost function $\Gamma(\cdot)$.

C.6 Estimation accuracy

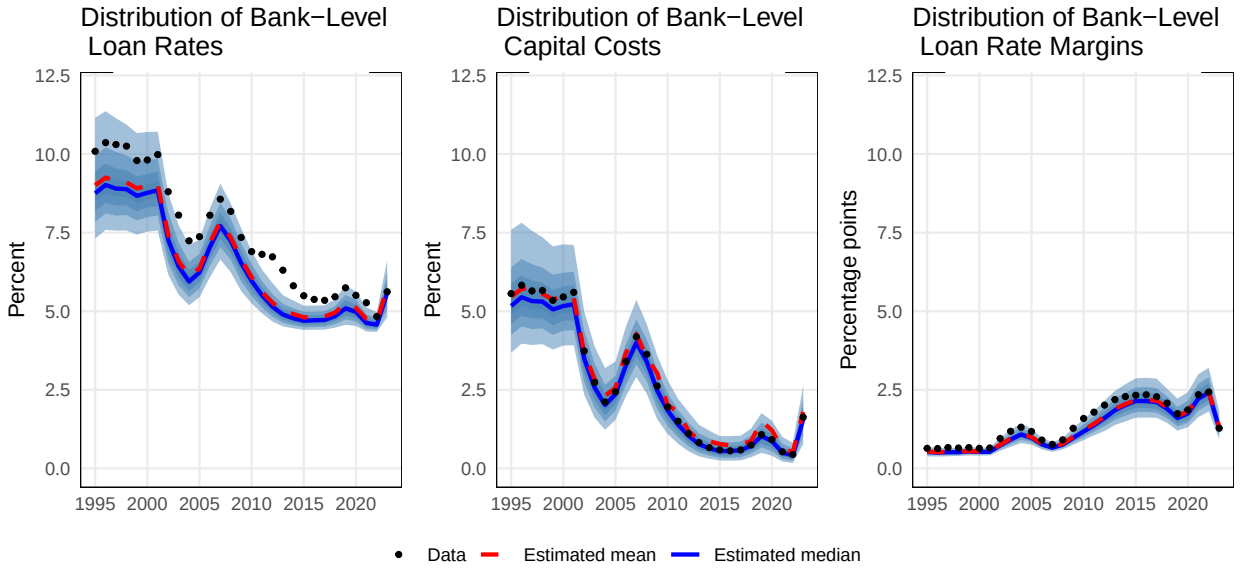
I lastly characterize the accuracy of my estimation algorithm based on its ability to recover bank-level loan rates. While the estimation algorithm uses variation in realized returns, it does not directly leverage information on loan rates, making them an untargeted moment for comparison.

Figure 26 shows the distribution of the empirically estimated bank-level (loan-weighted) loan rates. The blue bands mark the first through ninth deciles of the distribution, while the red dashed line marks the estimated mean. The figure further decomposes loan rates into their marginal costs and net interest margin components. The average observable loan rate and its component parts are

well matched their estimated counterpart. This is evidence that the empirical estimation is not only successfully reproducing observable loan rates, but is also correctly estimating its marginal costs and markup components.

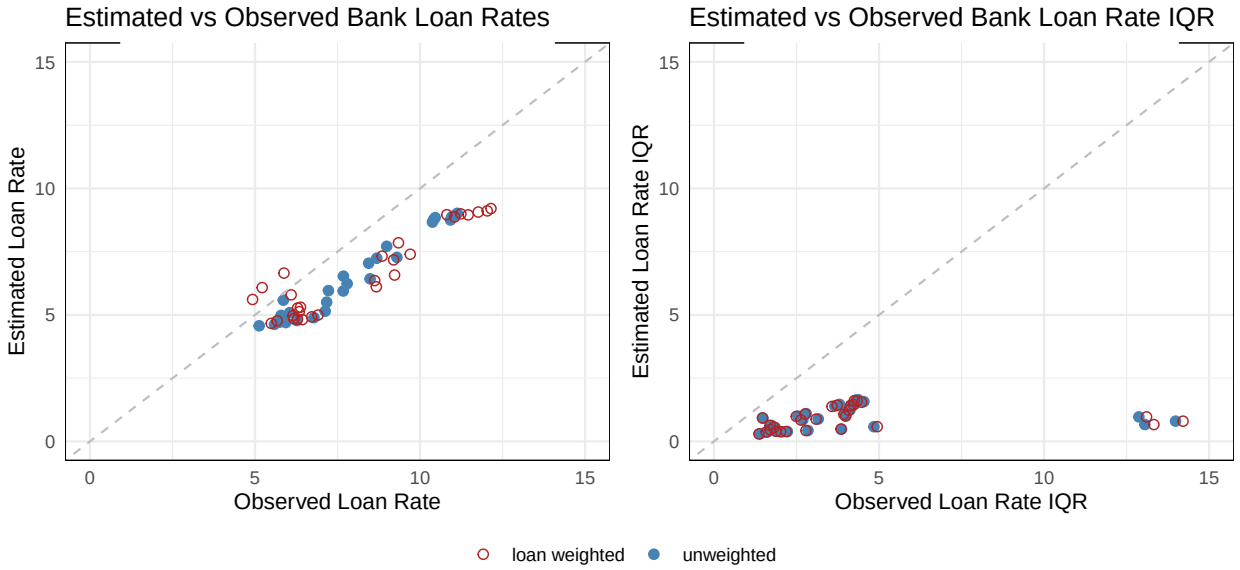
I then additionally consider the accuracy of the estimation procedure by comparing the mean and inter-quartile range of the unweighted and loan-weighed year-by-year loan rate distributions. Figure 27 shows these comparisons, and makes it clear that the mean of the distribution is well predicted across both unweighted and loan weighted treatments. In comparison the spread of loan rates is not well matched, with the estimation consistently producing too narrow of a loan rate distribution.

Figure 26: Distribution of estimated loan rate



Notes: This plot describes the estimated loan rates, marginal capital cost, and net interest margins, comparing them to observable data. The blue bands mark the first through ninth deciles, while the solid blue line indicates the median. The dashed red line is the mean, and the black dots are the average observable loan rate, cost, or spread. Data are annual from 1995 through 2024.

Figure 27: Comparing estimated and observable loan rates



Notes: This plot compares the mean and inter-quartile ranges of estimated and observable loan rates. The solid blue dots marks moments from the unweighted distribution while the red clear dots mark moments from the loan weighted distribution. The dashed gray line marks the forty-five degree line; when dots are on the forty-five degree line, then the model has perfectly reproduced the data. Data are annual from 1995 through 2024.

D Describing fundamentals

this appendix describes the regulatory costs, regional loan market frictions, and non-financial fundamentals estimated in Section 4, before turning to exercises mean tot help interpret the estimated capital cost premium and screening ability.

D.1 Regulatory costs, frictions, and non-financial fundamentals

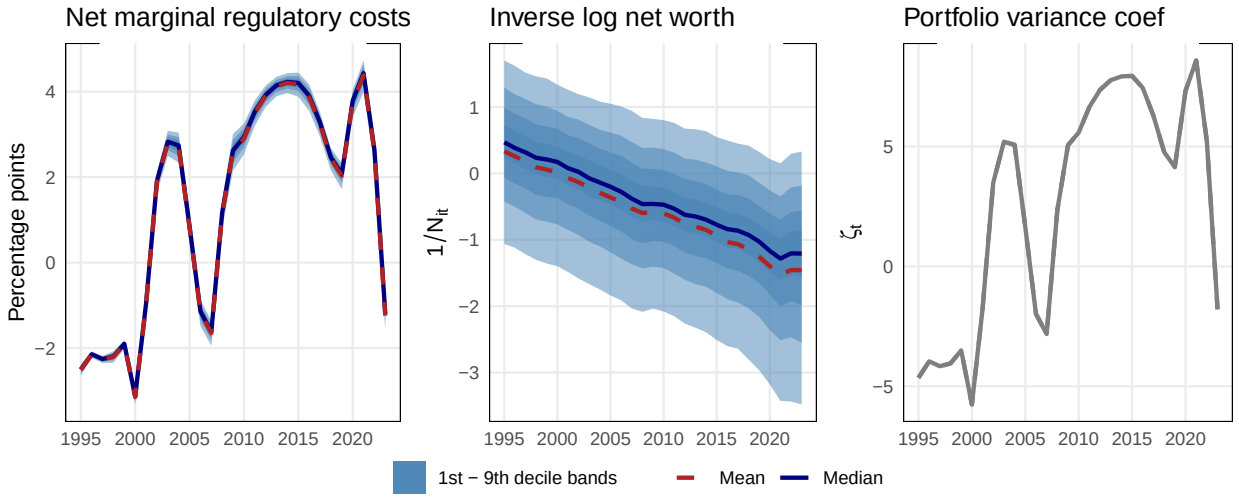
Regulatory costs. The regulatory cost is a significant, counter-cyclical, driver of loan premia. Figure 28 shows the net regulatory cost is negative, that is, acting as a subsidy, as banks seek to acquire risk during the late 1990s and the mid-2000s. These periods also coincide with commonly labeled Dot-com and mortgage bubbles. In comparison, the regulatory cost increases significantly during the 2001 recession (precipitated by the Dot-com bust) and the GFC (precipitated by the mortgage bubble bursting). The regulatory cost remains elevated through the 2010s and does not show signs of recovery until 2015. Dispersion in regulatory costs is narrow throughout the sample, though slightly increasing at peaks and troughs.

The counter-cyclical characterization of the regulatory cost points to the interpretation that the cost is capturing, on net, the banks demanded compensation for holding risky assets rather than a stable regulatory cost of doing business. Figure 28 shows the decomposition of the banks regulatory costs into the key components, log (inverse) net worth and the regulatory cost coefficient, which can be interpreted as a time-varying risk aversion parameter. Because the distribution of inverse net worth has been in secular decline since 1995, it is clear that the counter-cyclical behavior of the regulatory cost is being driven by the estimated proxy for time-varying risk aversion.

Regional loan market frictions. The regional loan market frictions have been increasing over time as the elasticity of loan substitution has been falling. Figure 29 shows that the elasticity of loan substitution has fallen from approximately 1.08 to 1.05, or three percent, on average across regions since 1995. As a result, the exogenous markup on loans has been increasing over time, rising from approximately 2.65 percentage points to 3.04 percentage points in 2023.

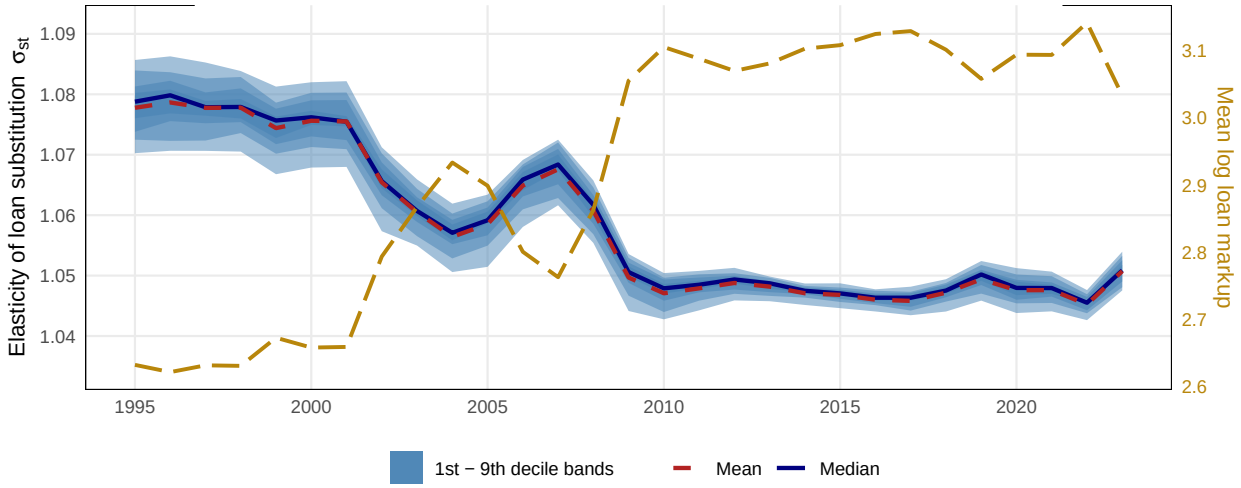
The increase in loan market frictions coincides with the decline in the number of both banks and branch locations in the United States (see figure 13 and Figure 14 respectively). This is consistent with the interpretation that the market frictions, which are theoretically frictions that make loans less substitutable, are reduced-form search and matching frictions that generate sticky borrower-lender relationships. It is documented that borrowers still need to go to branches (see for example Sakong and Zentefis 2023), so the decline in branch locations is making it more expensive to substitute loans from one bank to another since it is likely that there is a limited number of other banks with local branches close to borrowers (if any).

Figure 28: Banks regulatory costs



Notes: This figure shows the evolution of the estimated regulatory cost and its underlying factors. The blue bands mark the first through ninth deciles of each distribution. The dashed red line marks the equally weighted mean while blue marks the median. Data are annual from 1995 through 2023.

Figure 29: Regional loan market frictions



Notes: This figure shows the evolution of the estimated regional loan market frictions and the implied loan rate markup. The left axis marks the elasticity of loan substitution, the direct proxy for loan market frictions. The blue bands mark the first through ninth deciles of each distribution. The dashed red line marks the equally weighted mean and the blue line marks the median. The right axis marks the mean log loan rate markup from frictions, $\log(\frac{\sigma_{s,t}}{\sigma_{s,t}-1})$, which is represented by the dark gold dashed line. Data are annual from 1995 through 2023.

Non-financial fundamentals. Figure 30 shows the distribution of regional borrower quality (one minus the regional firm exit rate) and the log non-financial productivity wedge. The distribution of borrower quality mirrors the distribution of observable firm exit rates shown in Figure 20. Borrower quality is pro-cyclical, was stable from 1995 through the mid-2000s, but began a secular rise after the GFC. In comparison, the non-financial wedge has been largely increasing since 1995, with cyclical downturns in 2001 and the GFC. The non-financial productivity distribution becomes notably dispersed during periods of rapid growth, namely the mid-1990s and the recovery directly after the 2001 recession. However its widest dispersion occurs during the 2009 recession. There is more consistent dispersion in borrower quality, and Figure 31 shows the average borrower quality by state.

D.2 Interpreting capital costs

The estimated capital cost premium, $\kappa_{i,t}$, is persistently dispersed across banks. This appendix examines one candidate source of that heterogeneity: the composition of the liabilities a bank uses to fund its loan portfolio. Banks fund loans with a mix of deposits and wholesale funding. Deposits are cheap and rate-insensitive, owing in part to banks' market power in local deposit markets (Drechsler et al., 2017). Wholesale funding (including federal funds purchased, repurchase agreements, other borrowed money, and trading liabilities) is priced at or above the policy rate and reprices with it. A bank that funds a larger share of its balance sheet with deposits should therefore face a lower capital cost premium.

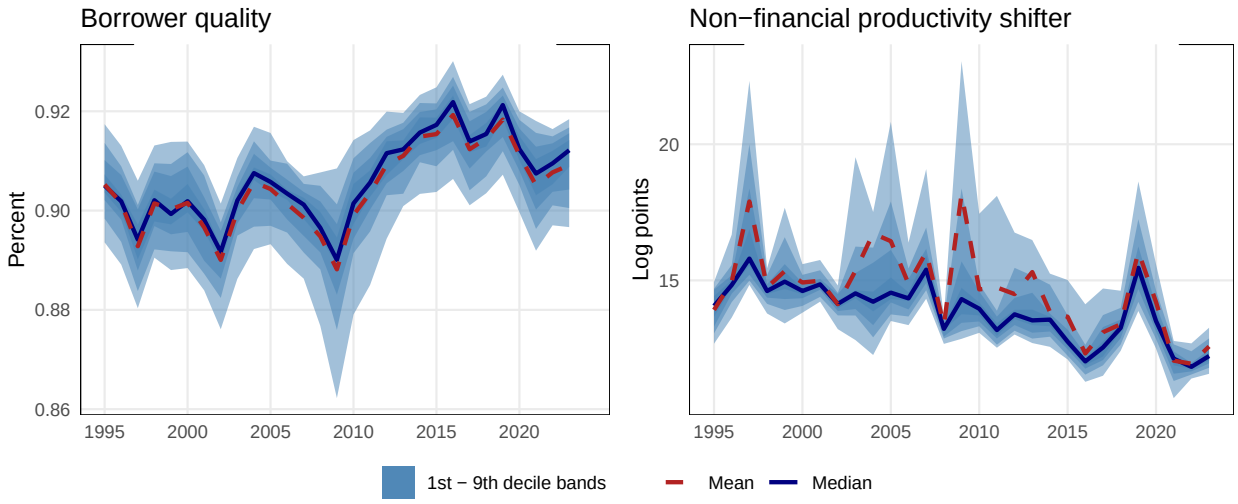
This appendix proceeds in three steps: 1) it states the identity linking funding composition to the measured premium, 2) it documents the evolution of banks' funding mix, and 3) it estimates the relationship between funding composition and $\kappa_{i,t}$, both between and within banks.

Accounting framework. The capital cost premium is measured from the ratio of a bank's total interest expense, $X_{i,t}$, to its loan portfolio, $L_{i,t}$, net of the policy rate. Decompose total liabilities, $F_{i,t} = D_{i,t} + W_{i,t} + O_{i,t}$, into deposits, wholesale funding, and all other liabilities. Let $s^j_{i,t} = j_{i,t}/F_{i,t}$ denote the funding share of each liability type and $r^j_{i,t}$ the average rate the bank pays on it, for $j \in \{D, W, O\}$. Since $s^D_{i,t} + s^W_{i,t} + s^O_{i,t} = 1$, the measured cost of funding a unit of loans satisfies

$$\frac{X_{i,t}}{L_{i,t}} = \frac{F_{i,t}}{L_{i,t}}\left[r^W_{i,t} - \left(r^W_{i,t} - r^D_{i,t}\right) s^D_{i,t} - \left(r^W_{i,t} - r^O_{i,t}\right) s^O_{i,t}\right]. \tag{37}$$

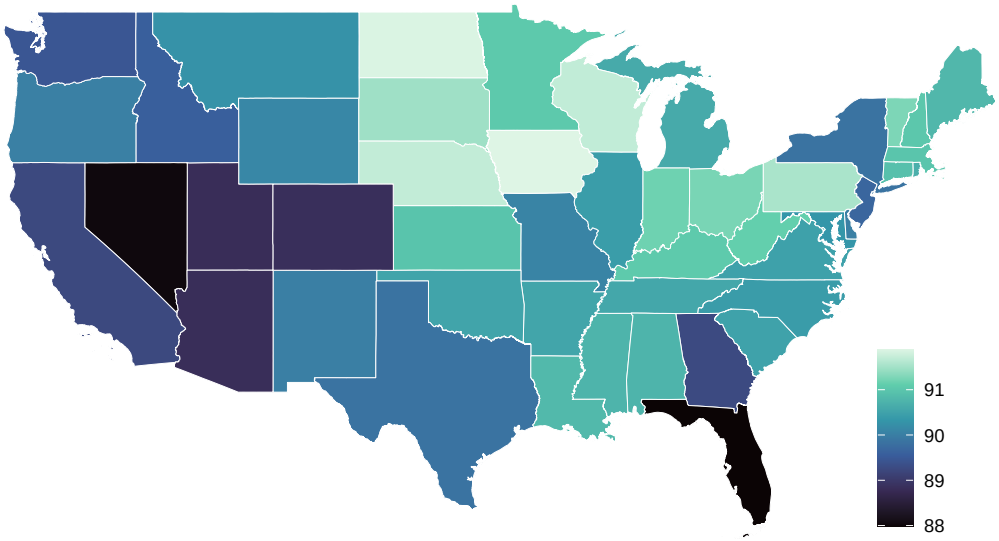
Because the policy rate is common to all banks in a given year, the premium inherits the cross-sectional variation in Equation (37) one-for-one.

Figure 30: Regional fundamentals and frictions



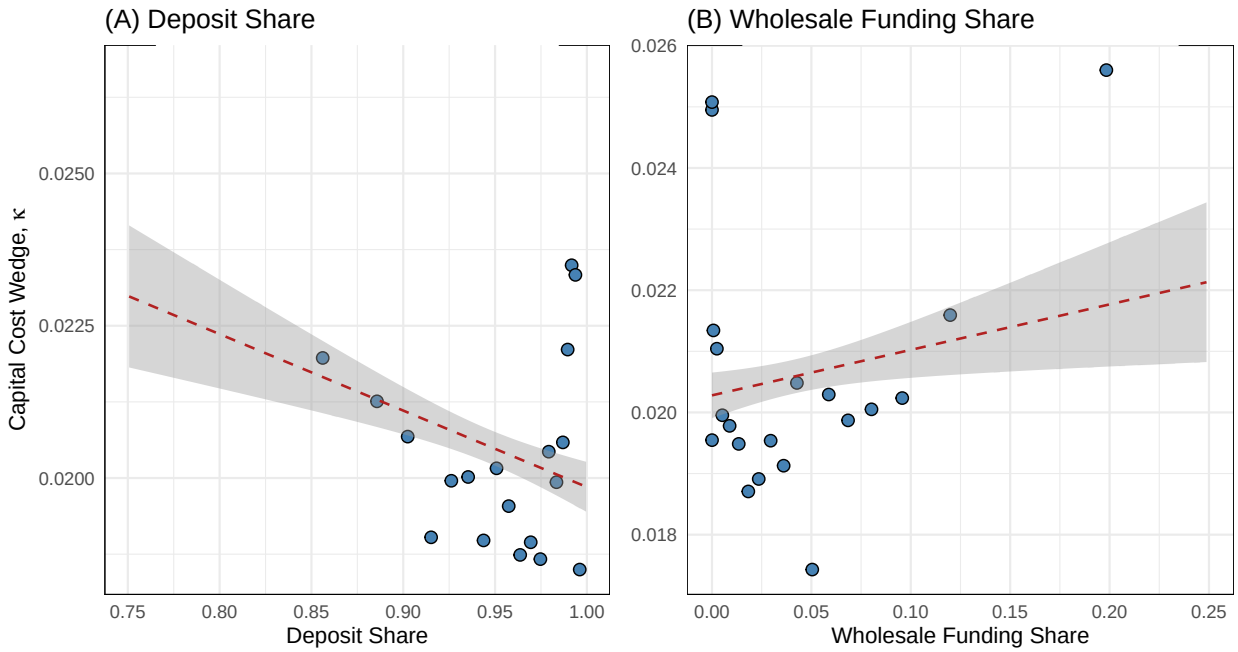
Notes: This plot shows the estimated distribution of region-specific borrower quality and non-financial productivity. The solid blue line marks the median and the dashed red line the unweighted mean. The blue banks mark the first through ninth deciles in each period. Data are annual from 1996 through 2023.

Figure 31: Average borrower quality by state



Notes: This plot shows the sample average borrower quality by state. Data are annual from 1995 through 2023.

Figure 32: Capital cost premium by deposit and wholesale funding share



Notes: This figure plots the relationship between bank-level capital cost premium, κ, deposit share and wholesale funding share. Each bank's premium and funding shares are first averaged over the years the bank appears in the sample. Banks are then sorted into twenty equal-sized bins by the relevant funding share, so that each dot plots the mean premium and mean funding share of five percent of banks. The dashed red line is the OLS fit estimated on the underlying bank-level averages, and the gray band marks the 95 percent confidence interval. The premium and funding shares are in decimal form.

Equation (37) has three implications for the analysis that follows.

First, holding funding prices, the other-liability share, and $F_{i,t}/L_{i,t}$ fixed,

$$\frac{\partial\left(X_{i,t}/L_{i,t}\right)}{\partial s^{D}_{i,t}} = -\frac{F_{i,t}}{L_{i,t}}\left(r^{W}_{i,t} - r^{D}_{i,t}\right),$$

which is negative whenever deposits are cheaper than wholesale funding. The negative relationship between the deposit share and $\kappa_{i,t}$ is therefore mechanical.

Second, a shift from deposits to wholesale funding at a given $s^{O}_{i,t}$ raises the premium by the same magnitude. The two funding shares thus carry nearly the same information. The regressions below therefore include only the deposit share.

Third, funding composition is only one of three channels through which a bank's liabilities enter $\kappa_{i,t}$. Dispersion in funding prices conditional on the mix also generates dispersion in the premium among banks with identical funding shares. The leading example is cross-bank differences in deposit rates arising from deposit market power. So does dispersion in the scale of liabilities relative

Table 8: Determinants of the capital cost premium

	Capital cost premium, $\kappa_{i,t}$			
	(1)	(2)	(3)	(4)
Constant	0.0789***			
	(0.0033)			
Deposit share$_{i,t}$	-0.0602***	-0.0369***	-0.0176***	-0.0156**
	(0.0034)	(0.0054)	(0.0039)	(0.0063)
log $N_{i,t}$	-0.0028***	-0.0149***	-0.0006***	-0.0015
	(0.0001)	(0.0003)	(0.0002)	(0.0015)
Bank fixed effects		✓		✓
Year fixed effects			✓	✓
Observations	167,648	167,024	167,648	167,024
R^2	0.04748	0.25280	0.27887	0.37224
Within R^2		0.10722	0.00332	0.00164

Notes: This table presents estimates from linear projections of the bank-level capital cost premium, $\kappa_{i,t}$, onto the deposit share and bank size. The dependent variable is in decimal form. The deposit share is total deposits divided by total liabilities, and $N_{i,t}$ is bank net worth. Column 1 is estimated by pooled OLS; columns 2–4 absorb the indicated fixed effects. Regressions are unweighted. Standard errors are clustered at the bank and year level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Data are annual from 1995 through 2024.

to loans, $F_{i,t}/L_{i,t}$. A regression of $\kappa_{i,t}$ on $s^D_{i,t}$ therefore recovers the composition effect, evaluated at the average spread and scale, plus the covariance between funding composition and these price and scale terms. Banks with strong deposit franchises, for example, both raise a larger share of funding from deposits and pay less for them. That covariance pushes the estimated coefficient away from zero relative to the pure composition effect. The estimates below are accordingly descriptive rather than causal: they measure how much of the observed dispersion in $\kappa_{i,t}$ aligns with funding composition, not the effect of an exogenous change in a bank's funding mix.

Data. A bank's funding structure can be observed on its balance sheet and income statements. The deposit share, $s^D_{i,t}$, is total deposits divided by total liabilities. The wholesale funding share, $s^W_{i,t}$, is the sum of federal funds purchased and securities sold under agreements to repurchase, other borrowed money, and trading liabilities, divided by total liabilities.

Empirical approach. The analysis combines two exercises.

First, Figure 32 presents the between-bank relationship. Each bank's premium and funding shares are averaged over the years it appears in the sample. Banks are then sorted into twenty equal-

sized bins by the relevant funding share. This exercise isolates persistent, long-run differences in funding structure across banks.

Second, Table 8 reports estimates of

$$\kappa_{i,t} = \alpha_i + \gamma_t + \beta\, s^D_{i,t} + \delta \log N_{i,t} + \varepsilon_{i,t}. \tag{38}$$

where α_i is the bank fixed effect and γ_t is the year fixed effect. Year fixed effects absorb the policy rate, the aggregate component of the premium, and the common trend in deposit shares. Bank fixed effects absorb persistent differences in funding costs and business models. With both included, β is identified from a bank's deviations in its deposit share from its own average, relative to other banks in the same year.

Results. Figure 32 shows that banks funding a larger share of their balance sheet with deposits pay a lower capital cost premium. Conversely, banks relying more heavily on wholesale funding pay a higher premium. The relationship is approximately linear for both funding shares. Moving from the bottom to the top ventile of the deposit-share distribution lowers the average premium by approximately 25 basis points.

Table 8 confirms the relationship in the bank-year panel and isolates its sources of variation. In the pooled specification, a 10 percentage point increase in the deposit share is associated with a 60 basis point lower capital cost premium.

Two features of the remaining columns discipline the interpretation of this estimate. First, the coefficient attenuates by roughly 70 percent once year fixed effects absorb aggregate variation. Within a year, the same 10 percentage point increase lowers the premium by 18 basis points. The pooled estimate therefore largely reflects the coincident secular rise in deposit shares and decline in aggregate funding costs, rather than cross-sectional differences in funding composition. Second, the within-year estimate is stable to the inclusion of bank fixed effects (-0.0156 versus -0.0176). The relationship therefore does not reflect persistent differences in banks' business models alone. A bank that raises its deposit share relative to its own average, and relative to other banks in the same year, experiences a commensurate decline in its premium.

D.3 Interpreting screening ability

The structural estimation in Section 4 recovers screening ability ψ_{it} as a latent state, identified from residual variation in bank-level loan default rates. If the estimated ψ_{it} measures a bank's capacity to distinguish good borrowers from bad, then banks that expand the resources devoted to that task should subsequently display higher estimated screening ability. This appendix tests whether they do with a simple in-sample forecasting exercise.

Data. The two variables of interest are bank-level screening ability and expenditure on screening operations and support. The latter is measured by Selling, General and Administrative expense (SG&A) from Compustat Fundamentals Annual (item `XSGA`). SG&A is the standard consolidated line item for information technology, data processing, occupancy, and personnel costs, including the compensation of loan officers. Because Compustat covers only publicly traded firms, the analysis is restricted to public banks. The test is therefore conducted on the large, listed tail of the bank size distribution. Its results should not be read as representative of the community banks that dominate the sample by count.

I specifically consider SG&A spending intensity rather than as a dollar amount:

$$x_{it} \;=\; \log\!\left(\frac{\mathrm{XSGA}_{it}}{\mathrm{AT}_{it}}\right), \tag{39}$$

where AT_{it} is total assets. Scaling by assets captures the resources a bank devotes to screening per unit of balance sheet. Unscaled expenditure, by contrast, largely tracks bank size.

Screening ability enters through the screening wedge $\log \chi^{\psi}_{it}$, where $\chi^{\psi}_{it} = (1-\psi_{it})/\psi_{it}$ is decreasing in screening ability. Both series are winsorized at the first and ninety-ninth percentiles in levels, differenced within bank, and winsorized again in differences. Banks are retained if they contribute at least eight usable bank-year observations.

Model. The baseline in-sample forecasting model is

$$\Delta \log \chi^{\psi}_{it} \;=\; \sum_{j=1}^{p} \alpha_j \, \Delta \log \chi^{\psi}_{i,t-j} \;+\; \sum_{j=1}^{p} \beta_j \, \Delta x_{i,t-j} \;+\; \mu_i \;+\; \lambda_t \;+\; \varepsilon_{it}, \tag{40}$$

with $p = 2$, bank fixed effects μ_i, and year fixed effects λ_t. The bank fixed effects absorb permanent differences in screening ability, business model, product mix, and reporting convention. The year fixed effects absorb aggregate credit conditions and the common price trend in nominal expenditure, so no external deflator is applied. No contemporaneous term Δx_{it} enters, so the specification is strictly predictive.

The null hypothesis of Granger non-causality is

$$H_0 : \; \beta_1 = \beta_2 = \cdots = \beta_p = 0. \tag{41}$$

It is tested with a Wald statistic constructed from the bank-clustered covariance matrix.

Expenditure may affect screening only with a delay. To allow for this, the dependent variable is also replaced by the cumulative change, holding the right-hand side of (40) fixed.

Table 9: Operating expenditures forecast screening ability

	SG&A → Screening			Screening → SG&A
	$h = 1$	$h = 2$	$h = 3$	
$\Delta\chi^{\psi}_{i,t-1}$	-0.1636***	-0.2488***	-0.3084***	0.0117***
	(0.0173)	(0.0209)	(0.0216)	(0.0026)
$\Delta\chi^{\psi}_{i,t-2}$	-0.1387***	-0.2119***	-0.3006***	0.0098***
	(0.0147)	(0.0188)	(0.0203)	(0.0026)
$\Delta x_{i,t-1}$	0.0642	-0.0986	-0.1967**	-0.1665***
	(0.0566)	(0.0808)	(0.0956)	(0.0175)
$\Delta x_{i,t-2}$	-0.1398**	-0.2883***	-0.3815***	-0.0763***
	(0.0592)	(0.0807)	(0.0953)	(0.0129)
Wald statistic	4.55	7.14	8.69	15.97
p-value	0.011	0.001	0.000	0.000
Bank fixed effects	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓
Observations	7,968	7,487	6,978	7,715
R^2	0.24287	0.40564	0.51664	0.12474

Notes: This table presents the estimated coefficients and Wald test of joint significance for in-sample forecasting models of bank-level screening ability and SG&A expenses. The dependent variable in columns 1–3 is the cumulative change in screening wedge χ^{ψ}. The final column reverses the direction. All specifications include $p = 2$ lags of both series, bank fixed effects, and year fixed effects. The Wald statistic tests $\beta_1 = \cdots = \beta_p = 0$. Standard errors are clustered by bank. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Data are annual from 1995 through 2023.

Finally, I reverse the predictive direction:

$$\Delta x_{it} \;=\; \sum_{j=1}^{p} \gamma_j \, \Delta x_{i,t-j} \;+\; \sum_{j=1}^{p} \delta_j \, \Delta \log \chi^{\psi}_{i,t-j} \;+\; \tilde{\mu}_i \;+\; \tilde{\lambda}_t \;+\; \tilde{\varepsilon}_{it}, \tag{42}$$

and test $H_0 : \delta_1 = \cdots = \delta_p = 0$. Table 9 reports both directions.

Identification. Identification comes from variation within a bank over time and within a year across banks. The exercise establishes predictive precedence, not causation. Both screening ability and expenditure are endogenous choices of the bank. At least two mechanisms other than the maintained interpretation generate the same lead-lag pattern. First, banks that expect an improved borrower pool may increase staffing in advance of it, which raises a reverse-causality concern.

Second, a bank-specific or regional shock to credit demand may move both series with different lags. Year fixed effects absorb only the aggregate component of such a shock. The remainder therefore constitutes a standard omitted-variable threat to identification.

Results. Table 9 reports the estimates. Lagged operating expenditure Granger-causes screening ability at every horizon. The null of non-causality is rejected at the 5 percent level in column 1 and at the 1 percent level in columns 2 and 3. Moreover, the sign is the one implied by the model's interpretation of the fundamental: growth in SG&A intensity predicts subsequent declines in χ^{ψ}, that is, improvements in screening ability.

The timing of this predictive content is informative. At $h = 0$, the rejection is driven entirely by the second lag. The first-lag coefficient is positive and insignificant, and the cumulative coefficient is indistinguishable from zero. Expenditure therefore carries no predictive content for screening within the following year. Its predictive content then accumulates with the horizon. The cumulative coefficient rises in magnitude from -0.076 at $h = 0$ to -0.387 at $h = 1$ and -0.578 at $h = 2$. At $h = 2$, SG&A intensity growing ten percent faster in each of the two preceding years predicts a cumulative decline in χ^{ψ} of approximately 5.8 log points.[33]

The reverse direction is also significant, but it is opposite in sign and small. A deterioration in screening ability predicts higher subsequent expenditure growth, with a cumulative elasticity of 0.021. Taken together, the two directions describe a feedback pattern. Banks raise operating expenditures following a deterioration in screening ability, and higher expenditure is followed, with a lag of roughly two years, by improvement.

[33] Since $d \log \chi^{\psi} = -d\psi/[\psi(1-\psi)]$, this corresponds to an increase in screening ability of approximately $0.058\,\psi_{it}(1-\psi_{it})$.

E Quantitative calibration and simulations

This appendix provides a detailed description of the quantitative model calibration, solution methods, and simulation procedures used to analyze bank dynamics and their macroeconomic effects.

E.1 Model calibration

The quantitative model combines empirically estimated laws of motion for bank and regional fundamentals with standard parameter values from the macroeconomic literature. This section details the calibration strategy for each component.

E.1.1 Estimated Fundamentals

Table 10: Estimated Fundamental Laws of Motion

	ρ_κ	ρ_ψ	ρ_q	ρ_a	ρ_η	Var(κ)	Var(ψ)	Var(q)	Var(a)	Var(η)
Micro-level parameters										
Micro	−0.347	0.821	0.983	0.300	0.937	15.6	0.403	0.0604	5.09	9138
Macro-level parameters										
Macro	0.927	1.020	1.000	0.311	—	0.00542	0.159	0.0114	2.12	—

Notes: The table reports estimated autoregressive coefficients and innovation variances for micro- and macro-level fundamentals. Dashes indicate parameters not estimated at the aggregate level. The variance of ϵ^η is the variance of shocks before bank-region-level loan appeal are normalized.

Each of the three fundamentals (screening ability $\psi_{i,t}$, capital cost premium $\kappa_{i,t}$, and bank-region loan appeal $\eta_{i,s,t}$) is the sum of a constant, an aggregate component common to all banks or regions, and an idiosyncratic component specific to a bank, bank-region pair, or region. Let $x^j_{k,t}$ denote fundamental j for bank b in period t, expressed in its estimation units. Then

$$x^j_{k,t} = \bar{x}^j + X^j_t + \tilde{x}^j_{b,t}, \tag{43}$$

$$X^j_t = \rho^j X^j_{t-1} + \varepsilon^j_t, \qquad \varepsilon^j_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma^2\right), \tag{44}$$

$$\tilde{x}^j_{b,t} = \tilde{\rho}^j \tilde{x}^j_{b,t-1} + \tilde{\varepsilon}^j_{b,t}, \qquad \tilde{\varepsilon}^j_{b,t} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \tilde{\sigma}^2_j\right), \tag{45}$$

where aggregate component X^j_t, and idiosyncratic component $\tilde{x}^j_{b,t}$ are mutually independent, and are both subject to i.i.d. Gaussian shocks, ε^j_t and $\tilde{\varepsilon}^j_{b,t}$. The two components are separately identified by the restriction that idiosyncratic innovations are mean zero in the cross section. The aggregate

component therefore absorbs all common variation, and the idiosyncratic component captures deviations from it.

The estimated parameters for $\eta_{i,s,t}$ describe unnormalized appeal. In the simulation, appeal is normalized within each region every period, meaning loan appeal has no aggregate component. In the simulation exercises, I shock the components of $\log \chi^{\psi}_{i,t}$ directly and recover screening ability as $\psi_{i,t} = \left(1 + \chi^{\psi}_{i,t}\right)^{-1}$ when needed.

Each region s is characterized by borrower quality $q_{s,t}$ and non-financial productivity $A_{s,t}$, which decompose analogously:

$$\log \chi^{q}_{s,t} = \bar{x}^{q} + X^{q}_{t} + \tilde{x}^{q}_{s,t}, \qquad \chi^{q}_{s,t} = \frac{1 - q_{s,t}}{q_{s,t}}, \tag{46}$$

$$a_{s,t} = \bar{x}^{a} + X^{a}_{t} + \tilde{x}^{a}_{s,t}, \qquad a_{s,t} = \log A_{s,t}, \tag{47}$$

where X^{j}_{t} and $\tilde{x}^{j}_{s,t}$ follow (44) and (45) with region-level Gaussian shocks, $\tilde{\varepsilon}^{j}_{s,t}$.

The persistence parameters ρ^{j} and $\tilde{\rho}^{j}$ and the innovation variances σ^2 and $\tilde{\sigma}^2$ are taken directly from the M-step of the empirical estimation. Table 10 reports these estimates.

E.1.2 Frictions, regulatory and distance costs

Market frictions. Regional loan market frictions are captured by the time-varying elasticity of substitution across banks, $\sigma_{s,t}$. The elasticity evolves according to the AR(1) process:

$$\log \sigma_{s,t} = \mu^{\sigma} + \rho^{\sigma} \log \sigma_{s,t-1} + \varepsilon^{\sigma}_{s,t}, \tag{48}$$

which is recovered from the structural net interest income equation via the M-step. The $\mu^{\sigma} = 1.107$, $\rho^{\sigma} = 0.898$, and $Var(\sigma_{s,t}) = 1.394e-05$.

Regulatory costs. Bank i's regulatory cost at time t is:

$$\Gamma_{i,t} = \zeta_{1t} \log(\sigma^{R}) + \zeta_{2t} \log(1/N_{i,t}), \tag{49}$$

where $N_{i,t}$ is the bank's net worth, σ^{R} is the portfolio risk, and ζ_{1t}, ζ_{2t} are the time-varying aggregate regulatory cost coefficients. However I keep the regulatory cost coefficients fixed for simulations to isolate changes in banks' decision making to the effects of frictions, fundamentals, and net worth fluctuations.

The estimated regulatory cost function coefficients are $\zeta_2 = 0.0689$ and $\zeta_1 = -8.985$, and the portfolio return risk is $\sigma^{R} = 0.4820696$. This implies the portfolio risk term will contribute a

Table 11: Deep parameters and model dimensions [old]

Parameter	Symbol	Value	Description
Model dimensions			
Number of regions	n_s	50	Total number of regions in the economy
Banks per region	n_b	500	Number of banks operating in the economy
Simulation periods	T	30	Number of simulated time periods
Deep parameters			
Discount rate	β	0.99	Time discounting rate of banks and households
Labor share	α	0.66	Labor's share of value added in production
Inverse Frisch elasticity	ϱ	1.00	Responsiveness of labor supply to real wages
Rotemberg adjustment cost	θ	77.0	Cost of adjusting prices (nominal rigidity)
Retail elasticity of substitution	ϵ	1.20	Demand elasticity setting the steady-state markup
Monetary policy rule			
Inflation response coefficient	ϕ_π	1.5	Taylor-rule coefficient on inflation gap
Output response coefficient	ϕ_y	0.5	Taylor-rule coefficient on output gap
Interest rate smoothing	ρ_r	0.8	Persistence of nominal policy rate

constant 2.847 percentage points to loan rates.

Distance costs. I will set the fixed cost $f = 0$ for banks operating in their headquarters region. The cost of operating in region s' for a bank headquartered in region s is:

$$c(s, s') = 2.5 \times \log(d_{s,s'}) + 1, \tag{50}$$

where $d_{s,s'}$ is the geographic distance between the centroids of regions s and s'. The elasticity of 2.5 is calibrated so that as the log-distance doubles the distance cost doubles, matching the distance cost estimated in Oberfield et al. (2024). Distance between regions are drawn from a log-normal distribution with mean one and standard deviation 0.1, but could be calibrated to the population centroids of U.S. states.

E.1.3 Deep structural parameters

The deep parameters governing preferences, technology, and nominal rigidities are calibrated to standard values in the New Keynesian DSGE and macro-banking literature. Table 11 summarizes these parameters. The curvature of the production function $\alpha = 0.66$ matches the long-run U.S. labor share of value added ($\approx 2/3$), the standard Cobb-Douglas target in the macroeconomic literature. The inverse Frisch elasticity $\varrho = 1$ follows Nakamura and Steinsson (2014) and, while larger than typically used in microeconomic studies, is meant to capture the extensive margin of labor

supply and is standard in the macroeconomic literature. The retail elasticity of substitution $\epsilon = 1.20$ and the Rotemberg adjustment cost parameter $\theta = 77$ are estimated jointly by Ireland (2001), where ϵ pins down the steady-state markup while θ governs the curvature of the price-adjustment cost. This value of θ falls within the $\theta \in [55, 100]$ range implied by the Calvo-equivalent price stickiness estimated by Nakamura and Steinsson (2008) at markups μ of approximately 1.1 to 1.2, consistent with the markup implied by $\epsilon = 1.20$. The discount rate $\beta = 0.99$ implies a modest one percent annualized steady-state risk-free interest rate qualitatively consistent with the sample period. Results are qualitatively robust under the capital-share convention $\alpha = 0.33$ or $\varrho = 5$.

The monetary policy rule parameters follow the standard specification in Taylor (1993), with $\phi_\pi = 1.5$ and $\phi_y = 0.5$ satisfying the Taylor principle and ensuring determinacy of the equilibrium. These are the canonical Taylor (1993) coefficients and serve as the standard benchmark for a determinate, stabilizing rule. The interest rate smoothing parameter $\rho_r = 0.8$ captures the observed persistence in the federal funds rate and is consistent with the degree of interest-rate smoothing documented in empirical Taylor-rule estimates, for example Clarida et al. (2000).

E.2 Simulation and solution details

This section describes the steps to solving and simulating the economy. Note that given the structure of the economy, most region-level outcomes can be solved analytically after solving for the banks' entry/exit decisions with a bounded rationality rule à la Krusell and Smith (1998) and Weintraub et al. (2008).

Simualtion. I simulate economies with 500 banks and 50 regions for 30 periods. For each of these simulations I take the following steps:

- Initialization:
  - Create S ex ante identical regions and generate an $S \times S$ matrix determining the relative distance of regions (for trade cost calculations). Distances are drawn from a log-normal random distribution with mean 1, standard deviation 0.1.
  - Create B ex ante identical banks. Uniformly distribute banks across the regions.
- Each Period:
  - Banks decide where to operate based on expected profits
    - Banks forecast each regions banking sector average fundamentals, market friction, its own appeal, and total loan demand to calculate its expected market share and

profits if it enters. Banks decide to enter a region if its forecasted profits satisfy condition 11.

- Banks draw fundamentals $G_{i,t}$, $\eta_{i,s,t}$ (conditional on entering region s), and forecast $\mathbb{E}q_{s,t}$, $\mathbb{E}\psi_{i,t}$. Regions draw $\sigma_{s,t}$ and $A_{s,t}$.
- Banks decide on $\{L_{i,s,t}\}$. Firms and households solve $Y_{s,t}$, $M_{s,t}$, $W_{s,t}$, given fundamentals and $\{L_{i,s,t}, R_{i,s,t}\}$.
  - Loan rates are solved analytically using Equation 7. With loan rates and fundamental shocks realized, regional planned output, employment, loan demand, and market clearing wages can all be solved analytically with Equations 30, 28, 29, and 27 respectively.
- Production occurs and firm exit rate shocks are realized. Loan losses are realized and bank net worth are updated. The Central Bank determines the policy rate for the start of the next period.
- Banks are hit with exit shocks, new banks enter. Update banking sector distribution.

The regional entry forecasting rule. The banks use a set of seemingly unrelated AR(1) models to forecast regional banking sector average fundamentals, their own appeal, and regional frictions before deciding to enter a region in a given period. These forecasting rules are estimated following the standard Krusell-Smith algorithm. I instantiate the forecasting rules as random walks, then (1) simulate the economy and (2) use the simulated paths to update the forecasting rules, repeating steps (1) and (2) until the forecasting rules have converged. Because the forecasting rules are at the bank, region, or bank-region level, the estimation preserves the panel structure of the simulated data, meaning the number of observations per simulation is quite large, making the algorithm computationally efficient in terms of the number of simulations required to converge.

Solution method and computational considerations. After solving the bank's regional entry problem with a Krusell-Smith bounded rationality forecasting approach, all other economic decisions can be solved analytically given the structure of the economy. This was done intentionally to keep the computational problem of simulation the economy tractable enough to be feasible with a large panel of banks and regions.

E.3 Simulated moments and data

This section compares the simulated economy to real observable data. While the simulated economy cannot capture the full dispersion in bank-level outcomes, cross-regional variation and aggregate

business cycles are well matched in the quantitative setting.

Banking sector moments. The cross-sectional variation in the simulated banking sector is a muted representation of the observable data. The simulated economy demonstrates approximately 20 percent of the across-bank variation in net worth growth as observed on average in the U.S. economy. This is because the empirical model used to estimate the bank-level fundamentals is restricted in its ability to capture the full dispersion in loan rates (as shown in Appendix C.6). However, this is a well-known limitation of heterogeneous firm models that only allow for normally distributed shocks, and, as I will discuss next, this does not impair the model's ability to qualitatively capture cross-regional variation or aggregate business cycle dynamics. Moreover, acknowledging this limitation of the quantitative framework, I will also turn to an empirical decomposition of observable loan spreads and market shares to show that the fundamentals driving bank dynamics analyzed in the simulated economy qualitatively match the dynamics observed in the data.

Regional moments. The cross-sectional variation in regional outcomes is well matched by the model. For example, the regional variation in loan rates is almost perfectly matched, at 0.0021 in the model and 0.0022 in the data. In comparison, the within-region dynamics are shorter-lived in the model than in the data: for example, the first-order autoregressive coefficient on regional loan rates is only 0.178 in the model versus 0.951 in the data, or almost 19 percent of the observable persistence. This is expected, however, given the lack of modeled mechanisms that would generate persistence. For example, there is no sticky investment or accumulated regional capital.

Aggregate moments. The aggregate business cycle is also well matched by the model. For example, the unconditional standard deviation of GDP growth is 1.72 in the simulated economy, which is less than a 10 percent difference from the observable 1.84 standard deviation. Fluctuations in inflation, as well as both the level and variation in the policy rate, are similarly well matched. Moreover, the persistence of the business cycle consistent across simulated and observed data. While simulated within-region dynamics are unrealistically short-lived, the medium level of persistence in aggregate conditions demonstrated in the simulated economy is in line with observable data. This reflects the fact that several aggregate variables are well known to have smoother and more transitory dynamics in the aggregate than at the micro-level (see Bloom (2009) for a well-known discussion of this pattern applied to capital investment).

Table 12: Simulated versus empirical moments

Panel A: Aggregate			*Panel B: Banks*			*Panel C: Regions*		
Statistic	Model	Data	Statistic	Model	Data	Statistic	Model	Data
Unconditional			*Cross-bank dispersion*			*Cross-region dispersion*		
SD inflation	0.162	0.999	SD net worth growth	0.038	0.191	SD wage growth	0.594	0.472
SD GDP growth	1.722	1.846	SD lending growth	0.008	0.199	SD loan growth	0.250	0.183
Avg. policy rate	3.438	2.423	SD loan rate	1.002	39.001	SD loan rate	0.210	0.220
SD policy rate	4.674	2.219						
Dynamics			*Dynamics*			*Dynamics*		
Growth AR(1)	0.517	0.109	Loan rate AR(1)	-0.033	0.897	Loan rate AR(1)	0.178	0.951
Inflation AR(1)	0.540	0.746	Lending growth AR(1)	0.958	0.318	Loan growth AR(1)	-0.010	0.120
Policy rate AR(1)	0.989	0.828	Net worth growth AR(1)	0.857	0.129	Wage growth AR(1)	-0.012	-0.241

Notes: This table compares select model simulated and real data moments describing the U.S. banking sector, state-level economies, and aggregate business cycles. Model moments are computed with 30 simulated periods across 50 regions and 500 banks. Aggregate inflation, GDP growth, the policy rate, and loan rates are reported in percent. AR(1) denotes the first-order autoregressive coefficient, estimated either within-bank or within-region for Panels B and C.

F Empirical bank dynamics

This appendix presents empirical counterparts fo the quantitative exercises characterizing bank dynamics.

F.1 The transmission of bank-level fundamentals

This section presents an empirical model of how bank-level fundamentals *Granger*-cause changes in bank-level outcomes, analogous to the quantitative analysis presented in Section 5. The following will introduce the data, model, and identification, before presenting the estimated effects and discussion.

Data. The three bank fundamentals are collected in

$$f_{it} \equiv \left(\log \chi_{it}^{\psi},\ \kappa_{it},\ \log \eta_{it}\right)'. \tag{51}$$

where $\log \eta_{it}$ is the unweighted average bank-region-level appeal for bank i across all regions it is operating in at time t. All fundamentals are standardized with mean-zero and a unit variance.

Then four bank-year outcomes of interest are considered, indexed by $m \in \{1, 2, 3, 4\}$:

$$y_{it}^{1} = \frac{R_{it}}{1 - \lambda_{it}}, \qquad y_{it}^{2} = \Delta \log s_{it}, \tag{52}$$

$$y_{it}^{3} = \Delta \log L_{it}, \qquad y_{it}^{4} = \Delta \log N_{it}, \tag{53}$$

where R_{it} is the observed loan interest rate, s_{it} the bank's average market share, L_{it} total loans, N_{it} net worth, and $\Delta \log x_{it} \equiv \log x_{it} - \log x_{i,t-1}$.

All micro-level data are winsorized within year at the first and ninety-ninth percentiles. Banks must have at least eight observations.

Model. The baseline within-bank, within-sample, one-period-ahead, forecasting model is

$$y_{it}^m = \beta_m' f_{i,t-1} + \alpha_i^m + \delta_t^m + \varepsilon_{it}^m, \qquad m = 1, \ldots, 4, \tag{54}$$

where α_i^m is a bank fixed effect and δ_t^m a year fixed effect. An augmented specification adding one lag of the standardized log bank-level net worth, $\log N_{i,t-1}$ is additionally estimated and reported. Standard errors for both specifications are two-way cluster-robust in the bank and year dimensions.

Identification. Identification comes from variation in the fundamentals within a bank over time and within a year across banks. However, this exercise does not establish a causal relationship between bank-level fundamentals and outcomes, only a predictive role for fluctuations in fundamentals to anticipate one-period-ahead changes in outcomes. This is because, unlike the quantitative setting in which omitted variable bias and reverse causality can be ruled out by construction, this empirical estimation is potentially subject to both.

Results. Screening ability is the only fundamental that has a consistent economically and statistically significant effect on bank-level outcomes. Moreover, an increase in χ^ψ, which corresponds to a decrease in screening ability, is correlated with bank-level deteriorating outcomes as predicted by both theory and the quantitative model. In particular, a one standard deviation decrease in χ^ψ increase the loan spread by approximately 2.2 percentage points, while simultaneously decreasing market share growth, lending growth, and net worth growth by 2.9, 2.6, and 2 percent respectively. When one controls for the bank's pre-existing net worth, the screening ability coefficients remain almost identical to the baseline (except for the model of net worth growth).

In comparison, the capital cost premium has a statistically significant conditional correlation with increasing market share, lending, and net worth growth. However, following the discussion in Appendix D.2, this may be a case of reverse causality —a bank's capital cost premium increases as it relies more heavily on wholesale funding, which it may do in an effort to expand its loan portfolio beyond what its current deposit base can support, which in turn expands its net worth.

Loan appeal is correlated with an increase in loan rates, in line with theory. However, there is no statistically significant correlation between loan appeal and bank-level growth. This is in line with the interpretation that bank loans are treated as homogeneous goods that are primarily

Table 13: An empirical model of bank-level fundamentals and outcomes

	Loan rate		Market share growth		Lending growth		Net worth growth	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\log \chi^{\psi}_{i,t-1}$	2.22***	2.21***	-0.029***	-0.026**	-0.029***	-0.028**	-0.020**	-0.017
	(0.321)	(0.315)	(0.009)	(0.010)	(0.010)	(0.012)	(0.009)	(0.011)
$\log \kappa_{i,t-1}$	-0.533	-0.508	0.080*	0.071**	0.057	0.052*	0.057	0.050*
	(0.582)	(0.650)	(0.041)	(0.034)	(0.037)	(0.028)	(0.040)	(0.028)
$\log \eta_{i,t-1}$	1.16*	1.14**	0.023	0.028	-0.0003	0.004	0.001	0.007
	(0.582)	(0.545)	(0.018)	(0.018)	(0.011)	(0.010)	(0.012)	(0.011)
$\log N_{i,t-1}$		1.03		-0.493***		-0.301*		-0.417**
		(2.62)		(0.126)		(0.154)		(0.165)
Bank fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Observations	196,576	196,576	214,815	214,815	229,144	229,144	229,132	229,132
R^2	0.354	0.354	0.291	0.324	0.194	0.212	0.214	0.251
Within R^2	0.025	0.025	0.021	0.067	0.016	0.038	0.014	0.060

Notes: This table presents the estimated empirical effects of fluctuations in bank-level fundamentals on bank-level outcomes. All regressors have been standardized with mean zero, unit variance. Standard errors are clustered at the bank and year level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Data are at the bank-year-level from 1995 through 2023.

differentiated by price rather than non-pecuniary features.

Taken together, these results provide empirical support of the quantitative finding: bank-level fluctuations in screening ability are conditionally correlated with bank-level outcomes in an economically and statistically significant manner.

F.2 Empirical level and variance decomposition

I have so far relied quantitative simulations of the economy to measure the causal effect of shocks on bank dynamics. However, the distribution of bank sizes and lending footprints in the U.S. has been heavily shaped by regulations governing the maximum size and geographic expansion of banks, as well as by ad hoc government interventions during financial crises. These historical institutional features are necessarily absent from the structural model. Consequently, the simulated data cannot fully characterize how banks' pre-existing size, measured by net worth, affected their realized market shares through regulatory costs or their loan rates through oligopolistic profits. To complement the structural variance decomposition, this section appeals directly to the data and decompose observed loan spreads and regional market shares into contributions from variation in the

underlying estimated fundamentals and frictions. This empirical decomposition captures the full realized distribution of bank outcomes and thus accounts for historical institutional features that are not present in the structural model. While this accounting decomposition does not necessarily identify causal relationships, it provides a complementary characterization of the sources of observed variation in bank outcomes. Taken together, the two exercises provide a more complete picture of the sources of growth and heterogeneity in the banking sector: the structural decomposition isolates causal effects in an environment free of historical institutional constraints, while the empirical decomposition characterizes how those forces are reflected in the realized distribution of banks.

Methodology. I decompose the level, within- and across-bank variation of the observable bank-level loan spreads and market shares. For the level decomposition, I compute the gross contribution of each pricing factor i to the level of the outcome y for each bank, or formally:

$$\mu_{y|i} = \frac{|b_i x_i|}{|y|}, \qquad |y| = \sum_j |b_j x_j|,$$

where y denotes the outcome of interest (the loan spread or regional market share) and

$$x = \left\{ \kappa_{i,t}, \chi^{\psi}_{i,t}, \chi^{q}_{s,t}, \eta_{i,s,t}, \log\left(\frac{\sigma_{s,t}}{\sigma_{s,t}-1}\right), N_{i,t} \right\}$$

collects the model-implied pricing components which are scaled by the appropriate coefficient, b_i, determined by the structural equations characterizing the objects of interest. The object $\mu_{y|i}$ measures the gross share of factor i in the level of y, abstracting from offsetting effects across components.[34]

To decompose variance, I apply the total variance formula following Hottman et al. (2016) and Eslava et al. (2024). The share of variance in y attributable to factor i is given by

$$\sigma_{y|i} = \frac{\mathrm{Var}(x_i) + \sum_{j \neq i} \mathrm{Cov}(x_i, x_j)}{\mathrm{Var}(y)},$$

where x is the same vector of fundamentals and frictions used in the level decomposition. Across-bank variance decompositions are computed within each period across banks, while within-bank variance decompositions are computed for each bank over time. Covariance terms are attributed

[34] Two notes on the computational details of the decomposition. First, regulatory costs are not included as a separate component because they are endogenous functions of bank net worth, borrower quality, and screening ability through the portfolio risk term (see Appendix A.5 for details). I instead first decompose regulatory costs into these primitives and incorporate their contributions indirectly through the corresponding elements of x. Second, the $\chi^{\psi}_{i,t}$ and $\chi^{q}_{s,t}$ terms enter the loan spread in a single term multiplying one another, so I calculate the contribution of the term and then allocate the contribution to either individual factor based on the estimated ration of $\chi^{\psi}_{i,t}$ to $\chi^{q}_{s,t}$.

Table 14: Decomposition of observed loan spreads

	Regional Frictions	Borrower Quality	Net Worth	Bank Ability	Capital Cost	Loan Appeal
Loan-weighted avgerage loan spread	23.14	0.68	3.07	18.18	54.87	0.05
Equally-weighted avgerage loan spread	33.58	0.65	2.95	25.73	37.08	0.02
Across-bank loan spread variation	0.97	4.26	29.70	40.10	24.40	0.53
Within-bank loan spread variation	4.56	2.35	0.95	82.50	9.56	0.06

Notes: Each row decomposes the variance (or level) of banks' loan spreads into percent contributions by fundamentals and frictions. The difference between the sum of each row and 100 is the residual variation or level.

Table 15: Decomposition of observed regional market shares

	Regional Frictions	Borrower Quality	Net Worth	Bank Ability	Capital Cost	Loan Appeal
Loan-weighted average market share	-	1.36	14.82	71.18	9.13	3.51
Equally-weighted average market share	-	1.46	18.08	67.98	8.99	3.49
Across-bank market share variation	-	0.02	3.29	95.54	1.05	0.10
Within-bank market share variation	-	0.02	0.53	98.95	0.46	0.05

Notes: Each row decomposes the variance (or level) of bank-region market shares into percent contributions from demand, ability, cost, borrower quality, and net worth components. The difference between the sum of each row and 100 is the residual variation or level.

evenly to the two co-variates.

Loan Spreads. Table 14 shows the level and variance decompositions of loan spreads. Approximately 73 percent of the activity-weighted average loan spread in the economy is accounted

for by bank-level fundamentals. However, capital costs are by far the largest factor driving the level of loan spreads, accounting for 54.87 percent of observed loan spreads. In comparison, screening ability accounts for 18.18 percent and loan appeal for 0.05 percent. This qualitative pattern remains true when considering the equally-weighted average loan spread in the economy, although the relative contribution of capital costs falls by approximately 18 percentage points to 37.08 percent while screening ability contributes 25.73 percent. The shift in weight between banks' capital funding premia and screening ability is consistent with the empirical relationships documented above. Screening ability is positively correlated with lending and banks' net worth, indicating that weaker screening ability elevates expected defaults and, in turn, loan spreads. Capital funding premia are negatively correlated with bank size, as large banks rely on inter-bank lending markets to fund balance sheet expansion. External factors, including regional market frictions and borrower quality, in turn account for a combined 23.82 percent of observed loan spreads in the loan-weighted decomposition. The bank's pre-existing net worth then makes up an economically modest 3.07 percent of observed loan spreads after accounting for bank-level fundamentals and external factors.

Across-bank variation. The across-bank variance decomposition, by contrast, characterizes the variation in loan rates offered by different banks in the same time period—and it is here that the empirical decomposition most directly speaks to the structural baseline above. Bank screening ability accounts for the largest share of across-bank variation in observed loan spreads, contributing 40.10 percent, followed by net worth at 29.70 percent and capital costs at 24.40 percent. Net worth's substantial share is notable given the motivation for this exercise: because banks are ex ante identical, pre-existing bank size is absent from the structural model's causal channels. Net worth's 29.70 percent contribution to cross-sectional spread variation is therefore the clearest empirical sign of the regulatory-cost and oligopolistic-profit channels through which size operates on loan pricing outside the model. Screening ability's cross-sectional share is smaller here than in the structural decomposition above (40.10 versus 54 percent), consistent with net worth absorbing variation that the structural model, which holds pre-existing size fixed, attributes to screening ability alone. The remaining variation is accounted for by borrower quality, regional frictions, and loan appeal, which together contribute less than 6 percent. Thus, when we observe variation in the loan rates offered by banks across the country, the largest share of those differences is accounted for by differences in banks' ability to screen risk, followed by differences in pre-existing bank size and capital costs, rather than by differences in the risk profiles of the borrowers those banks serve.

Within-bank variation. Lastly, the within-bank variance decomposition characterizes the variation in loan rates charged by the same bank over its lifetime. The pattern reverses: screening ability accounts for 82.50 percent of within-bank fluctuations in loan spreads over a bank's life, while

net worth's contribution falls to a negligible 0.95 percent. Screening ability therefore accounts for an even larger share of within-bank variation in the data than in the structural decomposition, where it accounts for 69.6 percent. The change in loan-rate spreads a bank charges over its life is consequently driven overwhelmingly by changes in its own ability to screen risk, not by changes in its size, funding costs, or the risk profile of its borrowers. Pre-existing size therefore shapes how banks are priced relative to one another at a point in time, but has little bearing on how an individual bank's pricing evolves over its own life.

Market share. Table 15 shows the corresponding level and variance decompositions for a bank's regional market share. Screening ability similarly dominates variation in market shares, both across banks (95.54 percent) and within banks over time (98.95 percent), mirroring the structural decomposition of lending and market-share growth above, though the empirical shares are considerably larger than their structural counterparts (44.5 and 72.9 percent, respectively). Net worth's role here is far smaller than for spreads—3.29 percent across banks and 0.53 percent within banks—indicating that pre-existing size shapes how a bank prices its loans considerably more than how large a share of the market it ultimately captures. Borrower quality, capital costs, and loan appeal together account for less than 5 percent of both the across-bank and within-bank variation in market shares.

Taken together, the empirical and structural decompositions tell a consistent story: screening ability determines who is lending. Banks that grow large are disproportionately those best able to screen and price credit risk—not simply those with access to cheap deposits, a superior capacity to attract borrowers, or an inherited size advantage. The empirical accounting decomposition, computed directly from observable data rather than simulated moments, confirms that this pattern is not an artifact of the model's functional form or calibration, while also revealing a role for pre-existing bank size in cross-sectional loan pricing that the structural model cannot speak to directly. More broadly, the two decompositions are complementary: the structural decomposition isolates the causal role of screening ability in an environment without historical institutional constraints, while the empirical decomposition shows how those forces are reflected in the realized distribution of banks, including the role of pre-existing size that is absent from the model.

G Productivity analysis

This appendix extends the discussion of how aggregate fluctuations in bank fundamentals are transmitted to the macroeconomy, presents proofs of the analytical propositions relating heterogeneity in screening ability to labor productivity costs, and a test of the screening ability-productivity relationship under varying degrees of bank credit reliance among firms.

G.1 Extended transmission analysis

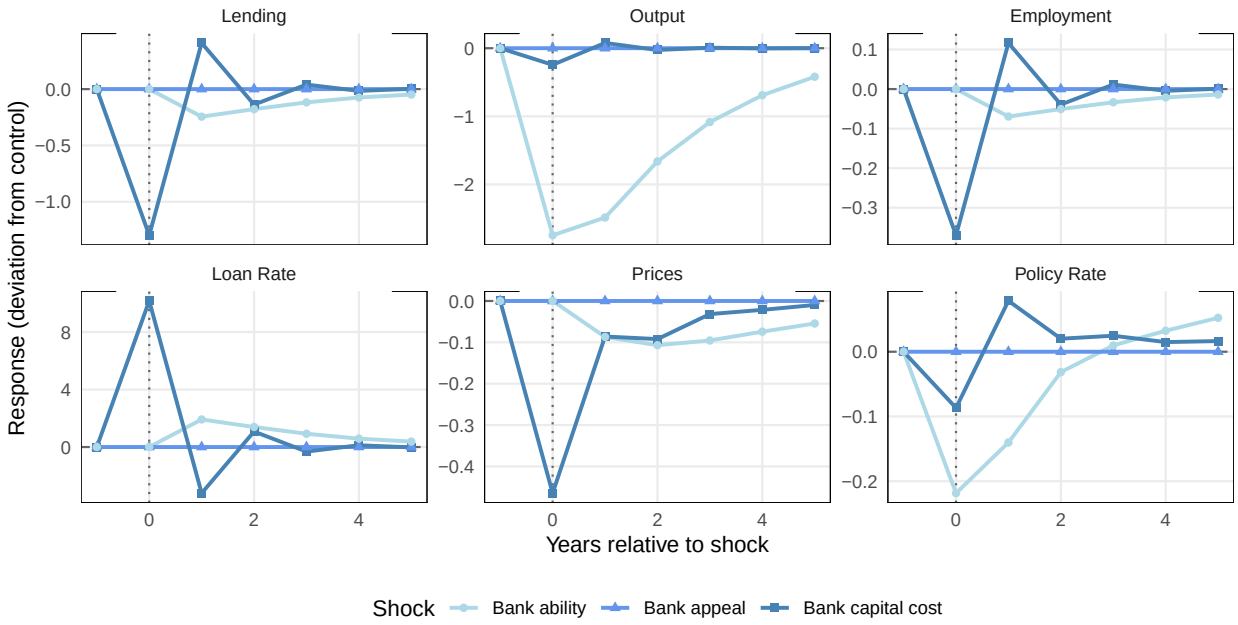
This appendix complements Section 6 and compares the effects of capital premium and loan appeal shocks on the aggregate economy alongside a screening ability shock. To do so, I estimate the IRFs of aggregate output, employment, prices, and the policy rate to aggregate one standard deviation bank fundamental shocks.[35] Figure 33 shows these IRFs based on the mean response of 500 simulated shocks.

Capital cost shocks. First, the capital cost shock is known at the time loans are priced, thus are passed through to borrowers on impact. The increase in loan rates makes firms' inputs more expensive, decreasing the optimal level of production. As planned production falls, then so does the demand for labor, pushing employment and wages down, and in general equilibrium, final goods prices as well. The coinciding decline in prices and output follows the contours of a standard demand shock, and the Central Bank steps in with accommodative policy by lowering the policy rate going into the next period. The easing policy then helps the economy recover starting one period after the deleterious bank funding productivity shock: prices recover approximately 77.8 percent, while output fully recovers and modestly increases relative to its pre-shock level. However, the economic significance of a one standard deviation decrease in bank's capital funding productivity is asymmetric in affect across production inputs and outputs. Lending falls by more than one percent, which pushes employment down by approximately 35 basis points and prices by approximately 45 basis points. However, the decrease in labor inputs is mitigated by curvature in the decreasing returns to scale technology and dampens output by less than one-third of a percent.

Loan appeal shocks. Second, an aggregate loan appeal shock has no material effect on the aggregate economy. This is because loan appeal is a bank's ability to attract customers *relative*

[35] I estimate the IRF of output, employment, and prices as percent deviations from the economies counterfactual evolution that would have been realized in the absence of the shock. Because these variables have a stationary ergodic mean in the simulated economy, averaging over a large number of simulations results in a response that marks the variables deviation from its stationary steady state. This is in contrast to using growth rates for bank-level variables, because an individual bank's net worth can grow unbounded over time, so their pricing, lending, and net worth outcomes are not stationary in level but instead must be presented in growth rates to achieve a stationary time series for analysis.

Figure 33: The macroeconomic response to bank fundamentals



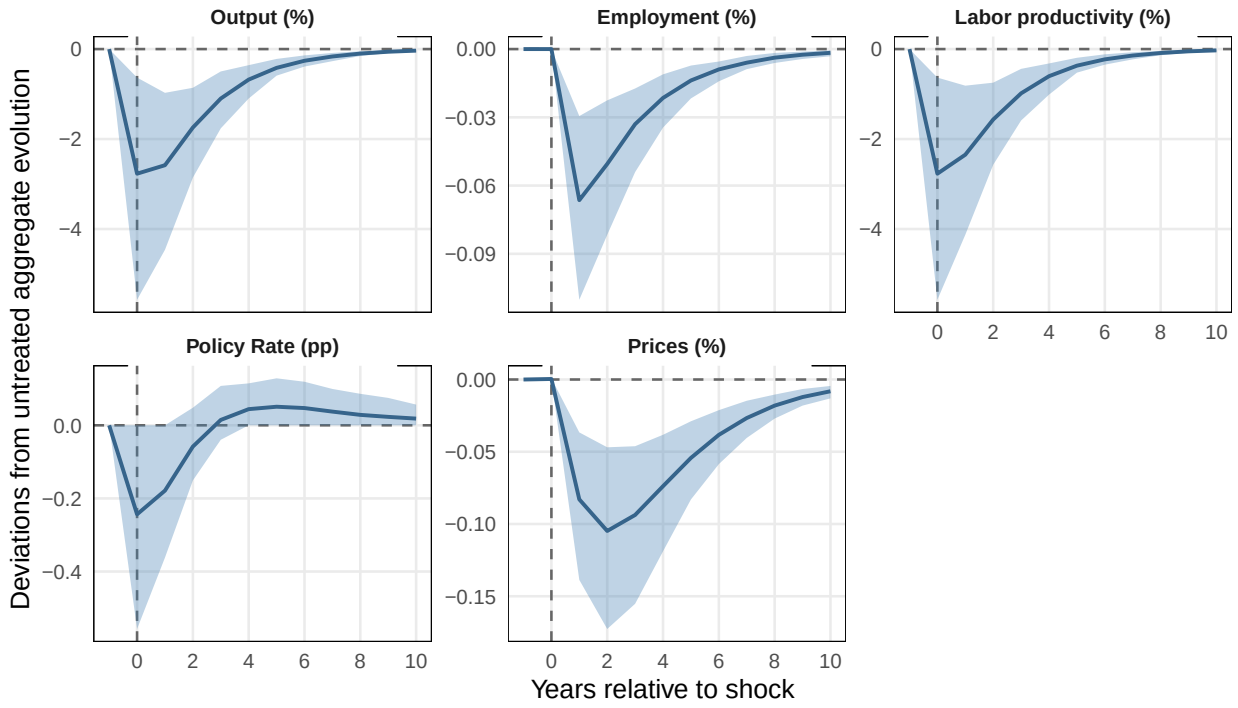
Notes: This figure presents the response of aggregate economic conditions to a one-standard-deviation deleterious shock to screening ability, the capital cost premium, or loan appeal. IRFs are based on the mean response of aggregate conditions after a common shock to bank fundamentals relative, to the counterfactual evolution of the same economy as if no shock had occurred. Outcomes are presented as percent deviation from the control, except for the spread and policy rate which are in percentage points. Responses are based on 500 simulations of the economy.

to its competitors. However, an additive shock to every banks' loan appeal increases no banks' loan appeal relative to its competitors. Thus there is no change in the aggregate economy and the estimated IRFs are zero across all endogenous outcomes and horizons.

Screening ability shocks. Third, the negative screening ability shock dampens output on impact, but not prices. This is because the screening ability shock is not priced into loan rates until they move the banks expected screening ability. As the loan rates are not initially moved by the shock, planned production, hiring, wages, and prices do not change. The broad deterioration in these macro outcomes begins one period after the shock is realized and is being priced into loan rates. Conversely, the decline in screening ability decreases output on impact. This is because when screening ability falls, then more defaulting firms are given loans all else equal, and the incident of exits among producing firms increase, increasing the gap between planned and realized production. The Central Bank observes this fall in output and steps in with accommodative policy, which then eases the decline in activity and prices in the next period.

Figure 34 shows the 90 percent confidence intervals around output, employment, labor productivity, the policy rate and prices, expanding on Figure 8.

Figure 34: The macroeconomic response to an aggregate deterioration in banks' screening ability



Notes: This figure presents the response of aggregate output, employment, labor productivity, prices, and policy to a one standard deviation deleterious shock to screening ability. Solid blue lines are the average response and the blue bands mark the 90 percent confidence intervals. IRFs are based on the mean response of aggregate conditions after a common shock to screening ability relative to the counterfactual evolution of the same economy as if no shock had occurred. Outcomes are presented as percent deviation from the control, except for policy which is in percentage points. Responses are based on 500 simulations of the economy.

G.2 Proofs

The following presents the proofs of Propositions 6.1 and 6.2. Throughout, $\mathcal{B}_{st}$ denotes the set of banks operating in region s at date t, $n_{st} = |\mathcal{B}_{st}| \geq 1$, and $\chi_{it} \equiv \chi_{it}^{\psi} = (1-\psi_{it})/\psi_{it}$. Because $\psi_{it} \in (0,1)$, $\chi_{it} \in (0,\infty)$ for all i and t, so $\log \chi_{it}$ is well defined and all three aggregates χ_{st}^{w}, $\bar{\chi}_{st}$, and χ_{st}^{g} are strictly positive.

Proposition 6.1 (The cost of dispersion)**.** *For any region s and period t, $\chi_{st}^{g} \leq \bar{\chi}_{st}$, with equality if and only if χ_{it} is degenerate across i. If in addition $\log \chi_{it} \sim \mathcal{N}(\mu_{st}, \varsigma_{st}^{2})$ in the cross-section, then*

$$\bar{\chi}_{st} \;=\; \chi_{st}^{g} \exp(\varsigma_{st}^{2}/2), \qquad \bar{\chi}_{st} - \chi_{st}^{g} \;=\; \chi_{st}^{g}\big[\exp(\varsigma_{st}^{2}/2) - 1\big] \;\approx\; \tfrac{1}{2}\,\bar{\chi}_{st}\,\varsigma_{st}^{2}.$$

The proof follows almost exactly from Silver and Heravi (2007), which is the price index measurement work that establishes the connection between the Dutot and Jevons price indices upon

which I base my counterfactual exercises.

Proof. The proof proceeds in three steps. First, I establish the inequality and its equality condition for an arbitrary cross-section. Second, I derive the exact relationship under cross-sectional lognormality. Third, I characterize the approximation and bound its error.

Step 1: The inequality. Fix (s,t) and suppress these subscripts. Define $y_i = \log \chi_i$ for $i \in \mathcal{B}$ and $\bar{y} = n^{-1} \sum_{i \in \mathcal{B}} y_i$, so that $\chi^g = \exp(\bar{y})$ and $\bar{\chi} = n^{-1} \sum_{i \in \mathcal{B}} \exp(y_i)$. The exponential function is strictly convex, so it lies weakly above its tangent line at $\bar{y}$:

$$\exp(y_i) \;\geq\; \exp(\bar{y}) + \exp(\bar{y})\,(y_i - \bar{y}) \qquad \text{for all } i \in \mathcal{B}, \tag{55}$$

with equality if and only if $y_i = \bar{y}$. Averaging (55) over $i \in \mathcal{B}$ and using $n^{-1} \sum_i (y_i - \bar{y}) = 0$ gives

$$\bar{\chi} \;=\; \frac{1}{n} \sum_{i \in \mathcal{B}} \exp(y_i) \;\geq\; \exp(\bar{y}) \;=\; \chi^g.$$

Equivalently, $\bar{\chi} - \chi^g = n^{-1} \sum_i D_i$, where $D_i \equiv \exp(y_i) - \exp(\bar{y})[1 + (y_i - \bar{y})] \geq 0$.

For the equality condition, note that $\bar{\chi} = \chi^g$ if and only if $\sum_i D_i = 0$. Because each D_i is nonnegative, this holds if and only if $D_i = 0$ for every i. By strict convexity, $D_i = 0$ if and only if $y_i = \bar{y}$. Hence equality holds if and only if $\log \chi_i = \bar{y}$ for all $i \in \mathcal{B}$, that is, if and only if χ_i is degenerate across i. (If $n = 1$, the distribution is trivially degenerate and $\chi^g = \bar{\chi} = \chi_1$.)

Step 2: The exact relationship under lognormality. Suppose that the cross-sectional distribution of $Y \equiv \log \chi$ is $\mathcal{N}(\mu, \varsigma^2)$. Then the equally weighted aggregates are the corresponding cross-sectional moments:

$$\chi^g = \exp\big(\mathbb{E}[Y]\big) = \exp(\mu), \qquad \bar{\chi} = \mathbb{E}\big[\exp(Y)\big].$$

The moment-generating function of a normal random variable is $\mathbb{E}[\exp(\tau Y)] = \exp(\tau\mu + \frac{1}{2}\tau^2\varsigma^2)$. Evaluating it at $\tau = 1$ gives

$$\bar{\chi} \;=\; \exp\big(\mu + \varsigma^2/2\big) \;=\; \chi^g \exp\big(\varsigma^2/2\big),$$

which is the first display. Two consequences follow. Subtracting χ^g from both sides gives $\bar{\chi} - \chi^g = \chi^g[\exp(\varsigma^2/2) - 1]$. Taking logarithms gives

$$\log \bar{\chi} - \log \chi^g \;=\; \varsigma^2/2,$$

so the log gap between the arithmetic and geometric aggregates equals exactly one half of the cross-sectional variance of log screening ability. This is the same identity that links the Dutot and

Jevons elementary indices under lognormality.[36]

Step 3: The approximation and its error. Let $x \equiv \varsigma^2/2 \geq 0$. Using $\chi^g = \bar{\chi}\exp(-x)$ from Step 2, the gap can be written in two equivalent forms:

$$\bar{\chi} - \chi^g \;=\; \chi^g\big[\exp(x) - 1\big] \;=\; \bar{\chi}\big[1 - \exp(-x)\big].$$

For $x \geq 0$, the elementary inequalities $\exp(x) - 1 \geq x$ and $x - x^2/2 \leq 1 - \exp(-x) \leq x$ hold. Applying the first inequality to the first form, and the second to the second form, yields the bounds

$$\tfrac{1}{2}\chi^g\varsigma^2 \;\leq\; \bar{\chi} - \chi^g \;\leq\; \tfrac{1}{2}\bar{\chi}\varsigma^2, \qquad \bar{\chi} - \chi^g \;\geq\; \tfrac{1}{2}\bar{\chi}\varsigma^2\Big(1 - \tfrac{\varsigma^2}{4}\Big).$$

Therefore $\bar{\chi} - \chi^g = \frac{1}{2}\bar{\chi}\varsigma^2 + O(\varsigma^4)$. The approximation in the proposition is first order in the variance of log screening ability, and it is an upper bound on the exact gap. Its relative error is at most $\varsigma^2/4$. ■

Proposition 6.2 (The gain from allocation)**.** *For all regions s and any period t with $\sum_{i\in\mathcal{B}_{st}} s_{ist} = 1$,*

$$\chi^w_{st} - \bar{\chi}_{st} \;=\; \sum_{i\in\mathcal{B}_{st}} \Big(s_{ist} - \tfrac{1}{n_{st}}\Big)\,(\chi_{it} - \bar{\chi}_{st}) \;=\; n_{st}\,\widehat{\mathrm{cov}}\big(s_{ist},\, \chi_{it}\big),$$

where $\widehat{\mathrm{cov}}$ denotes the unweighted cross-bank sample covariance.[37] *The identity is exact and requires only that the shares sum to one.*

The proof follows almost exactly from Olley and Pakes (1996), which first establishes this standard covariance representation of aggregate productivity arising from production sector with uneven market share and firm-level productivity.

Proof. The proof proceeds in three steps. Fix (s,t) and suppress these subscripts.

Step 1: The gap as a deviation-weighted sum. By the definitions of χ^w and $\bar{\chi}$,

$$\chi^w - \bar{\chi} \;=\; \sum_{i\in\mathcal{B}} s_i\,\chi_i \;-\; \sum_{i\in\mathcal{B}} \tfrac{1}{n}\,\chi_i \;=\; \sum_{i\in\mathcal{B}} \Big(s_i - \tfrac{1}{n}\Big)\chi_i.$$

[36] If $\mathcal{B}$ is a finite set of n independent draws from $\mathcal{N}(\mu,\varsigma^2)$, then χ^g and $\bar{\chi}$ are sample statistics, and the identities in Step 2 hold as probability limits as $n \to \infty$ by the law of large numbers and the continuous mapping theorem. In finite samples, $\bar{y} \sim \mathcal{N}(\mu, \varsigma^2/n)$, which implies $\mathbb{E}[\chi^g] = \exp(\mu + \varsigma^2/(2n))$ and $\mathbb{E}[\bar{\chi}] = \exp(\mu + \varsigma^2/2)$. Hence $\mathbb{E}[\bar{\chi}]/\mathbb{E}[\chi^g] = \exp(\frac{n-1}{n}\,\varsigma^2/2)$, which converges to the population expression at rate $1/n$. Step 1, by contrast, holds exactly for every finite cross-section.

[37] Here $\widehat{\mathrm{cov}}(s_{ist}, \chi_{it}) \equiv n_{st}^{-1}\sum_{i\in\mathcal{B}_{st}}(s_{ist} - \bar{s}_{st})(\chi_{it} - \bar{\chi}_{st})$, with $\bar{s}_{st} \equiv n_{st}^{-1}\sum_{i\in\mathcal{B}_{st}} s_{ist}$. If the covariance is instead defined with divisor $n_{st} - 1$, the factor n_{st} in the identity is replaced by $n_{st} - 1$. Nothing else changes.

Step 2: Centering. The share deviations sum to zero, because $\sum_i (s_i - 1/n) = \sum_i s_i - 1 = 0$. It follows that $\bar{\chi} \sum_i (s_i - 1/n) = 0$. Subtracting this term from the expression in Step 1 gives

$$\chi^w - \bar{\chi} \;=\; \sum_{i \in \mathcal{B}} \Big(s_i - \tfrac{1}{n}\Big)\big(\chi_i - \bar{\chi}\big),$$

which is the first equality.

Step 3: The covariance form. Because $\sum_i s_i = 1$, the unweighted mean share is $\bar{s} = 1/n$. Therefore $s_i - 1/n = s_i - \bar{s}$ for every i, and

$$\sum_{i \in \mathcal{B}} \big(s_i - \bar{s}\big)\big(\chi_i - \bar{\chi}\big) \;=\; n \cdot \frac{1}{n} \sum_{i \in \mathcal{B}} \big(s_i - \bar{s}\big)\big(\chi_i - \bar{\chi}\big) \;=\; n\, \widehat{\mathrm{cov}}(s_i, \chi_i),$$

which is the second equality. ■

G.3 Sensitivity to working-capital financing intensity

The productivity results of Sections 6.1 and 6.3 rest on two assumptions that jointly maximize the exposure of real activity to the banking sector. First, the firm's problem requires that the entire wage bill be financed with a working-capital loan, so the regional loan rate passes one for one into the marginal cost of labor. Second, the realized exit rate applies the screening wedge to every unit of regional production, so all output is exposed to the screening ability of the bank that funded it. In the data, firms finance only part of production externally, and only part of external finance is bank credit. Both assumptions therefore dampen the role of banks in influencing real activity.

This appendix relaxes the two margins with a single parameter and recomputes the two headline productivity objects: the transmission of an aggregate screening ability shock to labor productivity, and the productivity cost and gain of cross-bank heterogeneity in screening ability.

G.3.1 Two margins of exposure

Intensive margin: the cost of labor. Let $\phi \in [0, 1]$ denote the share of the wage bill that the representative regional firm pays in advance out of bank credit. The residual share $1 - \phi$ is paid out of contemporaneous revenue and carries no financing cost. The effective gross loan rate on the wage bill is:

$$\tilde{R}_{st}(\phi) \;\equiv\; (1 - \phi) + \phi\, R_{st}, \tag{56}$$

which replaces R_{st} in the firm's labor demand condition and in the labor-market-clearing wage. No other equilibrium condition of the regional block changes. The specification is a partial working-capital constraint, nesting the full-financing case. At $\phi = 1$ the entire wage bill must be financed

in advance, while at $\phi = 0$ no portion of the wage bill must be financed in advance and the loan rate no longer enters the allocation of labor. Note that $\tilde{R}_{st} - 1 = \phi\,(R_{st} - 1)$, so ϕ scales the net financing premium on labor exactly, independently of the level of the loan rate.

Extensive margin: selection. Let $\theta \in [0, 1]$ denote the share of regional production whose survival depends on the screening of the bank that funded it. The residual share is funded without screening and exits at the unconditional borrower exit rate. The realized regional exit rate becomes the mixture

$$\lambda_{st}(\theta) \;=\; \theta\,\chi^{q}_{st}\,\chi^{\psi}_{st} + (1-\theta)\,\chi^{q}_{st} \;=\; \chi^{q}_{st}\big[\,1 - \theta\,(1-\chi^{\psi}_{st})\,\big]. \tag{57}$$

Unscreened production is equivalent to production funded by a lender with $\chi^{\psi} = 1$, that is, $\log \chi^{\psi} = 0$ and $\psi = 1/2$: a lender whose portfolio fails at the population rate because its approval decision carries no information about borrower quality. Because the estimated wedge satisfies $\chi^{w}_{st} < 1$, the exit rate is decreasing in θ. Lowering θ therefore both weakens the exposure of output to screening and raises the level of exit, since the unscreened share forgoes the selection that banks provide.

Throughout, I impose $\theta = \phi$: the share of production whose survival depends on the lender is the share the lender finances. Under this restriction a calibration is a single number from which both margins are read, and the family nests the two limits of interest. At $\phi = \theta = 1$, the model of the main text obtains exactly. At $\phi = \theta = 0$, the banking sector is severed from real allocation: employment does not respond to the loan rate, and all production exits at the unscreened rate χ^{q}_{st}.

G.3.2 Exercises

I run two exercises for each calibration. The first studies the transmission of an aggregate screening shock to labor productivity. The second studies the steady-state cost of screening ability heterogeneity. It measures the productivity cost of cross-bank dispersion in screening ability and the gain from the allocation of credit across banks, evaluated at the empirical dispersion $\varsigma^{*} = 0.51$ (as in Section 6.3). Together, the two exercises show how the short-run and long-run productivity consequences of bank screening change as the exposure of real activity to the banking sector falls.

Calibration. I report three calibrations, $\phi = \theta \in \{1.00, 0.35, 0.20\}$. The first, $\phi = \theta = 1$, is the baseline model, in which firms must finance their entire wage bill before production can begin. The second, $\phi = \theta = 0.35$, is the calibrated bank credit reliance scenario, in which firms must finance only part of their wage bill in advance. I calibrate it to two estimates. First, Ajello (2016) estimates that U.S. public firms fund 35.89 percent of quarterly investment in fixed capital by accessing financial markets rather than using current operating cash flow. This estimate captures

firms' ability to finance production with internal funds. Public firms, however, also have access to bond and equity markets, so the estimate measures reliance on external finance broadly and overstates reliance on bank lending specifically.[38] Second, Doerr (2021) and Doerr et al. (2026) use the 2007 U.S. Census Bureau Survey of Business Owners to measure reliance on bank credit to start or expand young firms, which implies a reliance of 35.27 percent.[39] Because young firms have limited access to public debt markets and play a relatively small role in total employment (i.e., economic activity), this estimate overstates the role of bank credit in financing employment-weighted activity across the economy. The two estimates are close, and I set $\phi = \theta = 0.35$. Because both likely overstate the role of bank credit in financing productions, I add a third, conservative calibration, $\phi = \theta = 0.20$. In the absence of microdata that decompose firms' working capital into bank and nonbank sources, this value is not pinned down precisely, and is included for its value in drawing out a qualitative pattern.

Methodology. The sensitivity exercise differ from the baseline exercise only where necessary. There are three substantive changes made to the economy: the labor-market-clearing wage, the labor demand schedule, and the realized regional exit rate. The aggregate forecasting rules are re-solved for each calibration.

The exercise rests on three maintained assumptions. First, the banking side of the model is left as in the main text. Firms still borrow the full wage bill, and banks still lose principal on all of it at their own default rates, even though firms now pay interest on only part of the bill and only part of production is exposed to bank screening. This keeps bank balance sheets and market structure fixed, so that the exercise changes only how strongly banks affect real activity. The cost is that, below full financing, what firms pay and what banks earn no longer match exactly. Second, production not financed by banks is treated as unscreened: it fails at the same rate as the average borrower. This choice affects the overall level of firm exit in the lower-ϕ calibrations but not the dispersion cost or allocation gain, which depend only on differences in screening across banks. Third, the cost and gain of heterogeneity are accounting exercises that hold employment, technology, and loan rates fixed while swapping the screening wedge. The calibrations also inherit the main text's assumptions that every bank operates in every region and that banks neither enter nor exit.

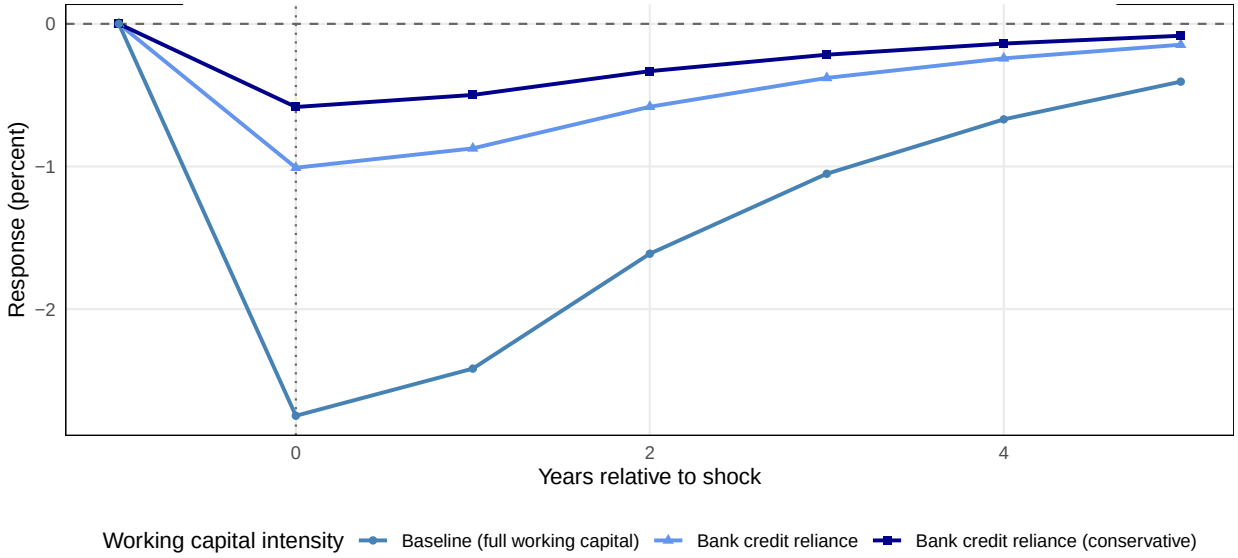
Results. The following summarizes the two sets of results.

Short-run transmission. Negative screening ability shocks have a qualitatively consistent deleterious effect on labor productivity across bank credit reliance calibrations. Figure 35 shows the

[38] This decomposition is in the spirit of Rajan and Zingales (1996), who study the analogous breakdown of public firms' funding sources.

[39] 35.27 percent is the employment-weighted two-digit NAICS industry average fraction of firms that report relying on bank credit to expand.

Figure 35: Labor productivity response to a screening ability shock, by working-capital intensity



Note: This plot shows the response of aggregate labor productivity to a one-standard-deviation aggregate deterioration in screening ability, in percent deviation from the control economy, for $\phi = \theta \in \{1.00, 0.35, 0.20\}$. Forecasting rules are re-solved for each calibration. Responses are means of 250 simulations.

Table 16: Dispersion Cost and Allocation Gain by Working-Capital Intensity

	$\phi = \theta$	Gain	Cost	Net
Full working capital (baseline)	1.00	0.039	5.008	−4.969
Calibrated bank credit reliance	0.35	0.014	1.839	−1.825
Conservative bank credit reliance	0.20	0.008	1.063	−1.055

Notes: This table shows the gain, cost, and net effect are in percent of aggregate labor productivity, evaluated at the empirical dispersion $\varsigma^* = 0.51$. The net effect is the gain less the cost, so negative values are productivity losses. ϕ is the working-capital financing intensity and θ the share of production exposed to bank screening. Estimates are means over 250 simulations.

response of labor productivity to a one-standard-deviation negative screening ability shock across calibrations. Under the full working-capital financing requirement scenario, labor productivity falls by approximately 2.7 percent on impact. In the calibrated bank credit reliance scenario ($\phi = 0.35$), transmission is muted, and labor productivity falls by approximately one percent on impact. Transmission weakens further as reliance on bank credit falls. In the conservative scenario ($\phi = 0.20$), labor productivity falls by approximately half a percent on impact.

The impact response therefore scales roughly in proportion to ϕ. This is expected: the selection channel accounts for almost the entirety of the baseline response, and under Equation 57 it operates only on the share of production exposed to bank screening. The short-run effect of bank screening ability on labor productivity is thus qualitatively robust to the economy's reliance on bank credit,

while its magnitude rises with that reliance.

Long-run cost of heterogeneity. Dispersion in screening ability has a qualitatively consistent deleterious effect on long-run labor productivity across bank credit reliance calibrations. Table 16 reports the gain from allocation, the cost of dispersion, and their net effect under each calibration. Under a full working-capital financing requirement, the net effect is a reduction in labor productivity of approximately five percent relative to an economy with no heterogeneity in screening ability. As reliance on bank credit falls, so does the effect of screening heterogeneity. The net reduction falls to 1.8 percent in the calibrated bank credit reliance scenario and to one percent in the conservative scenario.

This decline, however, largely reflects the smaller share of production exposed to bank screening rather than a weaker mechanism. Under Equation 57, dispersion and allocation affect exit only through the screened share θ of production, so both the cost and the gain scale with θ. Normalized by θ, the cost is nearly invariant across calibrations, and in fact rises modestly from 5.0 to 5.3 percent as θ falls.

H Empirical Heterogeneity in Monetary Policy Transmission

This appendix presents an empirical counterpart to the simulated policy transmission exercise in Section 7, which sorts economies on four banking-sector characteristics measured at the time of a monetary policy shock. The following introduces the data, model, and identification, before presenting the estimated state-dependent responses.

Data. The four banking-sector characteristics are collected in

$$Z_t \equiv \left(Z_t^{\chi},\ Z_t^{\sigma\chi},\ Z_t^{N},\ Z_t^{\sigma N}\right)', \tag{58}$$

namely the aggregate screening wedge, χ_t^{ψ}, its cross-sectional dispersion, aggregate bank net worth, N_t, and its cross-sectional dispersion. Each is oriented so that higher values correspond to worse or more dispersed screening ability and to larger or more dispersed net worth.

The three outcomes are log real GDP, the year-over-year log growth rate of aggregate bank loans, and core PCE inflation. The monetary policy shock ε_t^m is the high-frequency surprise in three-month-ahead federal funds futures (FF4) around FOMC announcements from Jarociński and Karadi (2020), summed within quarter and set to zero in quarters without an announcement.

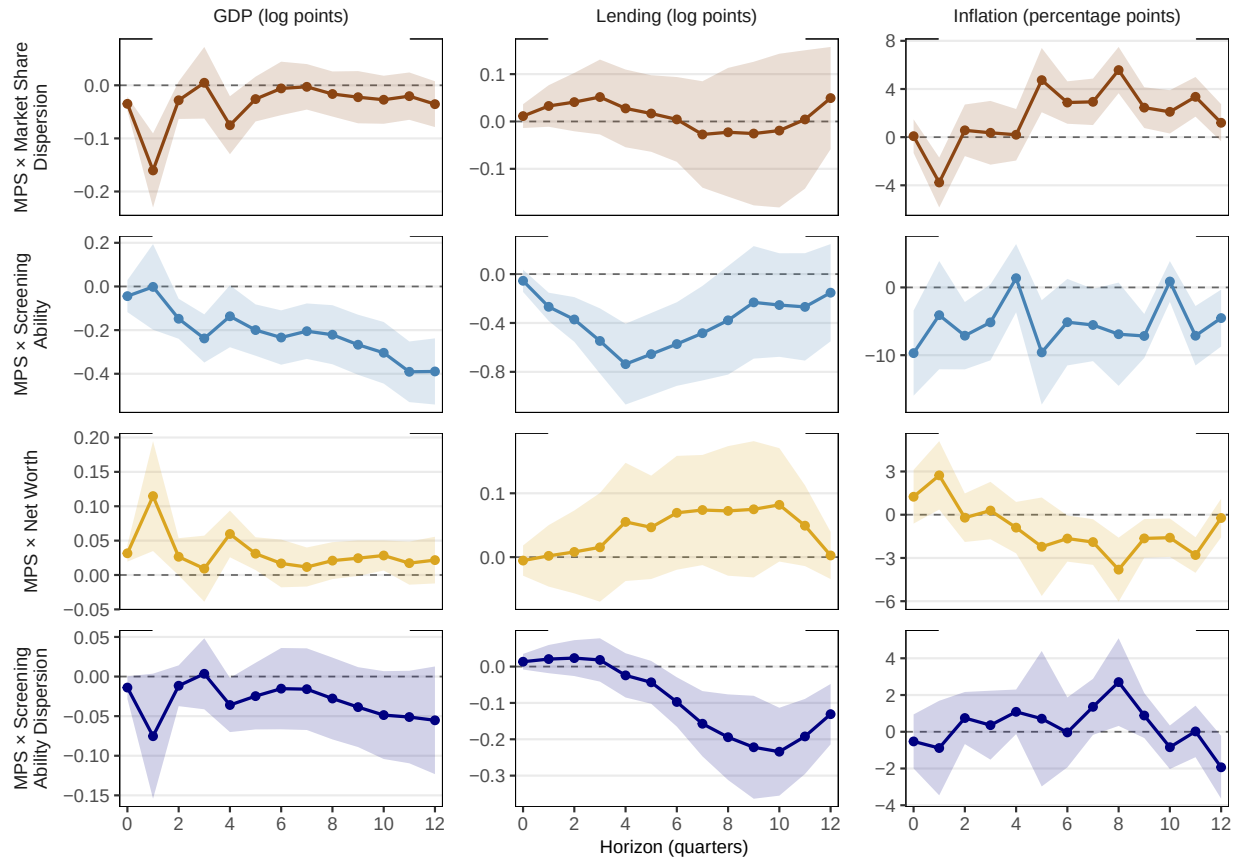
The bank characteristics are observed annually. Each annual value is anchored at the fourth quarter and interpolated to quarterly frequency, log-linearly for aggregate loans and linearly in levels otherwise.

Model. For each outcome y_t, characteristic $k \in \{\chi, \sigma\chi, N, \sigma N\}$, and horizon $h = 0, 1, \ldots, 12$ quarters, the state-dependent local projection (Jordà, 2005) is

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h\, \varepsilon_t^m + \gamma_h\, \tilde{Z}_t^k + \delta_h\left(\varepsilon_t^m \times \tilde{Z}_t^k\right) + \Gamma_h' X_t + u_{t+h}, \tag{59}$$

where $\tilde{Z}_t^k$ is the characteristic standardized to have mean zero and unit variance over the full sample, and X_t contains four lags each of year-over-year real GDP growth, core PCE inflation, the effective federal funds rate, and loans. Each characteristic enters a separate regression. The coefficient of interest, δ_h, is the change in the horizon-h response to a unit policy shock when the characteristic is one standard deviation above its sample mean; β_h is the response at the sample-average characteristic. Because (59) is estimated in cumulative-change form, δ_h measures the effect on the level of each outcome at horizon h relative to the quarter preceding the shock. The sign convention mirrors the simulated exercise: for a tightening shock with $\beta_h < 0$, a characteristic amplifies transmission if $\delta_h < 0$ and dampens it if $\delta_h > 0$.

Figure 36: Heterogeneous effects of monetary policy by banking-sector characteristics



Notes: This figure presents the estimated interaction coefficients δ_h from (59) for each banking-sector characteristic (rows) and outcome (columns) across zero through 12 quarters. Each characteristic is standardized with mean zero, unit variance, and enters a separate regression. For a tightening shock, $\delta_h < 0$ denotes amplification and $\delta_h > 0$ dampening. Bands denote 90 percent confidence intervals based on Newey–West standard errors with bandwidth $h+1$. Data are quarterly from 1995 through 2023.

Identification. Identification of β_h comes from high-frequency variation in policy surprises around FOMC announcements. However, the banking-sector characteristics are not randomly assigned, so δ_h establishes only that transmission is conditionally state dependent, not that the characteristic itself causes the differential response. This is because, unlike the quantitative setting in which the characteristic is varied holding all else fixed by construction, each Z_t^k may covary with other aggregate states that independently condition the transmission of monetary policy.

Results. Figure 36 reports δ_h for each characteristic and outcome. Screening ability has the most economically and statistically significant effect on the conditional transmission of monetary policy to output or lending. A one standard deviation increase in aggregate screening ability wedge —which is inversely related to the level of screening ability— amplifies the deterioration in lending

and output growth induced by a negative one standard deviation monetary policy tightening shock. In particular, a one standard deviation increase in the screening wedge further slows lending by approximately 80 basis points one year after impact, and slows GDP growth by approximately 16 basis points one year and up to 40 basis points three years after impact.

In comparison, the dispersion in screening ability has no statistically significant impact on lending until six quarters after impact, before slowing lending by approximately 22 basis points by 10 quarters after impact. It has no corresponding effect on the transmission to output.

Net worth has a short-lived but predictable effect on the transmission of policy. A one standard deviation increase in the aggregate net worth of the banking sector ameliorates the decline in output growth by 4 basis points. This is in line with the macro-banking literature mechanism linking bank's net worth to the transmission of policy. Likewise, the dispersion in net worth amplifies the transmission of the shock to output growth on impact, but this is a transitory effect that one lasts one period. However, neither banking sector size characteristic has an effect on the transmission o to lending outcomes, and no characteristic studied appears to have a meaningful effect on inflation dynamics.

Taken together, these results provide suggestive evidence in favor of the quantitative results that both the level and dispersion in screening ability determine the amplification of monetary policy.